

- pH 6.0, is used also for the broad pH gradient chromatofocusing method).
3. Make sure that the intended connections between the buffer reservoirs and the P-500 pumps will be correct: the start buffer (buffer A) to pump A and the eluent buffer (buffer B) to pump B. If two different eluent buffers are used, the first is connected to pump A and the second to pump B.
 4. Perform blank runs in order to check the equipment, programs, pH gradients, pressures, background absorbances, and baselines.

3.2. Sample Preparation

1. Lipid-containing samples should be delipidated. This is recommended for all sera, but can generally be omitted for CSF samples (unless they are known to be significantly contaminated by lipids). Depending on the available amount of material, delipidation may be performed before or after adjustment of the IgG concentration. The sample is mixed with an equal volume of chloroform, thoroughly agitated, and, 30 min later, centrifuged at 1400–1500g for 10 min.
2. The IgG concentration is adjusted to 1 mg/mL by either dilution with 0.15M sodium chloride or concentration, e.g., by vacuum ultrafiltration.
3. The samples are diluted 1:5 in start buffer (giving an IgG concentration of 0.2 mg/mL) and filtered through a sterile 0.2- μ m filter. There should not be any turbidity of the final material.

3.3. Sample Loading and Running

The equipment is switched on. It should be noted that the UV-1 single-path monitor will need about 2 h to

warm up. The buffer reservoirs are connected to their respective P-500 pumps.

3.3.1. *Chromatofocusing by Use of the Mono P Column and Polybuffer 96*

The basic parameters of the chromatofocusing program (volume based) are the following:

1. The Mono P column has been equilibrated with start buffer.
Buffers injected into the column after sample loading: 5 mL of 100% buffer A (start buffer) followed by 35–40 mL^a of 100% buffer B (10% Polybuffer 96, pH 6.0).^b
Flow rate 0.25 mL/min.
UV absorbance of the eluate measured at 280 nm (AU mode) and range selector set at 0.02.
Input to one channel of the recorder from the UV monitor, the second channel registers % buffer B or pH of the eluate (when a flow-through pH electrode is used); chart speed 0.25 cm/min.
Collection of consecutive 1-mL fractions.
2. Turn the V-7 valve to the "load" position. Check the positions of the V-8 valves if such are used to connect the column to the system.
3. Inject 0.5 mL of the prepared sample (having an IgG content of 0.1 mg) into the 0.5-mL sample loop.
4. Turn the V-7 valve to the "inject" position.
5. Enter the appropriate method number in instruction "Run method 0" on the display for the GP-250/LCC-500 unit and start the run by pressing "do store."
6. Run the separation.

3.3.2. *Chromatofocusing by Use of the Small (HR 5/5) Column and Polybuffer 96*

1. The same basic parameters are used with the following exceptions: Buffers injected into the column after sample loading: 2 mL of 100% buffer A (start buffer) followed by 15 mL of 100% buffer B (10% Polybuffer 96, pH 6.0).^b
Range selector set at 0.05–0.1.
- 2–6 The same procedures as in section 3.3.1.

3.3.3. *Chromatofocusing by Use of the Mono P Column and a Combination of Polybuffer 96 and 74*

1. The instrument set-up and basic parameters of the program deviate from those of the previous methods as follows:
A 10-mL Superloop is introduced between the P-500 pumps and the V-7 valve. The superloop is used as start buffer reservoir. Pumps A and B are used for the first and second eluent buffers, respectively. Buffers injected into the column after sample loading: 10 mL of 100% start buffer followed by 100% of the first (10% Polybuffer 96, pH 6.0) and second (10% Polybuffer 74, pH 4.0) eluent buffer for 25 and 45 mL, respectively. Range selector set at 0.02–0.1.
2. Fill the Superloop with 10 mL of start buffer.
- 3–7 The same procedures as step 2–6 in section 3.3.1.

^aA buffer B volume of 35 mL brings down the pH of the eluate to 6.5; an increment of the buffer B volume to 40 mL gives a final eluate pH of 6.0.

^bIn our previous applications of this chromatofocusing procedure (8,9,11,12), a final start buffer volume was injected after the eluent buffer in order to get a more well-defined state at the end of the separation. We have found this procedure to be unnecessary, however.

3.4. Column Washing and Regeneration

1. It is recommended to use a reversed-flow direction with these procedures. The 1.0 mL sample loop is used for the solutions.
2. Turn the V-7 valve to the "load" position, and inject 1 mL of 70% (v/v) acetic acid into the sample loop. Turn the V-7 valve to the "inject" position, and run the acetic acid through the column. Totally, three such procedures are performed.
3. Use the same technique to run three 1-mL volumes of 2M sodium chloride through the column.
4. Equilibrate the column with start buffer. This is accomplished by running start buffer through the column until the pH of the eluate is the same as that of the buffer.

3.5. Result Evaluation

1. The pH gradient as well as the reproducibility of the separations should be checked. Registration of the gradient is performed by either pH measurements on the collected fractions or use of a flow-through pH electrode. The reproducibility of the separations is checked by duplicate runs.
2. Evaluations of the standard of the separations are based on investigations of the final product, i. e., the material in the collected fractions, by suitable qualitative or quantitative methods. This step is especially important during the period of method development for a certain application, but should also be performed regularly when the technique is run more routinely.

Instructions for cleaning of heavily contaminated columns as well as for long-time storage of columns and equipment are available from the manufacturer.

4. Notes

1. As already stated in the introductory section, the present chromatofocusing methods were designed for applications on samples with relatively low protein amounts. Furthermore, in order to be suited for IgG separations, they were based on the use of quite broad pH gradients embracing the alkaline range. Figure 2 exemplifies the results from small-column chromatofocusing (section 3.3.2) of CSF with oligoclonal IgG and the paired serum sample with normal IgG. Figure 3 gives an example from chromatofocusing of normal CSF by use of the Mono P column and two eluent buffers (section 3.3.3). As shown by the figures, these two techniques give gradients covering the interval of pH 9.5–6.0 and 9.5–4.0, respectively. Figure 3 also illustrates the chromatofocusing profile obtained from normal IgG with the method in section 3.3.1. An eluent buffer volume of 35 mL for this latter method gives a gradient between pH 9.5 and 6.5; an increment of the eluent buffer volume from 35 to 40 mL will bring the acidic end of the gradient to pH 6.0.
2. The three chromatofocusing methods have been applied for separations of samples having an IgG content of 0.025–0.5 mg and a total protein amount in the range of 0.3–6 mg. These quantities are substantially lower than the protein capacity of the Mono P column: about 25 mg/column or 5 mg/single peak. The corresponding values for the small column are in the range of 5–10 and 1–2 mg, respectively. It is recommended to use relatively low protein loads, e.g., 0.1–0.2 mg IgG, during optimization of the method. Scaling up can then be performed by gradual increments of the protein load.

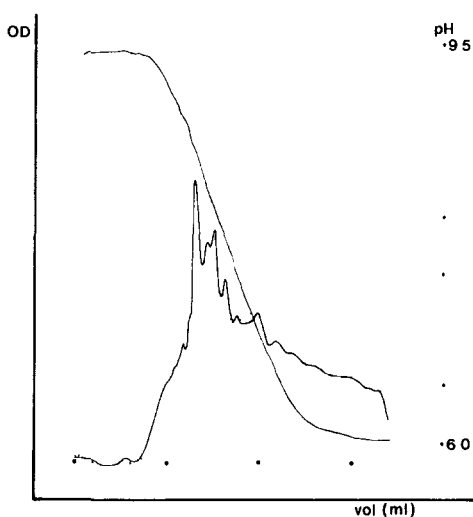


Fig. 2. Chromatofocusing (section 3.3.2) of CSF (continuous line) and serum (dotted line) from a patient with oligoclonal CSF IgG. The X-axis gives the initial start buffer volume (2 mL) followed by 15 mL of the eluent buffer; the Y-axis shows the UV absorbance of the eluate. The pH values are indicated to the right, and the continuous line across the graph shows the pH gradient. There are distinct peaks in the CSF profile that correspond to oligoclonal IgG (reproduced from ref. 14 with kind permission from Pergamon Press).

At the same time, signs of a deteriorating resolution must be looked for, since this will occur with too high-protein loads. Preparative separations for obtaining larger protein amounts are in that case performed by repetitive runs on submaximal protein amounts.

3. It is obvious that the choice of the pH gradient, i.e., the start and eluent buffers, has to be based on the *pI* of the protein(s) of interest. Naturally, the sample must also be compatible with the pH values and ionic strengths of the buffers. Proteins known to give well-focused bands at isoelectric focusing can be expected to give good results also at chromatofocusing. It may, however, be found that certain

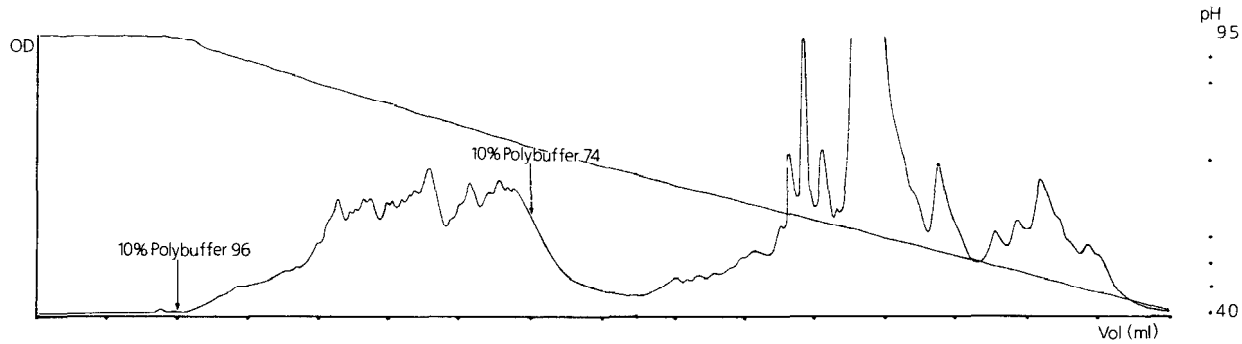


Fig. 3. Chromatofocusing (section 3.3.3.) of normal CSF. In this case, the X-axis gives the initial start buffer volume (10 mL) followed by 25 mL of the first and 45 mL of the second eluent buffer. The left part of the graph shows the IgG profile. The right part of the graph corresponds to proteins with more acidic pI values. albumin gives the broad prominent peak (partly outside the graph), and the three narrow peaks to the left correspond to transferrin, including the tau fraction (reproduced from ref. 10, with kind permission from Pergamon Press).

proteins that focus poorly on isoelectric focusing give more satisfactory results on chromatofocusing. In spite of the similarities between isoelectric focusing and chromatofocusing, it should be observed that the elution pH of a certain protein at chromatofocusing may deviate from its *pI*. This is probably related to displacement effects and/or disproportionation of buffer molecules between matrix pores and the bulk solution.

4. In principle, a separation program for a certain sample is a compromise between three factors: resolution, speed, and dilution of the material obtained in the eluate. The design of the pH gradient, the flow rate used for the separation, and the column type are of major significance for the aforementioned factors.
5. The characteristics of the pH gradient have profound effects on the resolution capacity and are defined by the buffer compositions. A flat gradient increases the resolution and is obtained by using a relatively diluted eluent buffer. This augments the necessary volume of this buffer, however, with consequences for the speed of the separation as well as the dilution of the material in the eluate. Too-dilute eluent buffers will also introduce risks of poor pH gradient shapes. Steeper gradients increase the speed and reduce the dilution of the material in the eluate, but the resolution is decreased. Furthermore, high-eluent buffer concentrations will give elevated background UV absorbances that may interfere with the detection of protein components. Mostly, 10% (v/v) Polybuffer solutions have proved optimal. Naturally, unnecessarily broad pH gradients should not be used. The gradient should also be designed in such a way that the components of interest elute in the inter-

mediate section of the pH interval and especially not in the terminal part where an impaired resolution may occur. The initial start buffer volume has to be large enough to eliminate proteins not bound to the column.

6. An increment of the flow rate increases the speed of the separation, but there will be some reduction of the resolution. We have, in spite of the relatively long separation times (about 3 and 5 h for the method in section 3.3.1 and 3.3.3, respectively), mostly used a flow rate of 0.25 mL/min. Flow rates of 0.5 or 1 mL/min have, however, also given quite good results.
7. A reduction of the column size makes it possible to form the gradient with a lower amount of eluent buffer. In turn, this decreases the separation time (about 75 min for the method in section 3.3.2) and the dilution of the material in the eluate. This latter fact also gives a more favorable ratio between sample and background UV absorbances. There is a decrease of the peak separation caused by the reduced length of the column. The resolving power is still quite satisfying, however.
8. Chromatofocusing may be run in analytical or preparative scale. In the first-mentioned case, the method is used to detect heterogeneities within a class or subclass of proteins. It has exhibited a high capacity to detect mono- or oligoclonal components of IgG in body fluids such as CSF and serum. Generally, electromigration techniques performed in gel media are more suitable for analytical purposes, but situations may be encountered in which an alternative method such as chromatofocusing could prove advantageous. The main applications will, however, probably be found in preparative work, e.g., when the technique is used as a first-step

procedure aimed at the isolation of relevant components for complementary investigations. Chromatofocusing has several promising features for such purposes: high resolution, relatively fast and easy performance, comparatively mild separation conditions, and methodological flexibility. Furthermore, the obtained material can be used directly, or after a simple concentration/dilution procedure, for examinations by, for example, isoelectric focusing, two-dimensional gel electrophoresis, or ELISA (8,9,11,12).

9. Troubleshooting with respect to components of the FPLC equipment is performed according to instructions in the respective manuals. Problems not referable to these parts are generally to be sought in the sample, the column, or the buffers (6). Column hygiene is of utmost importance for obtaining good and reproducible results, as well as for avoiding costs by a reduced life-time of the column. All samples, buffers, and other solutions run through the column should be carefully prepared, filtered, and free of air bubbles. There are three major groups of biomolecules that are prone to contaminate the column: polyions, lipids, and certain proteins (6). CSF samples and sera have been found to be well compatible with the column when the material is free of lipids. We have used a simple chloroform extraction for delipidation. This procedure could introduce denaturation of certain proteins. No significant problems have been observed, however, for IgG. Alternative methods for delipidation can also be applied, e.g., techniques based on the use of lipophilic dextran (13). A thorough elimination of residual material should be performed after each run, and we have found it advantageous to routinely use an acid as well as a salt solution. It may

be found for certain applications, however, that such a procedure is needed first after 5–10 runs. In that case, the column is washed only with 2M sodium chloride after each of the intervening runs. Column contamination is usually manifested by a declination of the reproducibility/resolution or an increment of the backpressure.

10. The present review on chromatofocusing is aimed at giving basic methodological data and indicating some advantages as well as applications of the technique. We have experienced few technical problems when the method is applied in a proper way and its limitations are considered. One drawback is the high price for the equipment. It should, however, be observed that chromatofocusing can be performed also by using an alternative stock of instruments (perhaps already available in your laboratory). Finally, you may find it advantageous to modify the methodology in order to optimize it for your special needs. Good luck!

Acknowledgments

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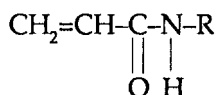
Chapter 15

Hybrid Isoelectric Focusing Using Mixed Synthetic-Carrier Ampholyte-Immobilized pH Gradient Gels

Michael J. Dunn and Ketan Patel

1. Introduction

An exciting recent innovation in isoelectric focusing (IEF) has been the development of Immobiline reagents (LKB) (1) for the preparation of polyacrylamide gels containing immobilized pH gradients (IPGs). The Immobilines are a series of seven acrylamide derivatives with the general structure:



where R contains either a carboxyl (acidic Immobilines) or a tertiary amino (basic Immobilines) group, forming

a series of buffers with different pK values (3.6, 4.4, 4.6, 6.2, 7.0, 8.5, 9.3). IGE IEF gels are prepared in the same manner as conventional gradient polyacrylamide gels using a concentration gradient, stabilized with a density gradient (e.g., glycerol), of the Immobiline reagents mixed with acrylamide and N,N' -methylene bisacrylamide (Bis) under the control of a suitable gradient forming device. Thus, during polymerization the buffering groups forming the pH gradient are covalently attached and immobilized via vinyl bonds to the backbone of the polyacrylamide matrix. This immobilization of the species forming the pH gradient results in the elimination of pH gradient drift (but *not* electroendosmosis), making it possible to use extremely narrow pH gradients spanning from 0.1 to 1 pH unit. Additional advantages of this technology are that the pH gradients are not disturbed by high salt concentrations, and separations can be carried out under conditions of controlled ionic strength and buffering power. Although this is a relatively new methodology, it has already been the subject of several review articles (2-6).

Initial efforts concentrated on the development of narrow and ultranarrow pH gradients to exploit the extremely high resolving power of such systems. It is claimed that a difference in pI of only 0.002 pH units can be detected. These gradients are generated using only two Immobelines, one acting as a buffer and the other as a titrant. Detailed formulations for a series of 58 such 1 pH unit-wide gradients are documented in refs. 6 and 7, and narrower pH gradients can be readily derived by linear interpolation.

Narrow and ultranarrow pH gradients have an excellent resolution capacity, but have limited application to samples containing proteins of widely varying pI values. Wider pH gradients are essential for the analysis

of such samples, but the formation of extended pH gradients, spanning greater than 1 pH unit, is complicated by the necessity to mix several buffering Immobililine species. This problem has been resolved by the creation of computer programs designed to calculate the mixtures required for this purpose (8,9). Using this approach, formulations have been derived for the generation of pH gradients spanning 2–6 pH units (9–11). Gradients extending from pH 4–10 can now be readily constructed, but IPGs cannot be extended to pH 3 without using two nonbuffering species, very acidic ($pK < 1$) and alkaline ($pK, 12$) Immobilines, as titrants. These reagents are not yet available commercially from LKB. It has also proved possible to generate nonlinear IPGs of any desired shape (12,13). This technique of pH gradient engineering can be used to optimize the shape of the pH gradient to obtain maximal resolution of the proteins in a particular type of sample.

Immobiline gels are best cast in standard vertical cassettes on GelBond PAG supports. Gels should generally be 1 mm or less in thickness. It is possible to form sample application wells in the gel at the time of gel casting using strips of adhesive embossing tape (e.g., Dymo) (6). Such depressions can accommodate 10–12 μL of sample, but it should be remembered that the depressions in the gel can disrupt the electric field during IEF. Once the appropriate recipe for the required pH gradient has been selected, the IPG gel can be formed using a simple two-chamber gradient-mixing device (6,7). Alternatively, a computer-controlled stepmotor-driven buret apparatus, commercially available from Desaga, can be used (14,15). This apparatus is expensive, but substantially improves reproducibility, flexibility, and documentation of the pH gradients. It is especially useful for the preparation of wide-range IPG

gels in which pH gradient engineering, for example, localized gradient flattening, will be used to enhance resolution.

In the standard recommended procedure (6,7) IPG gels are polymerized by heating at 50°C for 1 h. This procedure is used because at this temperature all the Immobililine reagents have been found to react at the same rate (16,17), with 84–88% incorporation of all seven Immobilines being obtained. These conditions are potentially disastrous for wide-range pH gradients, however, because of the instability of the pK 8.5 and pK 9.3 species at elevated temperatures (85% destruction of the former in less than 1 h at 60°C) (18). To overcome this problem, Righetti's group advises careful control of the pH of the mixture used for gel preparation, by suitable titration with acids or bases, to around pH 7.5–8 (18). An alternative strategy has been devised by Altland that permits IPG gels to be polymerized at room temperature (19). In this procedure the stock Immobililine reagents are titrated with Tris buffer to near neutrality and stored in aliquots at –20 to –70°C. Polymerization in neutral buffer at ambient temperature is a controlled and convenient procedure, but these advantages are gained at the expense of the necessity for removing the buffer from the gels by washing and prerunning steps prior to sample application (19).

After polymerization, the gel on its plastic support is removed from the cassette. IPG gels cannot be used immediately for IEF, but must first be washed to remove catalysts and unpolymerized Immobilines. During this process the gel swells so that it is then necessary to reduce the gel, with the aid of a fan, to its original weight (17). This process is difficult to control and rather time-consuming. It is, therefore, preferable to use a rehydratable gel system for IPG gels (20,21). In this tech-

nique, gels, after polymerization, are washed, dried under a fan at room temperature, and stored in sealed plastic bags at -20°C for future use. This has the added advantage that batches of gels can be prepared, minimizing inter-gel variability.

Two strategies have been devised for rehydration of IPG gels. In the first method, weight is used to control the process (20,21). The alternative approach, which is less time-consuming and more reproducible, is to use volume to control rehydration (19,22). In the latter procedure, the gel is reassembled into a cassette of the same dimensions as that in which it was cast and reswollen in a controlled amount of the required solution.

Although narrow IPGs function very well, it has become apparent that problems are associated with the use of wide pH gradients. These problems include: (a) slow entry of sample proteins into the gel, (b) lateral band spreading, (c) prolonged focusing times, and (d) increased electroendosmosis. These difficulties are associated with the inherently low conductivity of the Immobiline system. It has been found that improved separations can be achieved by addition of low concentrations, typically 0.5% (w/v), of synthetic carrier ampholytes of the appropriate range to the solution used for gel rehydration (19,23,24). This technique has been termed "hybrid isoelectric focusing" by Altland (19) or "mixed carrier ampholyte-IPG IEF" by Righetti (25). Here we describe a hybrid IEF gel system for the separation of human serum proteins in a pH 4–7 gradient.

2. Materials

1. Gel Bond PAG sheets
2. Two 5 mm thick plastic plates (17 x 13 cm) for the gel-casting cassette. One plate has a 3 mm hole

drilled in one corner, 10 mm from the two sides, to form a gel solution entry port.

3. Silicon rubber gasket (0.5 mm thick, 7 mm wide) to fit around the edge of the glass plates.
4. PVC sheet (17 x 13 cm) with a 3 mm-diameter hole punched in register with the entry port in the plastic plate.
5. Rubber roller.
6. Acrylamide stock solution: 14.4 g acrylamide; 0.6 g Bisacrylamide, made up to 50 mL with distilled water.

The solution is deionized with mixed-bed ion exchange resin (Duolite MB5113 mixed resin, BDH), 5 g/100 mL solution, for 1 h. After deionization, the solution is filtered. The stock solution is stored at 4°C for a maximum of 1 wk. This deionization step reduces ionic contamination (e.g., acrylic acid), which can adversely affect IEF.

7. Stock Immobiline solutions are prepared according to the method of Altland (19). This allows for polymerization of IPG gels under controlled conditions at room temperature. Acid Immobilines are titrated with 1M Tris to pH 6.8 and brought to a final concentration of 0.25M by adding 0.1M Tris-phosphate, pH 6.8. Basic Immobilines are titrated with 1M phosphoric acid to pH 6.8 and diluted to 0.25M in 0.1M Tris phosphate, pH 6.8. The volume required to obtain 0.25M immobiline solutions is calculated by the following formula:

$$V = M \times (1/C - 1/C_0 - V_0/M_0)$$

where V is the added total volume, M the amount of Immobiline in mmoles, M_0 the standard amount of Immobiline per bottle (5.11, 5.13, 5.14, 5.17, 5.19,

5.15, and 5.17 mmol for Immobelines of pK_a 3.6, 4.4, 4.6, 6.2, 7.0, 8.3, and 9.3, respectively), C the concentration of Immobiline to be adjusted for, C_0 the standard concentration of 0.2M in an Immobiline bottle after addition of the standard volume, V_0 , of 25 mL. The titrated and diluted Immobelines should be aliquoted and stored at -20 to -70°C .

8. Recipe for stock Immobiline reagents required to prepare the pH 4–7 IPG IEF gel (total gel volume, 30 mL) (10):

Immobiline pK_a	Acidic solution, μL	Basic solution, μL
3.6	578	301
4.6	110	738
6.2	449	151
7.0	—	269
9.6	—	876
	1137	2235

Note that the recipes in ref. 10 give ratios of the Immobiline reagents required. The ratios must be multiplied by 5 to obtain a working concentration (0.2M) expressed as $\mu\text{L}/\text{mL}$ of gel solution. Thus, for the 15 mL volumes of each solution used here (acidic, basic), the ratio value must be multiplied by a factor of 75.

9. The cassette will hold a total gel volume of 30 mL, so that 15 mL each of the acidic (light) and basic (dense) solutions are required to form the gradient. The final gel is T3C4 containing 4% (w/v) glycerol at the acidic end (top) and 15% (w/v) glycerol at the basic end (bottom) to stabilize the gradient until polymerization has occurred.

	Concentration	Acidic (light) solution	Basic (dense) solution
Acrylamide	T30C4	1.5 mL	1.5 mL
Glycerol	100%	0.6 mL	2.25 mL
Immobilines (acid)	0.2M	1137 μ L	—
Immobilines (basic)	0.2M	—	2235 μ L
TEMED	100%	7.5 μ L	8.5 μ L
Ammonium persulfate	10%	25 μ L	57 μ L
Distilled water	—	11.73 mL	8.85 mL
		15 mL	15 mL

10. Any gradient forming apparatus capable of delivering a linear gradient of the required volume can be used. The simplest device is a two-chamber gradient mixer (6,7), and an interesting modification has been published recently (26). We use an LKB Ultrograd electronic gradient former. Probably the best and most flexible approach is to use a computer-controlled, stepmotor-driven, high-precision buret system (14,15), which is available commercially from Desaga. The latter types of apparatus can be used to generate nonlinear IPGs of any desired shape that can be used to optimize protein resolution.
11. Rehydration solution: A solution of 8M urea containing 0.5% (w/v) ampholyte mixture (Servalyt pH 4–5, 5–6, 6–7 in a 1:1:1 mixture).
12. Any horizontal flat-bed IEF apparatus fitted with a cooling platten can be used. We currently use a Pharmacia FBE-3000 Flatbed system powered by a FBE-3000/150 powerpack fitted with a volt hour (Vh) integrator connected to a dual-channel chart recorder to give a continuous output of the voltage and current parameters.

13. Silicon fluid, Dow Corning 200/10 cs.
14. Electrode wicks (16 x 0.7 mm) cut from sheets of Whatman GF/B glass fiber paper. A stack of four wicks is used at each electrode position.
15. Electrolytes for the prerun: 0.5% (w/v) Servalyt 4–5, 5–6, 6–7 in a 1:1:1 mixture.
16. Silicon rubber sample application strip (available from Desaga), containing 27 slits for 20- μ L samples or 54 holes for 10- μ L samples.
17. Electrolytes for the focusing run.
Catholyte—0.1M NaOH
Anolyte—0.1M oxalic acid
18. Trichloroacetic acid (TCA) solution 20% (w/v) for gel fixation.
19. Staining solution: 0.5% (w/v) Coomassie Brilliant blue R250 in 45% methanol, 10% acetic acid.
20. Destaining solution: 45% methanol, 10% acetic acid.

3. Method

1. Cut a piece of GelBond PAG (17 x 13 cm). This material is specially treated so that the gel will adhere firmly to it. Only one surface (hydrophilic) is treated, the other surface being hydrophobic.
2. The IPG IEF gel is cast in a vertical cassette. The GelBond sheet is rolled onto a plastic plate with the hydrophobic surface of the GelBond in contact with the plate. A film of water is used between these surfaces to ensure good adherence, and air bubbles are removed with the roller. The U-shaped silicon rubber gasket is placed onto the GelBond/plastic plate assembly. This gasket provides a leak-proof seal without the use of grease. The PVC sheet is glued to the second plastic plate, ensuring the 3-

mm holes for entry of the gel solution are in register. Photographic "Spray Mount" adhesive is used to ensure good contact between the surfaces. The PVC sheet/plastic plate assembly is placed onto the gasket/GelBond/plastic plate assembly so that the gel solution entry port is at one of the lower corners of the cassette assembly. The cassette is held together along the two sides and the bottom edge with bulldog clips. The cassette is placed upright in a level, vertical position.

3. The gradient-forming device to be used is attached by tubing to the gel solution entry port at one lower corner of the cassette. The other side of the cassette is raised so that the bottom edge is at 45° to the horizontal. This forms a V shape with the solution input port at the lowest point. This ensures that the gel solution entering the cassette does so smoothly and is not subjected to excessive turbulence that would disturb the gradient. As the cassette fills, it is gently lowered to the horizontal position.
4. The gel is allowed to polymerize for 2 h at room temperature.
5. The cassette is dismantled, and the IPG gel on its GelBond support is carefully removed. The acidic (top) and the basic (bottom) ends of the gel are marked on the reverse side of the GelBond to ensure correct orientation of the gel during IEF.
6. The gel is washed twice, 30 min each, in 500 mL of 1% (w/v) glycerol. The presence of this low glycerol concentration in the gel prevents cracking and peeling of the gel during drying.
7. The gel is dried in a stream of air at room temperature. The drying time depends on the source of the air stream, but the gel must be completely dry before storage.

8. Gels can be stored indefinitely by sealing in plastic food-storage bags using a heat-sealer and storing in a dry container at -20°C .
9. Rehydration is performed using a cassette of the same dimensions as the one in which the gel was originally cast. The cassette is assembled as described previously using the dried gel on its support in place of the GelBond PAG sheet. A gasket of the same thickness is used. The solution used for rehydration can be introduced via the original input port or using a syringe in the top of the cassette. The gels are rehydrated with the rehydration solution containing 8M urea and 0.5% ampholytes for 2 h at room temperature. Rehydration is complete once all the solution in the cassette has been taken up by the gel.
10. The rehydrated gel is removed from the cassette and placed on the precooled (15°C) platten of the flat-bed IEF apparatus. A film of silicon fluid, which has excellent thermal conductivity and low viscosity, is used to provide good contact between the gel and the cooling plate. The electrode wicks wetted with prerunning electrolytes are applied to the ends of the gel.
11. The gel is prefocused at 3000 V, 5 mA, and 10 W limiting for 1 h. At this time the voltage will have risen to 3000 V whereas current will have dropped to below 5 mA. Refractive lines, running parallel to the electrode wicks, should be observed across the gel. This prerunning step removes any salts contained within the gel.
12. The electrode wicks are changed for others wetted with the running electrode solutions. The silicon rubber sample application strip containing sample application slits or holes is placed on the surface of

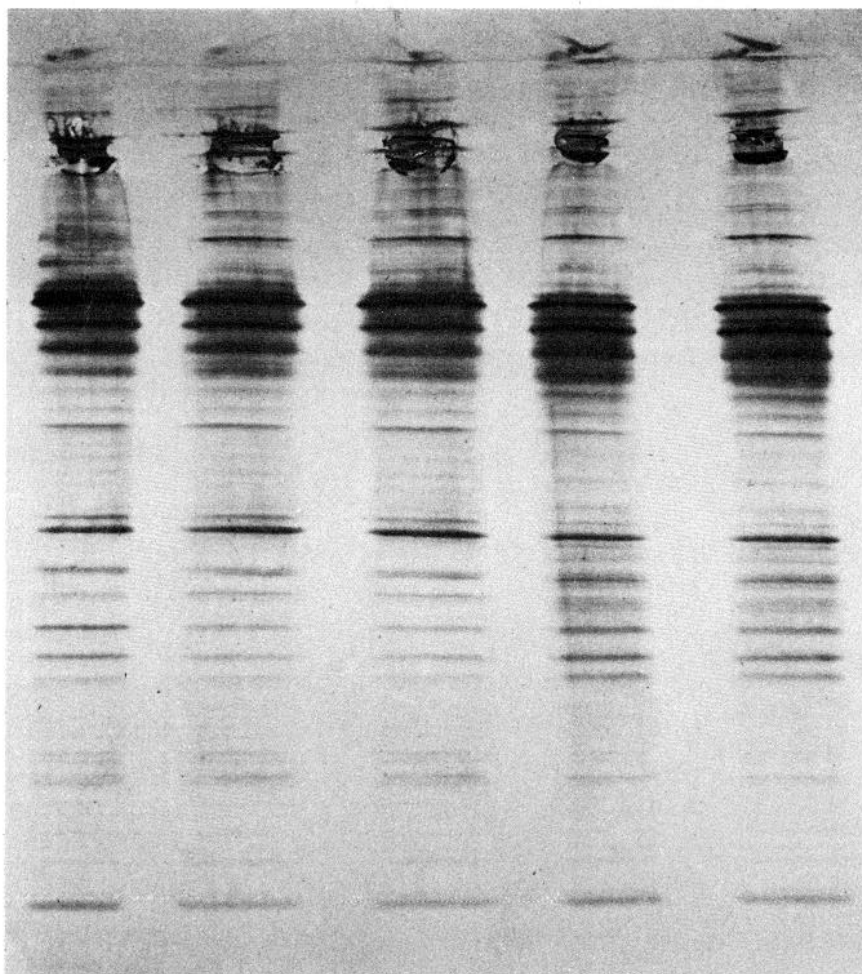


Fig. 1. Separation of human serum proteins using hybrid IEF in a pH 4-7 IPG gel.

the gel near the cathode. Serum samples (6 μ L) are applied in the wells.

13. The gel is focused at 3000 V, 5 mA, 10 W limiting for 9000 Vh.
14. After IEF is complete, the gel is fixed with 20% (w/v) TCA for 1 h.

15. The gel is washed for 30 min in water and stained for 30 min with Coomassie Brilliant blue R250. The gel is dried in air at 40°C. A typical separation of human serum proteins is shown in Fig. 1.

4. Notes

1. The catalyst levels used are chosen so that polymerization is complete 1 h after gel casting, allowing for maximum incorporation of Immobiline molecules into the polyacrylamide matrix without diffusion breaking down the gradient.
2. If nonionic (e.g., Triton X-100, NP-40) or zwitterionic (e.g., CHAPS) detergent is required in the gel to increase sample solubility, it should be added at the required concentration to the two solutions (acidic and basic) used to prepare the gel. [Levels of detergent between 0.5 and 2% (w/v) are compatible with the IEF gel system.] Detergent at the same concentration should also be included in the solutions used for washing the gel prior to drying. Detergent should not be added at the rehydration step, since it is already contained within the dried gel matrix. This method is used because it has been found to be difficult to rehydrate polyacrylamide gels containing *both* urea and detergent (21,27,28).

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Chapter 16

Two-Dimensional Electrophoresis Using Immobilized pH Gradients in the First Dimension

Michael J. Dunn and Ketan Patel

1. Introduction

The high-resolution capacity of two-dimensional polyacrylamide gel electrophoresis (2-D PAGE) based on the method of O'Farrell (1) makes it the method of choice for the analysis of complex protein mixtures (2–4). In the standard procedure, cylindrical tube isoelectric focusing (IEF) gels are used. Unfortunately this system suffers from severe cathodic drift resulting in pH gradients that do not extend above pH 8, resulting in concomitant loss of basic proteins from 2-D maps. Gradients can be extended to pH 10 by special treatment of the glass IEF tubes (5) or by using a horizontal flat-bed IEF (5–7). In spite of the improvement in resolution of basic proteins that can be achieved with these proce-

dures, difficulties are still associated with the first-dimension IEF gels as a consequence of the characteristic properties of the synthetic carrier ampholytes used to generate the pH gradients. These include problems of batch reproducibility of ampholytes, irreproducibility of separations, difficulty in control of pH gradient stability and shape (i.e., "pH gradient engineering"), and the possibility of artifacts caused by protein–ampholyte interactions.

Recent advances in IEF technology have provided alternative procedures that might overcome some of these problems. We have found one of these alternatives, buffer IEF, to be unsuitable or the first dimension of 2-D PAGE (8). The use of the Immobiline reagents (LKB), however, to generate immobilized pH gradients (IPGs) appears to be a much more promising approach for 2-D PAGE. The Immobilines are a series of seven substituted acrylamide derivatives with different pH values. IPG IEF gels are made by generating a gradient of the appropriate Immobiline solutions, so that during polymerization the buffering groups forming the pH gradient are covalently attached and immobilized via vinyl bonds to the polyacrylamide backbone. This immobilization results in the elimination of pH gradient drift (but *not* electroendosmosis) making pH gradients reproducible and infinitely stable—ideal properties for 2-D PAGE.

IPGs, reviewed in ref. 9 (and *see* chapter 15) were originally optimized for the generation of narrow and ultra-narrow pH gradients not generally suitable for 2-D applications. Recipes are now available, however for producing wide pH gradients (10), so that pH 4–10 gradients suited to 2-D PAGE can now be generated. Unfortunately, there are problems associated with the use of wide IPGs caused by the low conductivity of the

Immobiline system. These include slow entry of sample proteins, lateral band spreading, prolonged focusing times, and increased electroendosmosis. Improved separations can be obtained by the addition of low concentrations, typically 0.5% (w/v), of synthetic carrier ampholytes of the appropriate range to the IPG IEF gels (11–13). This technique has been termed "hybrid isoelectric focusing" by Altland or "mixed carrier ampholyte IPG IEF" by Righetti (*see* chapter 15).

IPG technology for 2-D PAGE is not fully developed, and it is true that the 2-D patterns obtained are generally inferior to those obtained using traditional methods. This appears to be caused not by problems with the IG IEF dimension *per se*, but by problems in elution and transfer of proteins from the first-dimension IPG gels to the second-dimension SDS-PAGE gels, resulting in straking of spots on 2-D maps (14–16). This phenomenon can probably be ascribed to the presence of fixed charges on the immobiline gel matrix, leading to increased electroendosmosis in the region of contact between the IPG IEF gel and the SDS-PAGE gel and resulting in disturbance of migration of proteins from the first to the second dimension gel.

Despite these problems, the ability of IPG technology to produce stable pH gradients whose shape can be engineered with ease makes this a method of great potential for increasing the reproducibility of 2-D maps. We will describe here a method of hybrid IEF using pH 4–10 IPG gels containing synthetic carrier ampholytes for the first dimension of 2-D PAGE.

2. Materials

1. GelBond PAG sheets.
2. Two 4-mm thick plastic plates (23 x 18 cm) for the

gel-casting cassette. One plate has a 3-mm hole drilled in one corner, 10 mm from each side, to form a gel solution entry port.

3. Silicon rubber gasket (0.5 mm thick, 7 mm wide) to fit around the edge of the plastic plates.
4. PVC sheet (23 x 18 cm) with a 3 mm-diameter hole punched in register with the entry port in the plastic plate.
5. Rubber roller.
6. Acrylamide stock solution: 14.4 g acrylamide; 0.6 g bisacrylamide made up to 50 mL with distilled water.

The solution is deionized with mixed-bed ion exchange resin (Duolite MB5113 mixed resin, BDH), 5 g/100 mL solution, for 1 h. After deionization, the solution is filtered. The stock solution is stored at 4°C for a maximum of 1 wk. This deionization step reduces ionic contamination (e.g., acrylic acid) which can adversely affect IEF.

7. Stock Immobiline solutions are prepared according to the method of Altland (11). This allows for polymerization of IPG gels under controlled conditions at room temperature. Acid Immobilines are titrated with 1M Tris to pH 6.8. Basic Immobilines are titrated with 1M phosphoric acid to pH 6.8 and diluted to 0.25M in 0.1M LTris-phosphate, pH 6.8. The volume required to obtain 0.25M Immobiline solutions is calculated by the following formula:

$$V = M \times (1/C - 1/C_0 - V_0/M_0)$$

where V is the added total volume, M the amount of Immobiline in mmoles, M_0 the standard amount of Immobiline per bottle (5.11, 5.13, 5.14, 5.17, 5.19, 5.15, and 5.17 mmoles for Immobilines of pK_a 3.6,

4.4, 4.6, 6.2, 7.0, 8.3, and 9.3, respectively), C the concentration of Immobililine to be adjusted for, C_0 the standard concentration of 0.2M in an Immobililine bottle after addition of the standard volume, V_0 , of 25 mL. The titrated and diluted Immobililines should be aliquoted and stored at -20 to -70°C .

8. Recipe for stock Immobililine reagents required to prepare the pH 4–10 IPG IEF gel (total gel volume, 30 mL) (10):

Immobililine pK_a	Acidic solution, μL	Basic solution, μL
3.6	1102	—
4.6	—	113
6.2	458	53
7.0	90	488
8.5	338	158
9.6	—	360
	1988	1172

Note that the recipes in ref. 10 give ratios of the Immobililine reagents required. The ratios must be multiplied by 5 to obtain a working concentration (0.2M) expressed as L/mL of gel solution. Thus, for the 15-mL volumes of each solution (acidic, basic) used here, the ratio value must be multiplied by a factor of 75.

9. The cassette will hold a total gel volume of 30 mL, so that 15 mL each of the acidic (light) and basic (dense) solutions are required to form the gradient. The final gel is T3C4 containing 4% (w/v) glycerol at the acidic end (top) and 15% (w/v) glycerol at the basic end (bottom) to stabilize the gradient until polymerization has occurred.

	Concentration	Acidic (light)	Basic (dense)
Acrylamide	T30C4	1.5 mL	1.5 mL
Glycerol	100%	0.6 mL	2.25 mL
Immobilines (acid)	0.2M	1988 μ L	—
Immobilines (Basic)	0.2M	—	1172 μ L
TEMED	100%	7.5 μ L	8.5 μ L
Ammonium persulfate	10%	25 μ L	57 μ L
Distilled water		<u>10.88 mL</u> 15 mL	<u>10.01 mL</u> 15 mL

10. Any apparatus capable of delivering a linear gradient of the required volume can be used. The simplest device is a two-chamber gradient mixer (9,17), and an interesting modification has been published recently (18). We use an LKB Ultrograd electronic gradient former. Probably the best and most flexible approach is to use a microcomputer-controlled, stepmotor-driven, high-precision buret system (19,20), which is available commercially from Desaga. The latter types of apparatus can be used to generate nonlinear IPGs of any desired shape that can be used to optimize protein resolution.
11. Rehydration solution: A solution of 8M urea containing 0.5% (w/v) ampholytes (Pharmalyte pH 3–10) and 15% (w/v) glycerol (16).
12. Any horizontal flat-bed IEF apparatus fitted with a cooling platten can be used. We currently use a Pharmacia FBE-3000 flat-bed system powered by a Pharmacia ECPS 3000/150 powerpack fitted with a VH-1 volt hour (Vh) integrator connected to a dual-channel chart recorder to give a continuous output of the voltage and current parameters.
13. Silicon fluid, Dow Corning 200/10 cs.
14. Electrode wicks (20 x 0.7 mm) cut from sheets of

- Whatman GF/B glass fiber paper. A stack of four wicks is used at each electrode position.
15. Electrolytes for the prerun: 0.5% (w/v) Pharmalyte pH 3–10.
 16. Silicon rubber (1 mm thick) sample application strip (available from Desaga) containing 27 slits for 20 μ L samples. From this strip sample application pieces with individual slits are cut.
 17. Electrolytes for the focusing run:
Catholyte—0.1M NaOH
Anolyte—0.1M oxalic acid
 18. Equilibration buffer containing 8M urea, 1% (w/v) sodium dodecyl sulfate (SDS), 20 mM dithiothreitol (DTT), 15% (w/v) glycerol in 150 mM Tris buffer, pH 8.6.

3. Method

1. Cut a piece of GelBond PAG (23 x 18 cm). This material is specially treated so that the gel will adhere firmly to it. Only one surface (hydrophilic) is treated. The reverse (hydrophobic) surface of the GelBond sheet is marked with a permanent marker pen into lanes (1.2 cm wide, 14 cm long) that are numbered to aid identification of sample strips for the second dimension.
2. The IPG IEF gel is cast in a vertical cassette. The GelBond sheet is rolled onto a plastic plate with the hydrophobic surface of the GelBond in contact with the plate. A film of water is used between these surfaces to ensure good adherence, and air bubbles are removed with the roller. The silicon rubber gasket is placed onto the GelBond/plastic plate assembly. This gasket provides a leak-proof seal without the use of grease. The PVC sheet is glued

to the second plastic plate ensuring that the 3-mm holes for entry of the gel solution are in register. Photographic "Spray Mount" adhesive is used to ensure good contact between the surfaces. The PVC sheet/plastic plate assembly is placed onto the gasket/Gel Bond/plastic plate assembly so that the gel solution entry port is at one of the lower corners of the cassette assembly. The gasket is held together along the two sides and the bottom edge with bulldog clips. The cassette is placed upright in a level, vertical position.

3. The gradient-forming device to be used is attached by tubing to the gel solution entry port at one lower corner of the cassette. The other side of the cassette is raised so that the bottom edge is at 45° to the horizontal. This forms a V shape with the solution input at the lowest point. This ensures that the gel solution entering the cassette does so smoothly and is not subjected to excessive turbulence that would degrade the gradient. As the cassette fills, it is gently lowered to the horizontal position.
4. The gel is allowed to polymerize for 2 h at room temperature.
5. The cassette is dismantled, and the IPG gel on its GelBond support is carefully removed. The acidic (top) and the basic (bottom) ends of the gel are marked on the reverse side of the GelBond to ensure correct orientation of the gel during IEF.
6. IPG gels cannot be used directly for IEF, but must first be washed to remove salts, catalysts, and unpolymerized Immobilines. The gel swells during this process so that it must be reduced to its original weight prior to IEF. This is best achieved using a rehydratable gel system. In this technique the gel is washed twice, 30 min each, in 500 mL 1%

- (w/v) glycerol. The presence of this low glycerol concentration in the gel prevents cracking and peeling of the gel during drying.
7. The gel is dried in a stream of air at room temperature. The drying time depends on the source of the air stream, but the gel must be completely dry before storage.
 8. Gels can be stored indefinitely by sealing in plastic food-storage bags using a heat-sealer and storing in a dry container at -20°C .
 9. Rehydration is performed using a cassette of the same dimensions as the one in which the gel was originally cast. The cassette is assembled as described previously using the dried gel on its support in place of the GelBond PAG sheet. A gasket of the same thickness is used. The solution used for rehydration can be introduced via the original input port or using a syringe into the top of the cassette. The gels are rehydrated with solution containing 8M urea and 0.5% ampholytes for 2h at room temperature. Rehydration is complete once all the solution in the cassette has been taken up by the gel.
 10. The rehydrated gel is removed from the cassette and placed on the precooled (15°C) platten of the flat-bed IEF apparatus. The silicon fluid, which has excellent thermal conductivity and low viscosity, is used to provide good contact between the gel and the cooling plate. The electrode wicks wetted with prerunning electrolytes are applied to the ends of the gel.
 11. The gel is prefocused at 3000 V, 5 mA, and 10 W limiting for 1 h. At this time the voltage will have risen to 3000 V, whereas the current will have dropped to below 5 mA. Refractive lines, running

dropped to below 5 mA. Refractive lines, running parallel to the electrode wicks, should be observed across the gel. This prerunning step removes salts contained within the gel.

12. The electrode wicks are changed for others containing the running electrode solutions. The individual silicon sample rubber application wells are applied to the surface of the gel in each lane near the cathode. Samples (10 μ L) are applied in the wells. Inclusion of Bromophenol blue in the samples helps in checking that the wells do not leak, since any mixing of adjacent samples is disastrous for 2-D PAGE.
13. The gel is focused at 3000 V, 5 mA, 10 W limiting for 15,000 Vh.
14. The gel is wrapped in plastic food-wrapping film and stored at -70°C .
15. For the second dimension, the gel is thawed and the appropriate sample lanes cut off with scissors using the lane markings on the back of the GelBond PAG sheet as a guide.
16. Strips are incubated at room temperature for 2 x 15 min in 7 mL of equilibration buffer.
17. Preparation of the second dimension SDS-PAGE gels will not be described, since they are of the normal Laemmli (21) type, as used in the standard 2-D PAGE procedure. The gels we use are linear 8–20% gradient SDS-PAGE gels (1 mm thick, 16 cm high, and 18 cm wide with a 2 cm high stacking gel) (*see* vol. 1 of this series).
18. The IPG IEF gel strips are applied to the SDS gels by filling the space in the cassettes above the stacking gels with equilibration buffer and gently sliding the strips into place. The strips should be in close contact with the top of the stacking gels. There is no

need to cement the strips in place with agarose.

19. The second-dimension SDS gels can be run in any suitable vertical slab gel apparatus. The gels are run at 12 mA/gel overnight until the Bromophenol blue tracking dye reaches the bottom of the gels.
20. After 2-D electrophoresis is complete, the gels can be subjected to any desired procedure to detect the separated components (e.g., electroblotting, fixation and staining, autoradiography and so on). An example of [³⁵S]-methionine-labeled human skin fibroblast proteins separated by this procedure is shown in Fig. 1.

4. Notes

1. The catalyst levels used are chosen so that polymerization is complete 1 h after gel casting, allowing for maximum incorporation of Immobililine molecules into the polyacrylamide matrix without diffusion resulting in breakdown of the gradient.
2. If nonionic (e.g., Triton X-100, NP-40) or zwitterionic (e.g., CHAPS) detergent is required in the gel to increase sample solubility, it should be added at the required concentration to the two solutions (acidic and basic) used to prepare the gel. The gel system is compatible with levels of detergent between 0.5% and 2% (w/v). Detergent at the same concentration should also be included in the solutions used for washing the gel prior to drying. Detergent should not be added at the rehydration step since it is already contained within the dried gel matrix. This method is used because it has been found difficult to rehydrate polyacrylamide gels containing *both* urea and detergent (14,22,23).



Fig. 1. Autoradiograph of a 2-D PAGE separation of human skin fibroblast proteins labeled with [^{35}S]-methionine using a pH of 4–10 IPG IEF gel in the first dimension.

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Chapter 17

Two-Dimensional Polyacrylamide Gel Electrophoresis Using Flat-Bed Isoelectric Focusing in the First Dimension

Michael J. Dunn and Ketan Patel

1. Introduction

Electrophoretic techniques are, perhaps, the most important group of procedures for the separation and analysis of complex protein mixtures because of their high resolution capacity. This resolving power is maximized in high-resolution, two-dimensional polyacrylamide gel electrophoresis (2-D PAGE) based on the technique developed by O'Farrell (1). This methodology involves a combination of a first-dimension separation in cylindrical isoelectric focusing (IEF) gels containing 9M urea and 2% (w/v) of the detergent NP-40 with

discontinuous gradient polyacrylamide gel electrophoresis in the presence of sodium dodecyl sulfate (SDS-PAGE) in the second dimension. Detailed reviews of the theoretical basis, methodological variations, and applications of this technology can be found in refs. 2-5.

The standard 2-D PAGE procedure based on the O'Farrell (1) method using rod IEF gels in the first dimension produces high-resolution protein separations (*see* vol. 1 in this series), but certain problems are associated with its use. The most important of these problems is severe cathodic drift during IEF, resulting in pH gradient instability. Thus, pH gradients rarely extend to above pH 8, leading to the absence of basic proteins from 2-D maps. Methylcellulose treatment of the walls of the glass tubes reduces electroendosmosis during IEF by masking the charges present on the glass. In this way, pH gradients can be extended to pH 10, but we have found that the basic proteins in the resultant 2-D maps are present as streaks rather than as discrete spots (6).

To overcome the problem of protein resolution at basic pH, a special system, termed nonequilibrium pH gradient electrophoresis (NEPHGE), using rod gels was developed (7). In this technique, proteins are loaded at the anodic end of the gel and electrophoresed for a relatively short time. The proteins are thus separated electrophoretically according to their mobilities in the presence of a rapidly migrating pH gradient. Their separation is dependent on the time of electrophoresis, and their final positions on the gel do not correspond to their isoelectric points. The non-steady-state nature of this procedure compared with conventional IEF makes it inherently rather irreproducible and very sensitive to small changes in experimental conditions.

Gradient stability can also be improved by using modified electrolytes (8,9). It should also be remem-

bered that polyacrylamide gels can contain agents that will contribute to cathodic drift: (a) trace impurities of acrylic acid, (b) covalent incorporation of polymerization catalysts as terminal groups in polyacrylamide chains, and (c) hydrolysis of amide groups to acrylic acid above pH 10. It is, therefore, essential to prepare freshly and deionize acrylamide solutions prior to IEF gel preparation (10).

In contrast to tube IEF gels usually used in 2-D PAGE systems, horizontal flat-bed gels are almost universally used for one-dimensional IEF separations. A significant advantage is that flat-bed apparatus is specifically designed for IEF applications, in contrast to the modified conventional electrophoresis apparatus normally used for tube-gel IEF. Thus, small electrolyte volumes can be used, and the electrodes can be positioned close to the ends of the gels in order to minimize pH gradient drift and loss of proteins from the gels (11). Perhaps most importantly, flat-bed IEF gels are capable of resolving basic proteins into discretely separated components rather than poorly resolved streaks.

Initial attempts to adapt flat-bed IEF technology to 2-D separations had limited success, probably because of the thick gels that were used (12–14). More successful procedures have been developed, however, using thin (6) and ultrathin (15,16) IEF gels.

In the method we have developed (6,17,18), 0.5 mm thick first-dimension IEF gel slabs are cast on a plastic support in a simple glass cassette with a silicon rubber gasket. The success of flat-bed IEF for 2-D separations depends on reliable binding of IEF gels to the plastic sheet, since they must be able to withstand the handling procedures involved in equilibration and transfer to the second-dimension SDS-PAGE gels. Originally (6) we treated polyester sheets with Dow Corning Prime Coat

1200, followed by reaction with silane A174. This method was not entirely satisfactory, but the commercial gel supports available at that time were not reliable for IEF gels containing both urea and nonionic detergent. By increasing catalyst concentrations two- to threefold, however, we are now able to reliably bind polyacrylamide IEF gels to GelBond PAG sheets in the presence of 8M urea and 0.5% (w/v) NP-40 or CHAPS detergents (17).

After polymerization is complete, the IEF gel is placed on a suitable flat-bed apparatus. The upper surface of the gel is covered with a plastic sheet, containing punched-out holes for sample application, to protect the gel from desiccation and the effects of the atmosphere. Small volumes of electrolyte can be used to advantage in flat-bed IEF, but they must be somewhat stronger than those used for rod-gel IEF. For the electrode wicks containing the electrolytes, we use strips of glass fiber paper (19). The gels should be run at a controlled temperature of 15°C, since at lower temperatures there is risk of crystallization of urea within the gel matrix. One of the major advantages of the flat-bed system is that there is a great flexibility in the choice of sample application position, thus avoiding problems associated with sample application under extreme acid or alkaline conditions that are unavoidable in the tube-gel IEF system.

After IEF is complete, the cover-sheet is removed and the gel is covered with plastic food-wrapping film. For nonradioactive samples, individual sample tracks are cut using a template of lane markings made during gel preparation. The sample strips are frozen rapidly and stored at -70°C. For appropriately radiolabeled samples, the wrapped gel can be placed in contact with a sheet of X-ray film and exposed overnight (or longer) at -70°C. The film is then developed and the resulting

autoradiograph used as a guide to excise the sample tracks.

Gel strips are equilibrated for 5–10 min in buffer-containing SDS prior to application to the second-dimension SDS-PAGE gels. Since the gels are firmly attached to a solid support, they are relatively easy to handle, and the stretching that often occurs with cylindrical IEF gels, resulting in distorted 2-D maps, can be avoided. The top of the SDS-PAGE gel cassette is filled with equilibration buffer, and the IEF strip is slid into place, avoiding entrapment of air bubbles. It is usually not necessary to cement the strips in place, but this can be done, if required, using buffered agarose. The second, SDS-PAGE dimension then proceeds in the normal manner as in the standard 2-D PAGE technique, and after 2-D electrophoresis is complete the resultant 2-D protein maps can be visualized using any appropriate detection procedure.

We will describe how this technique is used to separate human skin fibroblast whole-cell proteins that have been radiolabeled with [^{35}S]-methionine.

2. Materials

1. GelBond PAG sheets (17.5 x 23 cm).
2. Glass plates, 17.5 x 23 cm, for cassettes.
3. Silicon rubber gasket, 0.5 mm thick x 7 mm wide, to fit around the edge of the glass plates.
4. PVC sheets (17.5 x 23 cm).
5. Rubber roller.
6. Solution A: Acrylamide stock solution (T30/C4): 14.4 g of acrylamide; 0.6 g of *N,N'*-methylene bisacrylamide (Bis) made up to 50 mL with distilled water.

The solution is deionized with mixed-bed ion exchange resin (Duolite MB5113 mixed-bed resin, BDH), 5 g/100 mL solution, with stirring for 1 h. After degassing using a vacuum pump, the solution is filtered using a sintered glass filter under vacuum. The solution can be stored at 4°C for a maximum of 1 wk. The deionization step reduces ionic contaminants, which can increase cathodic drift and pH gradient instability.

7. Urea (BRL enzyme grade) should be stored dry at -20°C to reduce the rate of breakdown of urea with the formation of cyanate ions, which can react with protein amino groups to form stable carbamylated derivatives of altered charge.
8. Solution B: Acrylamide-urea solution: 5.32 mL acrylamide stock (solution A), 19.40 g urea, 14.48 mL distilled water.

After the urea is completely dissolved, the mixed-bed ion exchange resin (5 g/100 mL) is added, and the solution is deionized with stirring for 1 h. This removes charged contaminants from the urea, most importantly any cyanate ions that are present. The solution is filtered under vacuum.

9. Any synthetic carrier ampholyte preparations can be used, but we use a mixture (*see* section 3, step 3) of the following:

Pharmalyte 3-10 (Pharmacia)

Pharmalyte 5-8 (Pharmacia)

Servalyt 2-11 (Serva)

Servalyt 4-6 (Serva)

Ampholine 3.5-10 (LKB)

The appropriate ampholyte mixture for a particular type of sample must be established (*see* section 4, Note 5).

10. Solution C: 10% (w/v) 3-[(cholamidopropyl)-dimethylammonio]-1-propane sulfonate (CHAPS).

The product from CalBiochem is salt free, but rather expensive. The Sigma product is cheaper, but must be deionized using mixed-bed ion exchange resin (5 g/100 mL) with stirring overnight at 4°C.

11. *N,N,N',N'*-Tetramethylethylenediamine (TEMED).
12. Solution D: Ammonium persulfate, 10% (w/v), made fresh daily.
13. Silicon fluid, Dow Corning 200/10 cs.
14. Electrode wicks (20 x 0.7 cm) cut from sheets of Whatman GF/B glass fiber paper.
15. Solution E: Catholyte, 1M NaOH.
16. Solution F: Anolyte, 1M oxalic acid.
17. Solution G: Bromophenol blue, 0.1% (w/v) in distilled water (BDH).
18. Solution H; (Sample lysis buffer): 9M urea, 3% (w/v) CHAPS, 20 mM dithiothreitol (DTT).
19. Solution I: Equilibration buffer—10% (w/v) SDS (final concentration, 2.3%), 11.5 mL—1M Tris buffer, pH 6.8 (final concentration, 124 mM), 6.2 mL DTT (final concentration, 20 mM), 0.154 g. Sodium β -mercaptoacetate (final concentration, 5 mM) 0.114 g, made up to 50 mL with distilled water.

3. Method

1. Cut a piece of GelBond PAG measuring 17.5 cm x 23 cm. The gel will adhere only to the hydrophilic side of the sheet. The reverse side (hydrophobic surface) is marked using a permanent marker pen into individual lanes (13.5 x 1.2 cm). Fifteen lanes can be accommodated, and these are flanked on each side by a further 1-cm wide lane. The sample lanes are numbered (1–15) to aid in identification of the sample strips at the completion of the IEF run.

2. The IEF gel is cast in a vertical cassette fitted with a U-shaped gasket. The GelBond PAG sheet is rolled onto a glass plate (17.5 cm x 23 cm) with the marked, hydrophobic side in contact with the glass plate. A small film of water is used between the glass and plastic sheets to ensure good adherence. Small air bubbles must be removed using a roller. On a second glass plate of the same dimensions, a sheet of PVC (17.5 x 23 cm) is rolled. The U-shaped silicon rubber gasket, which provides a leak-proof seal without the use of grease, is placed onto the GelBond PAG/glass plate assembly. The PVC/glass plate is then placed onto the gasket/GelBond PAG/glass plate to form the complete cassette, which is held together with bulldog clips on the two sides and along the bottom edge.
3. Samples for 2-D PAGE are solubilized and denatured using high levels of urea and detergent (*see* solution H) so that the proteins can be separated as individual polypeptide chains. Thus the IEF gels also contain urea and zwitterionic detergent (CHAPS). To 17.5 mL of the acrylamide-urea mixture (solution B) are added the ampholytes (0.46 mL Pharmalyte 3–10, 0.46 mL Servalyt 2–11, 0.46 mL Ampholine 3.5–10, 0.12 mL Pharmalyte 5–8, 0.06 mL Servalyt 4–6). TEMED (15 μ L) is then added, and the solution is degassed under vacuum. After degassing, 1 mL of CHAPS (solution C) is added, followed by 60 μ L of ammonium persulfate (solution D). The solution is mixed and poured, avoiding the formation of air bubbles, into the cassette using a syringe barrel with a needle fitted with plastic tubing. The gel is allowed to polymerize at room temperature for 1 h. The resulting gel contains 8M urea, 0.5% (w/v) CHAPS, and 3.12%

- (w/v) ampholytes. The gel cannot be stored and must be used for IEF immediately.
4. The bulldog clips are removed, and the PVC/glass plate is gently removed. The gel on its GelBond PAG support can now be removed from the cassette.
 5. Any flat-bed IEF apparatus can be used. We are currently using a Pharmacia FBE-3000 flat-bed system powered by a Pharmacia ECPS 3000/150 powerpack fitted with a VH-1 volt hour (Vh) integrator. The Vh integrator has been fitted with outputs to monitor constantly V and mA output using a dual-channel chart recorder. The flat-bed apparatus is cooled to 15°C using a thermostatically controlled circulator.
 6. A thin film of silicon fluid is applied to the cooling plate. The gel on its plastic support is then placed onto the cooling plate. The film of silicon fluid, which has excellent thermal conductivity properties and a low viscosity, allows for good contact between the gel support and the cooling plate.
 7. A cover sheet (20 x 13.5 cm), having the same lane markings as the GelBond sheet and with slits (3 x 10 mm) punched for sample application, is applied to the surface of the gel. The lane markings should be superimposable, and care must be taken to eliminate air bubbles between the surface of the gel and the cover sheet, which is weighted with glass plates.
 8. A stack of four wicks is used at the anode and cathode. The wicks are moistened (not too wet) with the electrode solutions (solutions E and F). The wicks are then applied at either end of the cover sheet with the cathode arranged to be at the sample-application side of the gel. A 1 mm gap should

- remain between the wicks and the cover sheet to prevent electrical arcing. Bromophenol blue (solution G) (2 μ L) is applied at each sample application site, and the electrode assembly is placed on the gel.
9. The gel is prefocused at 600 V, 30 mA, 15 W limiting for 600 Vh. After prefocusing the Bromophenol blue should have migrated about half way to the anode, and all lanes should have migrated with equal velocity.
 10. The samples, solubilized in sample lysis buffer (solution H), are then applied in the holes in the cover sheet. We load 10^6 dpm of [35 S]-methionine-labeled proteins; it is best to limit the sample volume to 5 μ L. Only the central lanes (3–12) should be used for samples, with the remainder being loaded with 2 μ L of Bromophenol blue (solution G). The gel is focused at 700 V, 30 mA, 15 W limiting for a total of 14,000 Vh. The voltage is then increased to 1000 V (30 mA, 15 W limiting) to sharpen the bands to give a total of 15,000 Vh.
 11. After completion of IEF, the gel is removed from the apparatus. The wicks and cover sheet are carefully removed. A lane to which a sample was not applied is cut off for pH gradient determination. The remainder of the gel is covered with food-wrapping film and stored between two glass plates at -70°C . Autoradiography can be used at this stage to identify sample lanes and to monitor the quality of the separation (*see* section 1).
 12. The pH gradient is determined with a surface pH electrode (Ingold, Desaga) at 1-cm intervals along the length of the gel strip.
 13. If an autoradiograph has been made, sample tracks are excised using the X-ray film as a template. Otherwise, sample lanes must be cut off using the lane markings as a guide.

14. Strips are equilibrated for 7 min in 5 mL of equilibration buffer (solution I).
15. The preparation of second-dimensions SDS-PAGE gels will not be described here, since they are of the normal Laemmli (20) type, as used in the standard 2-D PAGE technique (*see* vol 1 of this series). These gels can be either of a suitable constant percentage concentration of polyacrylamide or of a linear or nonlinear polyacrylamide concentration gradient (*see* vol. 1 of this series). We currently use modified nonlinear (8–20%) gradient SDS-PAGE gels (21). The gels (0.75 mm thick) are 16 cm wide and 18 cm long. A short (2 cm) stacking gel is used.
16. The IEF strips are applied to the SDS gels by filling the space in the cassettes above the stacking gels with equilibration buffer (solution I) and gently sliding the strips into place. Good contact between the IEF strips and the tops of the stacking gels must be achieved. There is no need to cement the strips in place with agarose.
17. The second-dimension SDS-PAGE gels can be run in any suitable vertical electrophoresis apparatus. The gels are run at 12 mA/gel overnight until the Bromophenol blue reaches the bottom of the separating gels.
18. The gels can be subjected to any suitable procedure to detect the separated proteins (e. g., electroblotting, autoradiography, protein staining, and so on). A typical separation of [³⁵S]-methionine-labeled human skin fibroblast proteins using this technique is shown in Fig. 1.

4. Notes

1. Avoid touching the hydrophilic surface of GelBond PAG, since this reduces gel adherence.



Fig. 1. Autoradiograph of a two-dimensional separation of human skin fibroblast proteins labeled with [^{35}S]-methionine using flat-bed IEF in the first dimension.

2. Different sizes of IEF gel can be readily used in this technique within the constraints of the size of the cooling plate in the IEF apparatus being used.
3. The cassette must be standing upright and level during gel preparation.

4. We use the zwitterionic detergents CHAPS in the IEF gel, but it is also possible to use nonionic detergents such as Triton X-100 or NP-40 with equally good results.
5. We use mixtures of different commercial synthetic carrier ampholytes, since the greater diversity of ampholyte species increases resolution. The mixture has also been optimized for the separation of human skin fibroblast proteins. Such pH gradient engineering using synthetic carrier ampholytes is a rather empirical process, however. For any particular type of sample to be separated by 2-D PAGE, it is good practice to establish which ampholyte or mixture of ampholytes produces the best separation for that sample.
6. The IEF gel can be run without a cover-sheet, but the gel should then be exposed to a humid N₂ atmosphere, or an NaOH solution should be present within the apparatus to absorb CO₂.
7. If no cover sheet is used, samples should be applied using silicon rubber sample application strips (Desaga) or similar devices. It is not recommended to apply samples as droplets to the gel surface (they tend to spread) or to use pieces of filter paper on which the sample has been absorbed (some proteins bind irreversibly).

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Chapter 18

Preparative Aspects of Immobilized pH Gradients

Pier Giorgio Righetti and Cecilia Gelfi

1. Introduction

We will give here guidelines on the preparative aspects of immobilized pH gradients (IPGs), the latest and most powerful fractionation technique in electrophoresis. The load ability of IPG gels has been demonstrated to be at least 10 times higher than in conventional isoelectric focusing (IEF), thus approaching or even passing the load limit of isotachopheresis (1). The following aspects will be illustrated: (a) explorative runs aimed at defining the load capacity; (b) optimization of experimental parameters, such as ionic strength (I), pH gradient width, and gel thickness; (c) general considerations on the load ability and ionic strength of the milieu; (d) protein load as a function of %T in the matrix.

At the methodological level, we will present the following procedures: (a) casting of soft and thick Immobiline gels; (b) recovery of protein zones from an Immobiline gel by electrophoretic elution into hydroxyapatite beads; (c) recovery by electrophoretic transfer into ion-exchange resins.

2. Theory

Although in the past, in preparative electrophoretic or IEF runs, most separations have been performed by trial and error, with no real knowledge of the capacity of the system, it is now possible in IPGs to predict theoretically and from a few analytical runs the load capacity of the system compatible with a given degree of resolution. We will discuss here the various parameters needed for optimizing a preparative run.

2.1. Theoretical Prediction of Acceptable Protein Loads in IPGs

For practical preparative work, an equation has been derived correlating the maximum protein load in a single zone to the *pI* (isoelectric point) distance (ΔpI) with the nearest contaminant, to the gel cross-sectional area and to the slope of the pH gradient. The equation

$$M = [(\Delta pI) / ([d(pH)] / (dx) - L)]^2 C_M A \quad [1]$$

where *M* = protein load in a single zone (major component) in mg; ΔpI = *pI* difference between major component and nearest contaminant (in pH units); $d(pH)/dx$ = slope of the pH gradient along the separation track (pH units/cm); *L* = protein free space, between the major band and the impurity, that is needed to cut the gel

without loss of protein or without carrying over the impurity (in general 1 mm is an acceptable distance); C_M = average concentration in the focused zone of the major component (mg/mL); A = cross sectional area of the gel perpendicular to the focusing direction (in cm^2) (2).

It can be seen that protein load can be maximized by increasing A (the liquid volume available to the focused zone) and by decreasing the slope of the pH gradient (i.e., by focusing in narrow or ultranarrow pH gradients). As a practical guideline, a graph has been constructed correlating these three basic experimental parameters: protein load in a single zone, ΔpI between the band of interest and nearest contaminant and slope of the pH gradient along the separation axis [$d(\text{pH})/dx$]. This graph is essentially a plot of Eq. [1], taking as a concentration limit (C_M) a common upper limit experimentally found of 40 mg/mL (in reality, as shown below, this "solubility" limit depends on the %T; in a 2.5 %T matrix, the C_M value can be as high as 90 mg protein/mL gel phase). Figure 1 shows how the graph is laid out: the abscissa reports the ΔpI value (in pH units), and the ordinate the protein load (mg/ cm^2) for a given A value. The ΔpI vs. protein load plane is cut by lines of different slopes representing pH gradients of different widths along the IPG gel length. It is seen that ultranarrow pH gradients (e. g., 0.02 pH units/cm) allow for extremely high protein loads (up to 80 mg/ cm^2), while still retaining a resolution better than $\Delta pI = 0.01$. At the opposite extreme, broad pH gradients (e.g., 0.2 pH units/cm) would allow a resolution of only $\Delta pI = 0.1$ with a protein load of less than 40 mg/ cm^2 .

2.2. Optimization of Environmental Parameters

We have also performed a thorough study on the optimization of environmental parameters [I (ionic

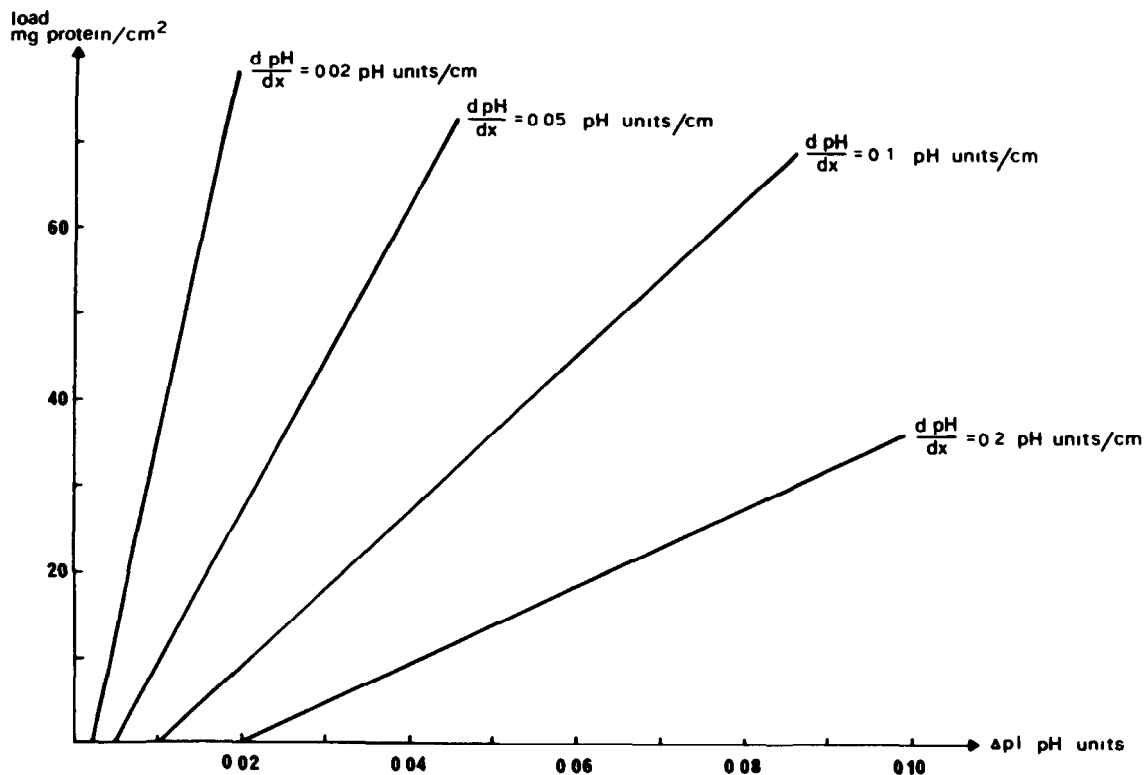


Fig. 1. Acceptable protein load as a function of ΔpI for different pH slopes plotted for a mean concentration of 45 mg/mL in the major protein zone. This is a graphical representation of Eq. [1] (ref 2).

strength), gel thickness, pH gradient width] for maximizing protein loads in Immobiline matrices. These aspects are summarized in Fig. 2. By increasing the ionic strength of the gel from 1.25 to 7.5 mEq/L, a fourfold increment in load capacity is obtained; above this level, a plateau is abruptly reached around 10–12 mEq/L. By increasing the gel thickness from 1 to 5 mm, a proportional fivefold increment in protein load ability is achieved. The system does not level off. A 5-mm thickness seems to be optimal, however, since thicker gels begin to develop thermal gradients in their transverse section, generating skewed zones. Finally, by progressively decreasing the width of the pH interval, there is a linear increase in protein load capability. Here too the system does not reach a plateau. Because of the very long focusing times required by narrow pH gradients, however, aggravated by the high viscosity of protein zones at high loads, it is suggested not to attempt to fractionate large proteing amounts in pH ranges narrower than 0.5 pH units (3).

2.3. General Considerations on the Load Ability and the Ionic Strength of the Milieu

We have seen above that IPGs have a load ability at least 10 times higher than conventional IEF. We believe this is mostly because of strong difference in ionic strength characteristic of the two systems. At the very low I values typical of IEF (approximately 1 mEq/L) (4), macromolecules will have a very low solubility minimum and will tend to aggregate and flocculate. An increase of I (approximately 7–10 mEq/L) as characteristic of IPGs, is thus beneficial since according to the Debye-huckel equation:

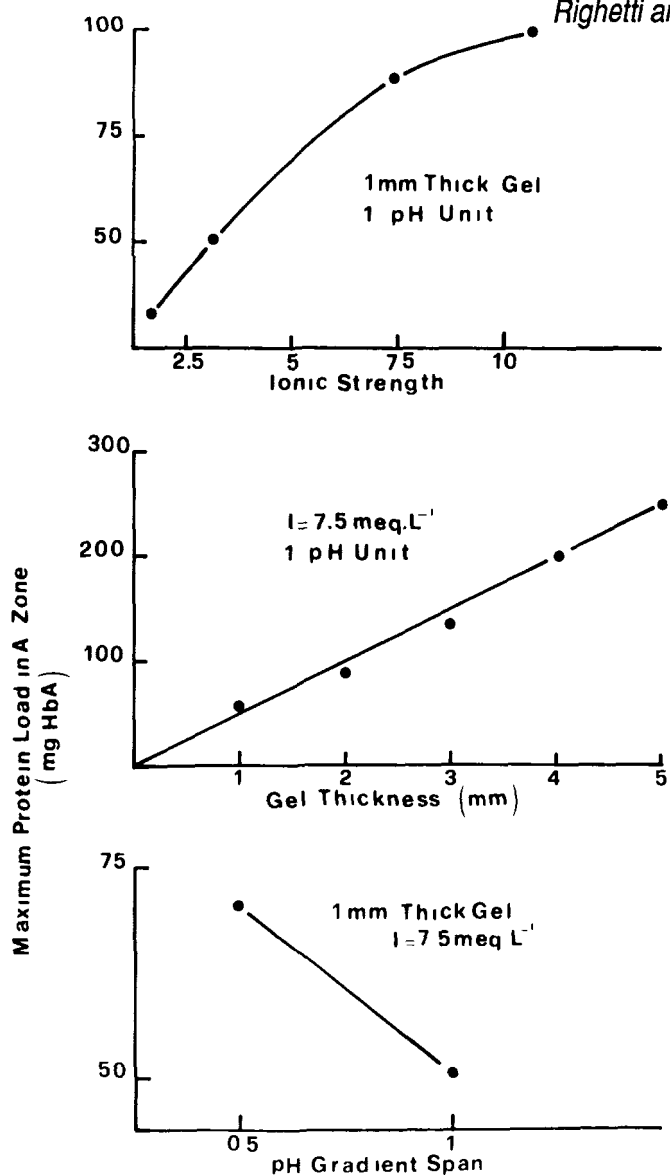


Fig. 2. Loading capacity of IPGs. The maximum load in a single protein zone is plotted: (top) as a function of ionic strength at constant gel thickness in a 1-pH unit interval; (middle) as a function of gel thickness at constant ionic strength in a 1-pH unit interval; (bottom) as a function of pH gradient width at constant ionic strength and constant gel thickness (ref. 3).

$$-\log \gamma = \log (S/S_0) = 0.51 Z^2 \sqrt{I} \quad [2]$$

where γ is the activity coefficient of an ion of charge Z , S and S_0 are the solubilities of a protein at the pI at a given ionic strength and as extrapolated to zero ionic strength, respectively. Thus, as the environmental I increases, and the γ values of the ions (both in solution and in the protein) decrease, the protein solubility increases: This is the well known "salting in" effect described already in 1936 by Cohn (5). We are tempted to equate IPG gels to "salting-in" media and IEF gels to "salting out" milieus. It might be argued that, as long as the proteins precipitate at their pI , and this material is confined in the isoelectric zone, this should not affect the load capacity in gel matrices, since the precipitated zone is gravitationally stable. The point of the matter is that in real cases this does not quite happen. As demonstrated by Gronwall (6), the solubility of an isoionic protein, plotted against pH near the isoionic point, is a parabola, with a fairly narrow minimum at relatively high I , but with progressively wider minima, on the pH axis, at decreasing I values.

This phenomenon can be appreciated in Fig. 3, taken from a 1942 classic. This graph has more than one degree of complexity, worth discussing. On the one hand, by increasing the environmental ionic strength from 1 to 20 mM NaCl, the solubility of β -lactoglobulin, at a given pH (approximately 5.3, very close to the pI value), increases by a factor of eight. As a second phenomenon, it can be seen that, at moderately high ionic strength (20 mEq/L), all the protein molecules tend to be isoelectric at a precise point on the pH axis, whereas at low I values (1 mEq/L) they are smeared over as much as 0.4 pH unit. For easier reading, we have replotted these physical phenomena in the two follow-

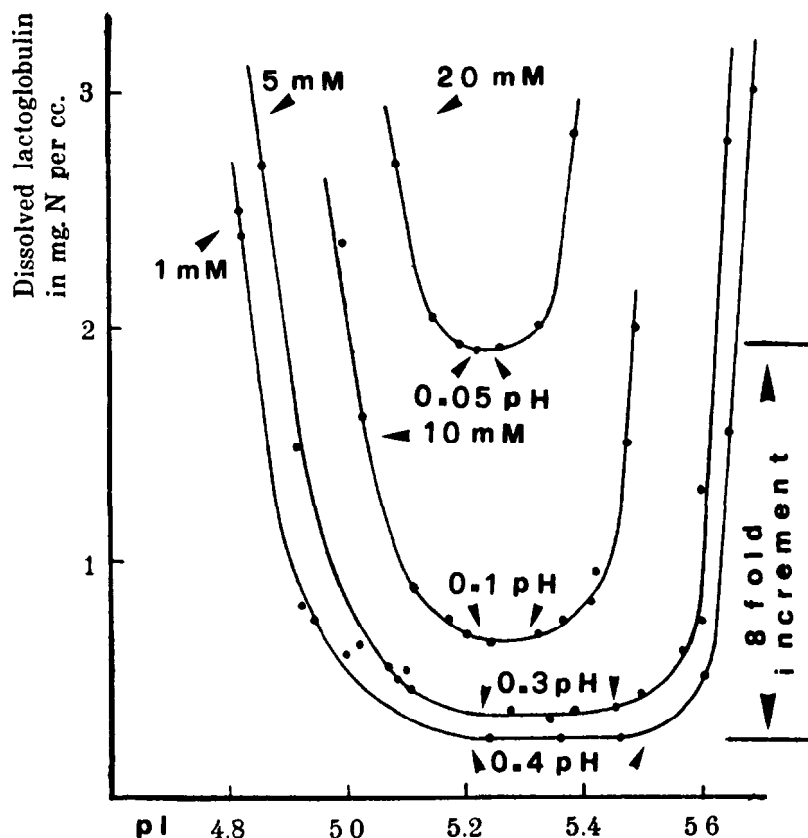


Fig. 3. Solubility of β -lactoglobulin in the pH range of 4.8–5.6 as a function of the ionic strength of the environment, from 1 to 20 mM NaCl. The amount of protein dissolved has been determined as Kjeldahl nitrogen. The pH values on the parabolas give the pH span of the solubility minimum, centered on a theoretical pI of 5.3 (modified from ref. 6).

ing graphs: the dependence of the solubility of the protein at pI from the environmental ionic strength is displayed in Fig. 4, and the width of the apparent pI zone on the pH axis, as a function of the I of the milieu, is plotted in Fig. 5, and is seen to vary in a funnel-shaped fashion. This last graph deserves further comments: it

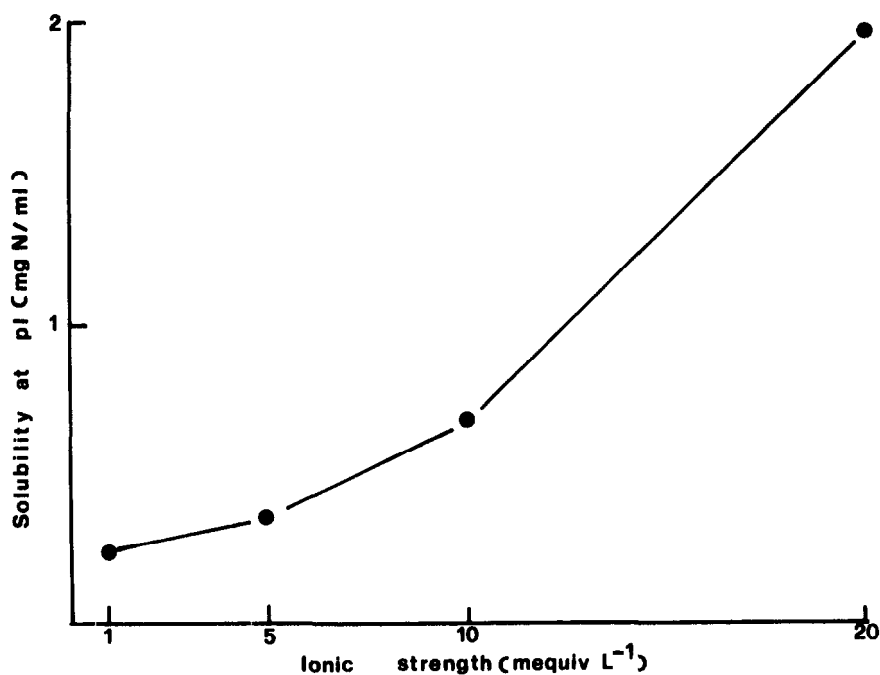


Fig. 4. Plot of solubility at pI vs. ionic strength of the milieu for β -lactoglobulin (data from Fig. 3). The solubility increases by a factor of 8 for a 20-fold increase in ionic strength.

can be seen that, at the prevailing I values typical of conventional IEF (1 mEq/L), the solubility minimum of β -lactoglobulin spans a 0.4 pH unit interval, whereas in IPGs (average $I = 10$ mEq charges, but easily adaptable to any other I value), the pH width of solubility minima is strongly decreased in a funnel-shaped fashion down to only 5/100 of a pH unit. In other words, what is detrimental in conventional, preparative IEF runs is not isoelectric precipitation, but near-isoelectric precipitation. The precipitate is not confined at the pI position, but is usually smeared over as much as 1/2 pH unit interval, thus being completely detrimental to the resolution of adjacent species (7).

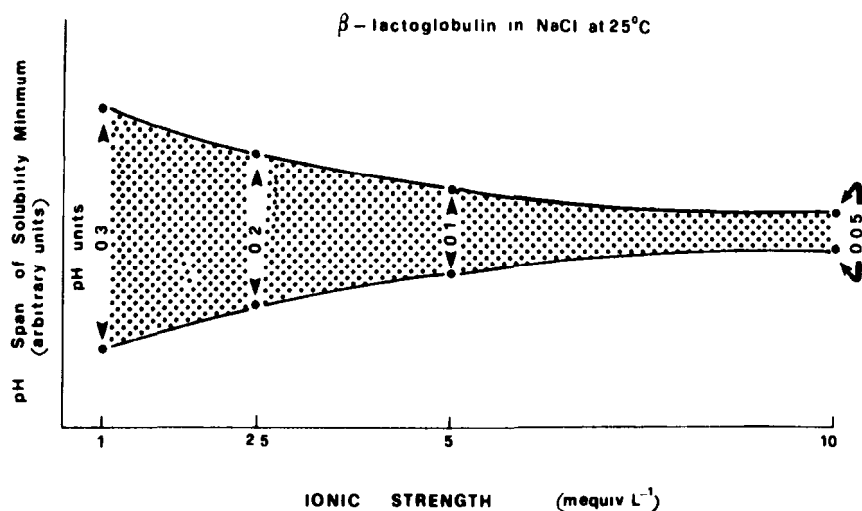


Fig. 5. Solubility of a protein in the neighborhood of its *pI* as a function of the ionic strength of the milieu. At 1 mEq/L (conditions prevailing in conventional IEF), β -lactoglobulin has a minimum of solubility over a span of 0.3 pH units; the width of the solubility funnel is markedly decreased at high *I* values (it is only 0.05 pH unit at 10 mEq/L, values typical of an IPG milieu) (from ref. 3).

2.4. Protein Load as a Function of %T in the Matrix

We have seen, so far, three parameters influencing a gel load ability (ionic strength, gel thickness, pH gradient width). In reality, there is a fourth, hidden parameter, perhaps even more important than the others, that we might have grasped immediately if we had paid more attention to a theoretical article by Bode (8) on the structure of hydrophilic matrices. We have seen above that, no matter how the experimental conditions are optimized and the ionic strength is increased, a common upper load limit for all proteins investigated has been found, (approximately 40 mg protein/mL gel solution). We have been intrigued by this barrier and have tried to find an explanation for it. The key to this apparent "solu-

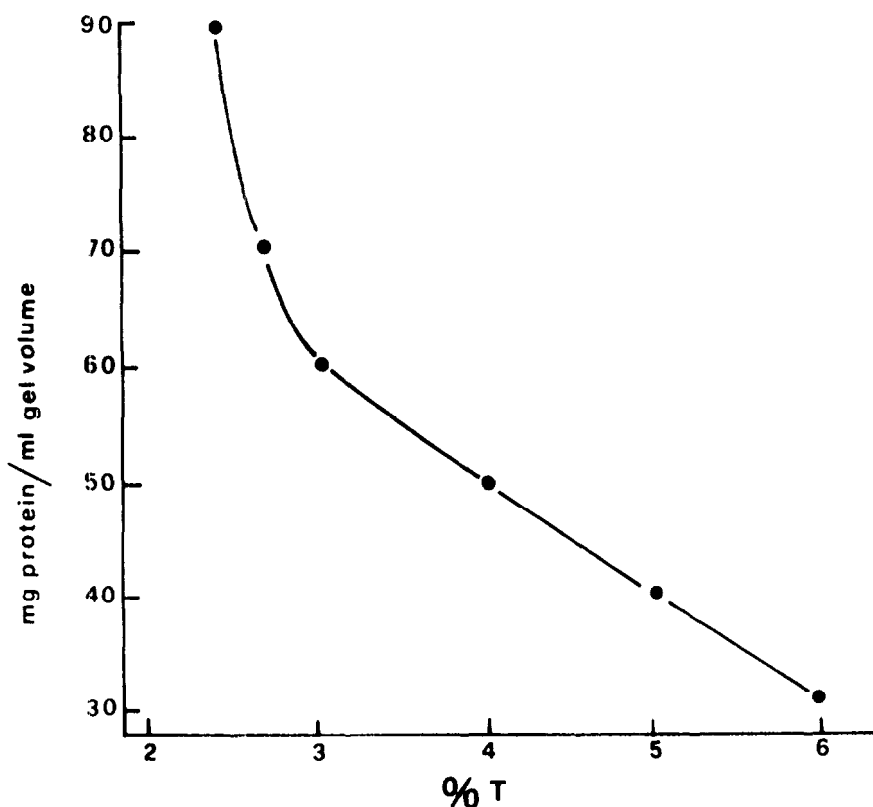


Fig. 6. Relation between loading capacity (in terms of mg protein/mL gel volume) and % T (T = grams of acrylamide and crosslinker per 100 mL gel volume) value of the gel matrix. Notice that, although in the range of 3–6% T , the protein load decreases linearly, in softer gels (3% T) it increases exponentially (from ref. 9).

bility limit" is to be found in Fig. 6: The amount of protein accepted by a gel matrix is directly related to its composition (% T). The limit of 40 mg protein/mL gel is only valid for a 5% T polyacrylamide matrix: as the number of fibers in the gel is decreased, progressively more protein can be loaded in the system, so that in a 2.5% T gel as much as 90 mg proteins/mL gel can be

applied (9). This has been interpreted as a competition for the available water between the two polymers, the polyacrylamide coils, and the protein to be fractionated. This is an extraordinary amount of material to be carried by a gel phase and renders IPG by far the leading technique in any electrophoretic fractionation.

3. Materials

DEAE- and CM-Sephadexes were from Pharmacia (Uppsala, Sweden), whereas low-gelling agarose (A-37) was from IBF (Villeneuve La Garenne, France). Immobiline chemicals (LKB brand of nonamphoteric buffers having the formula $\text{CH}_2=\text{CH}-\text{CONH}-\text{R}$), acrylamide, *N,N'*-methylene bisacrylamide (Bis), TEMED, persulfate, and GelBond PAG were purchased from LKB Produkter AB, Bromma, Sweden. The Immobiline buffers (five are powders and two, with pK values of 8.5 and 9.3, are liquid) are reconstituted to 0.2M stock solutions by adding 25 mL (by weight) of double distilled water. The solutions are dispensed in 5-mL aliquots that are stored frozen (only one 5-mL aliquot should be kept in the refrigerator at any given time as a working solution; stability at 4°C is not guaranteed after 3 wk, whereas in the frozen state, they are stable for at least 6 m. Acrylamide and Bis are mixed as a stock 30% T, 4% C solution that can be kept at 4°C in a brown bottle for up to a month (by slow hydrolysis, acrylic acid will form with time. *Use with care. Acrylamide, Bis, and Immobilines are neurotoxins; do not inhale powders, and handle liquids with gloves.* After polymerization, they are inert. A stock, 40% solution of persulfate is made fresh every week and stored at 4°C in a dark bottle (0.5–1 mL will be plenty as a week's supply). We routinely use a Mul-

tiphor II or Ultrophor chamber, coupled to a Macrodrive V power supply and to a Multitemp thermostat (LKB Produkter AB). In addition to our published recipes for 2–6 pH unit wide IPG gradients (10), formulations for 58 IPG intervals spanning 1 pH unit, at increments of 0.1 pH unit, are given in Table 1. Only these recipes are provided, since it is quite likely that in preparative runs only narrow IPG gradients will be utilized.

4. Methods

We will give here an example for the purification of normal human adult hemoglobin (HbA) over a 1-pH unit IPG range. The nearest contaminant is its glycosylated derivative (Hb A1c) with a ΔpI of only 0.04 pH unit. A preparative experiment in general requires a 2-d period, as follows: (a) day one: gel casting, washing, weight reconstitution, sample application, overnight run; (b) day two: excision of the major protein zone, casting of the agarose plate, preparation of a trench with either hydroxyapatite or DEAE (diethyl aminoethyl)-Sephadex, electrophoretic transfer and protein elution from the exchanger beads. For the materials check list, see section 3 above. In addition, we will use agarose M or, even better, the low-gelling type (Agarose A-37) and, for protein recovery, two different methodologies: adsorption on a weak [hydroxyapatite (HA)] or a stronger (DEAE-Sephadex) ion exchanger. For the preparative run, we will use a 2-mm thick gel, one half the size of the cooling block of the Ultrophor (11 x 11 cm), over a 1-pH unit interval (pH 6.8–7.8), of low matrix content (3% T), thus being able to fractionate at least 150 mg of total sample. The operative steps are described below. The gel composition of the IPG run is given in Table 2,

Table 1: Narrow pH Gradients^a

Control pH at 20°C	Volume (μL) 0.2M Immobililine pK acidic dense solution							pH range	Mid- point	Control pH at 20°C	Volume (μL) 0.2M Immobililine pK basic light solution						
	3.6	4.4	4.6	6.2	7.0	8.5	9.3				3.6	4.4	4.6	6.2	7.0	8.5	9.3
3.84 0.03	—	750	—	—	—	—	159	3.8–4.8	4.3	4.95 0.06	—	750	—	—	—	—	591
3.94 0.03	—	710	—	—	—	—	180	3.9–4.9	4.4	5.04 0.07	—	810	—	—	—	—	667
4.03 0.03	—	—	755	—	—	—	157	4.0–5.0	4.5	5.14 0.06	—	—	745	—	—	—	584
4.13 0.03	—	—	713	—	—	—	177	4.1–5.1	4.6	5.23 0.07	—	—	803	—	—	—	659
4.22 0.03	—	—	689	—	—	—	203	4.2–5.2	4.7	5.33 0.08	—	—	884	—	—	—	753
4.32 0.03	—	—	682	—	—	—	235	4.3–5.3	4.8	5.42 0.10	—	—	992	—	—	—	871
4.42 0.03	—	—	691	—	—	—	275	4.4–5.4	4.9	5.52 0.12	—	—	1133	—	—	—	1021
4.51 0.04	—	—	716	—	—	—	325	4.5–5.5	5.0	5.16 0.14	—	—	1314	—	—	—	1208
4.61 0.05	562	—	600	863	—	—	—	4.6–5.6	5.1	5.69 0.04	—	—	863	—	—	—	105
4.75 0.05	458	—	675	863	—	—	—	4.7–5.7	5.2	5.79 0.04	—	—	863	863	—	—	150
4.86 0.04	352	—	750	863	—	—	—	4.8–5.8	5.3	5.90 0.04	—	—	863	863	—	—	202
4.96 0.03	218	—	863	863	—	—	—	4.9–5.9	5.4	5.99 0.03	—	—	863	863	—	—	248
5.07 0.03	158	—	863	863	—	—	—	5.0–6.0	5.5	6.09 0.04	—	—	863	803	—	—	338
5.17 0.04	113	—	863	863	—	—	—	5.1–6.1	5.6	6.20 0.04	—	—	863	713	—	—	443
5.24 0.18	1251	—	—	1355	—	—	—	5.2–6.2	5.7	6.34 0.04	337	—	—	724	—	—	—
5.33 0.12	1055	—	—	1165	—	—	—	5.3–6.3	5.8	6.43 0.03	284	—	—	694	—	—	—
5.43 0.12	899	—	—	1017	—	—	—	5.4–6.4	5.9	6.53 0.03	242	—	—	682	—	—	—
5.52 0.09	775	—	—	903	—	—	—	5.5–6.5	6.0	6.63 0.03	209	—	—	686	—	—	—
5.62 0.07	676	—	—	817	—	—	—	5.6–6.6	6.1	6.73 0.03	182	—	—	707	—	—	—
5.71 0.06	598	—	—	755	—	—	—	5.7–6.7	6.2	6.82 0.03	161	—	—	745	—	—	—
5.81 0.06	536	—	—	713	—	—	—	5.8–6.8	6.3	6.92 0.03	144	—	—	803	—	—	—
5.91 0.05	486	—	—	689	—	—	—	5.9–6.9	6.4	7.02 0.03	131	—	—	884	—	—	—
6.01 0.05	447	—	—	682	—	—	—	6.0–7.0	6.5	7.12 0.03	120	—	—	992	—	—	—
6.10 0.04	416	—	—	691	—	—	—	6.1–7.1	6.6	7.22 0.03	112	—	—	1133	—	—	—
6.11 0.11	972	—	—	—	1086	—	—	6.2–7.2	6.7	7.21 0.03	262	—	—	—	686	—	—
6.21 0.09	833	—	—	—	956	—	—	6.3–7.3	6.8	7.31 0.03	224	—	—	—	682	—	—
6.30 0.08	722	—	—	—	857	—	—	6.4–7.4	6.9	7.41 0.03	195	—	—	—	694	—	—
6.40 0.07	635	—	—	—	783	—	—	6.5–7.5	7.0	7.50 0.03	171	—	—	—	724	—	—
6.49 0.06	565	—	—	—	732	—	—	6.6–7.6	7.1	7.60 0.03	152	—	—	—	771	—	—

6.59 ± 0.05	509	—	—	—	699	—	—	6.7–7.7	7.2	7.70 ± 0.03	137	—	—	—	840	—	—
6.69 ± 0.05	465	—	—	—	683	—	—	6.8–7.8	7.3	7.80 ± 0.03	125	—	—	—	934	—	—
6.78 ± 0.04	430	—	—	—	684	—	—	6.9–7.9	7.4	7.90 ± 0.03	116	—	—	—	1058	—	—
6.88 ± 0.04	403	—	—	—	701	—	—	7.0–8.0	7.5	8.00 ± 0.03	108	—	—	—	1217	—	—
6.98 ± 0.04	381	—	—	—	736	—	—	7.1–8.1	7.6	8.09 ± 0.03	103	—	—	—	1422	—	—
7.21 ± 0.06	1028	—	—	—	750	750	—	7.2–8.2	7.7	8.36 ± 0.05	548	—	—	—	750	750	—
7.31 ± 0.06	983	—	—	—	750	750	—	7.3–8.3	7.8	8.46 ± 0.05	503	—	—	—	750	750	—
7.41 ± 0.05	983	—	—	—	750	750	—	7.4–8.4	7.9	8.56 ± 0.05	458	—	—	—	750	750	—
7.66 ± 0.15	1230	—	—	—	—	1334	—	7.5–8.5	8.0	8.76 ± 0.04	331	—	—	—	—	720	—
7.75 ± 0.12	1037	—	—	—	—	1149	—	7.6–8.6	8.1	8.85 ± 0.03	279	—	—	—	—	692	—
7.85 ± 0.10	885	—	—	—	—	1004	—	7.7–8.7	8.2	8.95 ± 0.03	238	—	—	—	—	682	—
7.94 ± 0.08	764	—	—	—	—	893	—	7.8–8.8	8.3	9.05 ± 0.06	206	—	—	—	—	687	—
8.04 ± 0.07	667	—	—	—	—	810	—	7.9–8.9	8.4	9.14 ± 0.06	180	—	—	—	—	710	—
8.13 ± 0.06	591	—	—	—	—	750	—	8.0–9.0	8.5	9.24 ± 0.06	159	—	—	—	—	750	—
8.23 ± 0.06	530	—	—	—	—	710	—	8.1–9.1	8.6	9.34 ± 0.06	143	—	—	—	—	810	—
8.33 ± 0.05	482	—	—	—	—	687	—	8.2–9.2	8.7	9.44 ± 0.06	130	—	—	—	—	893	—
8.43 ± 0.04	443	—	—	—	—	682	—	8.3–9.3	8.8	9.54 ± 0.06	119	—	—	—	—	1004	—
8.52 ± 0.04	413	—	—	—	—	692	—	8.4–9.4	8.9	9.64 ± 0.06	111	—	—	—	—	1149	—
8.62 ± 0.04	389	—	—	—	—	720	—	8.5–9.5	9.0	9.74 ± 0.06	105	—	—	—	—	1334	—
8.40 ± 0.14	1208	—	—	—	—	—	1314	8.6–9.6	9.1	9.50 ± 0.06	325	—	—	—	—	—	716
8.49 ± 0.12	1021	—	—	—	—	—	1133	8.7–9.7	9.2	9.56 ± 0.06	275	—	—	—	—	—	691
8.59 ± 0.10	871	—	—	—	—	—	992	8.8–9.8	9.3	9.69 ± 0.06	235	—	—	—	—	—	682
8.68 ± 0.08	753	—	—	—	—	—	884	8.9–9.9	9.4	9.79 ± 0.06	203	—	—	—	—	—	689
8.78 ± 0.07	659	—	—	—	—	—	803	9.0–10.0	9.5	9.88 ± 0.06	177	—	—	—	—	—	713
8.87 ± 0.06	584	—	—	—	—	—	745	9.1–10.1	9.6	9.98 ± 0.06	157	—	—	—	—	—	755
8.97 ± 0.05	525	—	—	—	—	—	707	9.2–10.2	9.7	10.08 ± 0.06	141	—	—	—	—	—	817
9.07 ± 0.04	478	—	—	—	—	—	686	9.3–10.3	9.8	10.18 ± 0.06	129	—	—	—	—	—	903
9.16 ± 0.07	440	—	—	—	—	—	682	9.4–10.4	9.9	10.28 ± 0.06	119	—	—	—	—	—	1017
9.26 ± 0.07	410	—	—	—	—	—	694	9.5–10.5	10.0	10.38 ± 0.06	111	—	—	—	—	—	1163

*Volumes of Immobiline for 15 mL of each starting solution (two gels) From LKB Application Note No. 324. The pH range (given in the middle two lanes) is the one existing in the gel during the run at 10°C. For controlling the pH of the starting solutions, the values (control pH) are given at 20°C (LKB Produkter AB, Bromma, Sweden).

Table 2
Formulation for a Preparative pH 6.8–7.8 IPG Run^a

Chemicals	Acidic (heavy) sol.	Basic (light) sol.
Immobiline pK 3.6 ^b	387.5 μ L	103 μ L
Immobiline pK 7.0 ^b	569 μ L	781 μ L
Acrylamide stock (30 %T, 4 %C) for (T% = 3)	1.25 mL	1.25 mL
Glycerol (87%)	2.50 mL	
Water	7.79 mL	10.36 mL
Total volume	12.5 mL	12.5 mL
TEMED	9 μ L	9 μ L
Persulfate (40% sol.)	12 μ L	12 μ L

^aGel thickness, 2 mm; size, 11 x 11 cm.

^bFrom reconstituted, stock 0.2M solutions.

whereas two alternative formulations for the electrophoretic elution are listed in Table 3. The hemoglobin (Hb) lysate is prepared as follows: The red cells are washed three times in normal saline and lysed in 2 vol of distilled water (final concentration approximately 8% Hb). The stroma is removed by shaking the lysed cells with 1 vol of carbon tetrachloride and centrifuging at full speed in a bench centrifuge (approximately 3000g) for 20 min. We prefer not to add KCN, since it often disturbs the run. It is best to gas the sample with CO for 30 s and then stored in 35% ethylene glycol at -20°C. The stability is guaranteed for at least 1 yr.

4.1. First Day

1. Assemble the cassette (15 min).
2. Mix the IPG solutions (Table 2), and check the pH (30 min).

Table 3
Eluting Systems for Proteins from an Immobiline Matrix

	HA-Ultrogel		DEAE-Sephadex	
Agarose	M, 0.8 %	800 mg	A-37, 1%	1 g
Buffer	100 mM Tris-Gly ^a pH 9.1	12.1 g Tris + 4.1 g Gly in 1 L final volume (20°C)	100 mM Tris-acetate ^b pH 8.5	12.1 g Tris + 35 mL 1M CH ₃ COOH in 1L final volume (20°C)
Ion exchanger	HA-Ultrogel	15 mL Equilibrated in 150 mL Tris-Gly buffer pH 9.1 to fill a 12.5 x 1 x 0.5 cm trench	DEAE-Sephadex	500 mg swollen and equilibrated in 100 mL of Tris acetate buffer, pH 8.5, to fill a 12.5 x 1 x 0.5 cm trench
Elution buffer	0.2 M Phosphate buffer , pH 6.8	2.84 g Na ₂ HPO ₄ + 1.75 g NaH ₂ PO ₄ in 100 mL final volume (20°C)	0.2M Tris-Gly pH 9.5 in 0.2M NaCl	2.42 g Tris + 0.21 g Gly + 1.168 g NaCl in 100 mL) final volume (20°C)

^aElectrode buffer. double molarity, same pH.

^bElectrode buffer: same molarity and pH.

3. Pour the gradient, carefully rinse the apparatus, and dry the outlet with ethanol (20 min).
4. Polymerize the gel (1 h in a forced ventilation oven at 50°C or in a water bath at the same temperature) (11).
5. Weigh the gel and mark the weight on the plastic backing.
6. Wash the gel (2 x 1 h) in 1 L of distilled water each time.
7. Blot the gel surface with soft tissue (to prevent local gel swelling caused by droplets on the surface) and reconstitute to its original weight (1 h). (During washing the gel swells as it is a weak ion-exchanger. Eliminate adsorbed excess water by blowing warm air with a fan at approximately 1 m distance from the vertically standing gel; check periodically with a balance till weight reconstitution) (12).
8. Apply the sample in the trench precast in the gel with the aid of a Pasteur or a Gilson pipet (150 mg in 0.9 mL), and start the experiment (15 min). Run at low voltage (400 V maximum) until all the sample has entered the gel. The run will continue overnight at 2000 V/10 cm.
9. Reswell the ion-exchanger overnight (Table 3) (if DEAE-Sephadex is chosen for elution) (13).

4.2. Second Day

For all materials in this section, refer to Table 3.

1. Wash the ion-exchanger (30 min).
2. Boil the agarose in buffer (15 min).
3. Cut the IPG gel strip with the protein band of interest (the matrix stays bound to the plastic foil).
4. Pour the agarose into the tray and allow to cool at 42°C.

5. Embed the IPG gel strip into the melted agarose A-37 plate, with the plastic backing against the glass bottom of the tray (this and following operations are shown in Fig. 7).
6. Cut a trench (2 cm wide, 24 cm long) anodic to the IPG strip (*see* Fig. 7).
7. Fill this trench with a slurry of ion-exchanger (DEAE-Sephadex or with HA-Ultrogel, according to the elution system chosen);
8. Start the experiment (1 h at 30 W, approximately 300 V, 5°C).
9. Remove the ion exchanger and transfer to a syringe (5 min).
10. Wash with elution buffer (20 min; in case of DEAE-Sephadex, refer to Fig. 8).
11. For testing protein recovery, read the eluate at 546 nm (for Hb).

5. Notes

1. It is best to start the preparative run at low voltage (maximum of 300–400 V) until all the protein has entered the trench (1 h or more). The reason: If the sample contains salts, the conductivity in the sample zone will be very high, so that, if the experiment is started immediately at the maximum voltage, the protein will be heat-denatured. In addition, in moving from the free phase (the trench) into a restrictive medium, the protein concentrates against the wall of the trench, and there is a risk of aggregation and precipitation. We recommend this procedure also for analytical runs.
2. The advantage of the soft gels will be apparent during the run: The viscoelastic forces of the matrix are weakened, the osmotic force in the protein zone

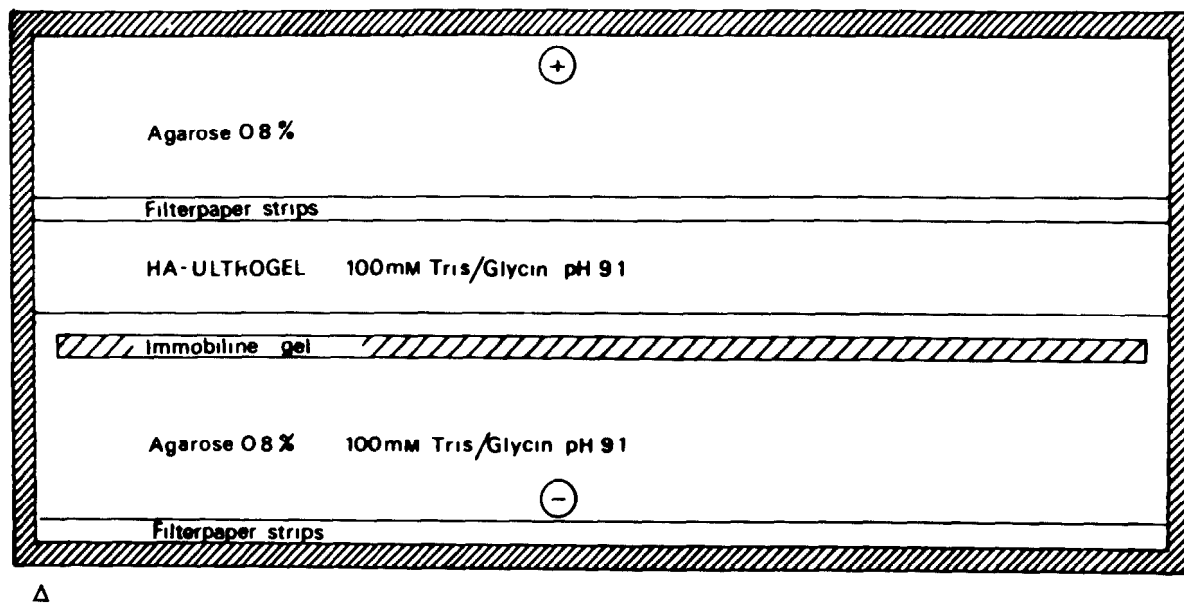


Fig. 7. Assembly for electrophoretic protein recovery from an immobilized strip. The glass tray (LKB 90.00.0157) is the same as that used for preparative IEF runs in granulated gel layers. It is filled with 100 mL of melted 1% agarose in appropriate buffer. Before gelation, the IPG strip is embedded in the agarose phase, to ensure optimum contact between the two matrices upon gel formation. Five millimeters in front of the IPG strip a trench is dug, to be filled either with HA-Ultrogel or with the DEAE-Sephadex, according to the chosen protocol. Electrical contact is ensured with filter paper strips stretching from the agarose gel surface to the electrolyte reservoirs of the Multiphor chamber (from ref. 2).

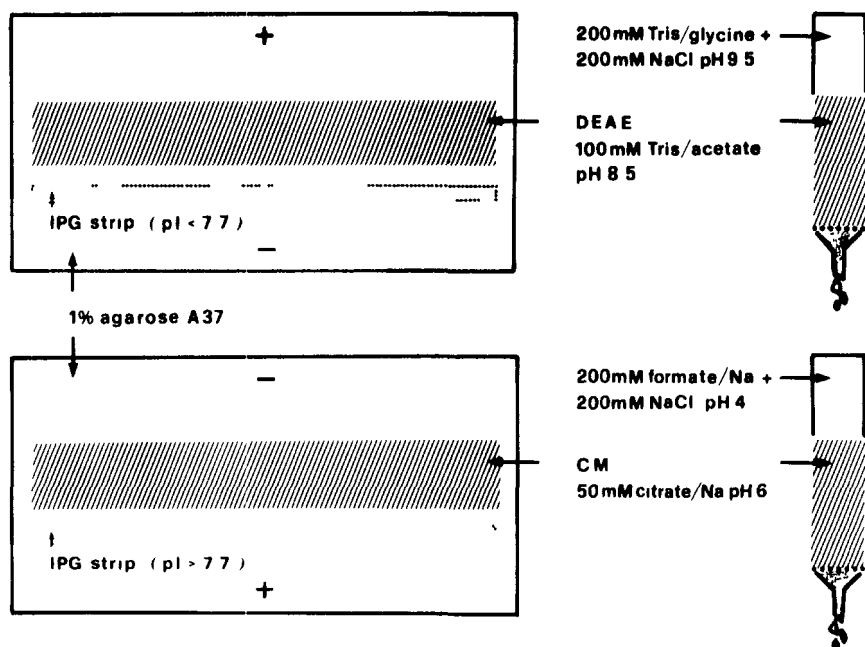


Fig. 8. Preparative set-up for elution into ion-exchange resins. The assembly is the same as in Fig. 7, except that the trench is filled with either DEAE-Sephadex (for acidic to neutral proteins, up to pI values of approximately 7.7) or CM-Sephadex (for decidedly alkaline proteins, $pI > 7.7$) (see the left part of the figure; notice that in the case of CM, the polarity is reversed). Upon electrophoretic transfer, the beads are transferred to a syringe or small column and the proteins eluted by titrating the resins with the two types of buffers indicated in the right part of the figure (upper formulations refer to the elution system; lower formulations to the buffers used during the electrophoretic recovery; these latter formulations are indicated by bidirectional arrows) (from ref. 15).

takes over so that, proportionally to the amount loaded in a single zone, the gel will stretch locally forming a peak in the focused protein band, the peak height representing an equilibrium between the two opposite forces. There is another important

feature of soft gels: The local density of charges is increased by a factor of 1.6 compared with a 5 %T matrix. This results in much sharper bands, as though one had increased the ionic strength of the bulk liquid. The explanation? We believe IPGs proteins are attracted by the charged polymer coil and focus next to it, so that they feel a local increase of charge. Thus, we believe that in IPGs, the protein is isoelectric, but not isoionic, i.e., it forms a salt (protein-Immobilinate or Immobiline-proteinate, in fact both types of salts) with the charged matrix, with the advantages we have seen in section 2.

3. A few more words on the elution systems. We have devised two types, an elution in HA-beads (*see* Fig. 7) for moderate protein loads. For very high protein levels (in the gram range) hydroxyapatite (HA) exchangers will be saturated and lose the protein during the electrophoretic step at the anode. Therefore, we have devised an elution in DEAE- (for proteins up to pI 7.7) or in CM-Sephadex (for alkaline proteins, $pI > 7.7$), as illustrated in Fig. 8. From these last systems, the proteins are eluted by titrating (i. e., deprotonating the DEAEs or protonating the CMs) the exchanger, with recoveries better than 90% protein in all cases tested (from HA beads, the classical eluant is 0.2M phosphate buffer, pH 6.8). As an alternative, for very small protein loads, one could resort to the canal-Immobiline system devised by Bartels and Bock (14). Finally, we have described a fourth elution system that, unlike the previous three, allows for direct recovery in a free liquid phase in a dialysis bag (16).

Acknowledgments

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Chapter 19

Isoelectric Focusing Under Denaturing Conditions

*Christopher F. Thurston
and Lucy F. Henley*

1. Introduction

The introduction of methods for the electrophoretic analysis of proteins under denaturing conditions (1,2) is something of a landmark in the development of methodologies for the analysis of proteins. Not only was analysis giving good correlation to molecular weight possible with small samples of complex mixtures of proteins, but in addition, analysis of normally insoluble proteins became possible in a relatively simple manner. These methods were based on solubilizing protein in sodium dodecyl sulfate (SDS) such that the intrinsic charge differences between proteins were overwhelmed by the relatively strong charge of the dodecyl sulfate ion, which resulted in very good corre-

lation (for most, but not all, proteins) between electrophoretic mobility and molecular weight (*see* Chapter 6 in Vol. 1 of this series).

It is precisely the property of proteins masked by dodecyl sulfate that is exploited in isoelectric focusing (IEF) analysis, and consequently if denatured protein is to be analyzed by IEF, an alternative method of solubilization is required. We describe here the IEF analysis of protein samples solubilized by the use of 8M urea, a powerful denaturant that is uncharged, based on the method of O'Farrell (3).

Complete denaturation of many proteins involves, in addition, the reduction of inter- and/or intramolecular disulfide bonds that is accomplished by inclusion of a thiol reagent such as 2-mercaptoethanol or dithiothreitol. This is simple to do for electrophoretic analysis because the dissociated/unfolded polypeptides are stabilized in this state by SDS, but here again the requirements of IEF are different. The redox state of a protein is affected by thiol reagents and so consequently is their behavior during IEF analysis. We therefore show an example of such effects and give conditions for maintaining protein in the fully reduced state during the focusing process (*see* Figs. 1 and 2).

2. Materials

1. Silane A174 (γ -methacryloxypropyltrimethoxysilane).
2. Glacial acetic acid, analytical grade.
3. Urea. Use the highest purity available. "Ultrapure" from Schwartz-Mann is suitable. Although urea solutions can be deionized with a mixed-bed ion-exchange resin, the procedure requires solid urea at

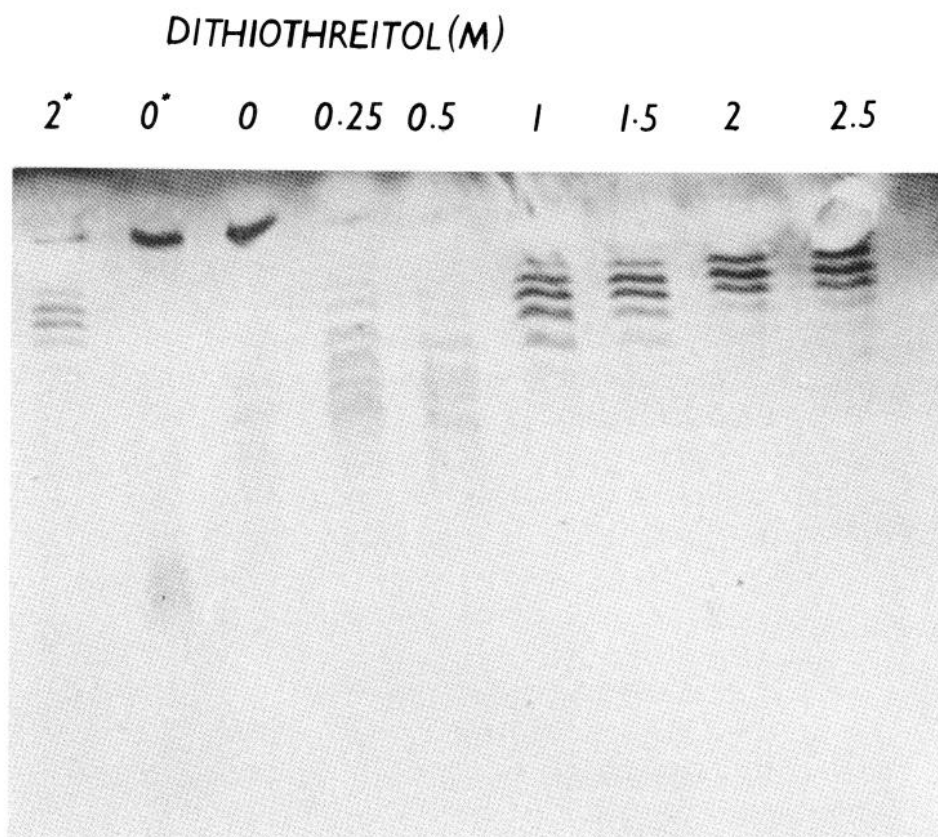


Fig. 1. Isoelectric focusing gel stained with Coomassie blue. All tracks were loaded with 10 μ g pure isocitrate lyase. The concentration of dithiothreitol in the samples is shown above each track. Asterisks indicate tracks loaded with immunoprecipitated isocitrate lyase (see Chapter 12). The anti-isocitrate lyase immunoglobulins present in these samples did not focus as distinct bands. The heavily stained bands near the top of the photograph in tracks containing no dithiothreitol are artifactual; in the absence of reducing agent, much of the protein commonly precipitates in the position of the applicator strip or along one end of the strip. The region of the gel shown is from about pH 6 (top) to about pH 4 (bottom) (from ref. 5, with permission).

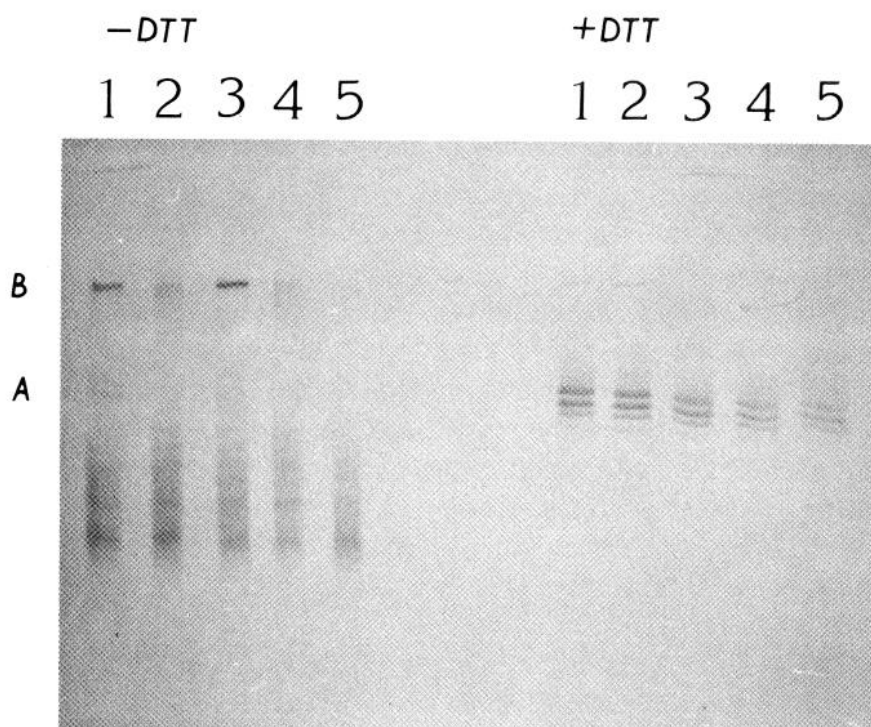


Fig. 2. Analysis of immunoprecipitated isocitrate lyase by isoelectric focusing under denaturing conditions. Samples taken from a culture labeled by growth on [^{35}S]-sulfate at 0, 2, 4, 7, and 11 h after transfer to "deadapting" conditions. With increasing time of sampling, the amount of radioactivity precipitable with anti-isocitrate lyase decreases from 11,000 cpm at 0 h (track 1) to 5000 cpm at 11 h (track 5). Fully reduced isocitrate lyase focuses as two main bands and one or two minor bands, average pI , 5.6, as seen when the samples were focused in the presence of 2M dithiothreitol (+DTT). In the absence of dithiothreitol (-DTT), only the initial sample contained detectable, fully reduced isocitrate lyase (marked A), and in successive samples the enzyme becomes progressively more oxidized focusing at progressively more acidic pI values. Bands marked B are precipitated protein at the position of the applicator strips. The pH range shown in the fluorograph is approximately from pH 8 (top) to pH 3.5 (bottom) (from ref. 5, with permission).

least for sample preparation, so it must be recrystallized if purification is attempted.

4. Nonidet NP40: 10% (v/v) solution in distilled water.
5. 30% Acrylamide: 28.38% (w/v) acrylamide and 1.62% (w/v) *N,N'*-methylene bisacrylamide, dissolved in distilled water, filtered through a 0.22- μ m pore-size membrane filter and stored in the dark at 4°C. This reagent can also be contaminated with charged impurities (acrylic acid), which can be removed by recrystallization or treatment with ion-exchange resin, but there are now highly purified grades available from most suppliers.
6. Ampholytes: Use premixed preparations to give the desired pH range, such as LKB "Ampholines" or Pharmacia "Pharmalyte."
7. *N,N,N',N'*-Tetramethylethylenediamine (TEMED).
8. Ammonium persulfate. Use analytical grade and store the solid at 4°C. The solid can decompose if heated, without change in appearance, so do not rely on a bottle of uncertain provenance.
9. 10 mM Phosphoric acid.
10. 20 mM NaOH.
11. Fixing solution: 0.7M trichloroacetic acid, 0.16M sulfosalicylic acid (TCA/SSA).
12. Destain solution: glacial acetic acid:ethanol:water; 1:2.5:6.5 by volume.
13. Staining solution: 2.5 g Coomassie blue R250, 100 mL glacial acetic acid, 500 mL ethanol, 400 mL distilled water. Filter through Whatman No. 1 paper on the day of use. (Dissolve Coomassie in ethanol component first.)
14. 1M Sodium salicylate or a commercial water-miscible fluorographic solution such as "Amplify"

from Amersham-Searle or "Enlightning" from New England Nuclear.

15. Flatbed isoelectric focusing apparatus with provision for cooling the gel-carrying plate.

3. Method

3.1. Casting the Gel

1. Adjust the pH of 1 L of distilled water to 3.5 with glacial acetic acid and mix in 4 mL of Silane A174 by stirring for 15 min. Soak a glass gel-plate in the solution for 1 h, rinse with distilled water, and dry in air (*see* Note 2 in section 4).
2. Use a few drops of ethanol to adhere a clean plastic sheet to a second glass plate (*see* Note 2 in section 4).
3. Mount the silanized plate and the plate-backed plastic sheet separated by a 1-mm casting gasket, with the open corner of the gasket uppermost.
4. Prepare 20 mL of gel-forming solution in a sidearm flask as follows: 9.6 g urea, 4 mL distilled water, 4 mL 10% nonidet, 2.66 mL acrylamide solution, and 0.5 mL ampholyte mixture are gently swirled to dissolve the urea (*see* note 1 in section 4). Place an oversize rubber bung over the top of the flask and apply a vacuum from a water pump via the sidearm for 10 min with occasional further swirling to thoroughly degas the mixture. Allow air to re-enter the flask slowly (*see* Note 3 in section 4).
5. Add 22 μ L of TEMED and 16 μ L of freshly prepared 10% (w/v) ammonium persulfate, swirl briefly to mix, take up into a disposable syringe, and deliver immediately into the gel-casting box from step 3 above. Completely fill the system and cover the opening in the casting gasket with parafilm. The

mixture should polymerize within about 2 h, but may conveniently be left overnight at this stage (*see* Note 4 in section 4).

3.2. Prefocusing the Gel

1. When the gel has polymerized, unclamp the plates, separate the plastic sheet from its backing plate, then very slowly peel off the plastic sheet from the acrylamide gel layer. Lift away the casting gasket and remove any liquid from the edge of the gel with clean tissue.
2. Mount the gel-plate on the cooling plate of the isoelectric focusing apparatus. Circulate water at 10°C through the system.
3. Apply electrode wicks soaked in 10 mM phosphoric acid (anode) and 20 mM NaOH (cathode) to the edges of the gel and connect the electrodes.
4. Using a dc power source, set to give constant voltage, apply 200 V for 15 min, 300 V for 30 min, and 400 V for 30 min, to establish the pH gradient within the gel.

3.3. Sample Preparation

To samples in solution, add solid urea to give a concentration of 8M then mix with an equal vol of 8M urea, 2% (w/v) Nonidet NP4D, 2% (w/v) ampholytes. Protein precipitates may be dissolved directly in this mixture. The sample mixture must contain a relatively high concentration of protein since the maximum volume that can be loaded is 35 μ L. The sample mixture may in addition contain 2M dithiothreitol (*see* Note 5 in section 4).

3.4. Focusing

Disconnect the power supply from the prefocused gel and place paper sample applicators near the cathode end of the gel (*see* Note 6 in section 4). Deliver the sample (preferably no more than 10 μ L) onto each applicator and focus for 30 min at 500 V. Disconnect the power supply and remove the paper sample applicators from the gel with flat-ended forceps. Focus for a further 90 min at 800 V.

3.5. Measuring the pH Gradient

When focusing is finished, remove the electrode wicks and cut a track of 5-mm square sections from the position of the anode to the position of the cathode. Soak each section overnight in 0.25 mL distilled water and measure the pH with a microelectrode without removing the slice of gel.

3.6. Staining

Fix for 1 h in TCA/SSA (this step also removes ampholytes that would otherwise stain with Coomassie blue), and wash for 30 min in destain solution. Stain with Coomassie blue with constant agitation for 2–16 h. Destain with repeated changes of destain solution until the background of the gel is cleared of stain.

3.7. Fluorography

For detection of radioactive bands on the gel, it is necessary to detach the gel from the silanized glass plate. By the time the gel is fully destained, it has usually started to detach at the edges. Hold the plate upside

down over a tray containing destain solution, and separate the gel from the plate with a sharp scalpel. Soak the gel in 1M sodium salicylate or a commercial fluor for 2 h, and dry down onto Whatman 3-mm paper and expose to X-ray film (see Chapters 16 and 17 in vol. 1 of this series).

4. Notes

1. The quantities given are for a gel 230 × 115 mm. Plates of this size, plastic sheets, and casting gaskets are obtained from Pharmacia. The Pharmacia manual *Isoelectric Focusing—Principles and Methods* is an extremely useful reference work for both the background and practicalities of isoelectric focusing.
2. The thickness of the gel (1 mm) is optimal for focusing, but makes handling difficult. The polymerized gel is covalently linked to the glass plate by the silane in order to prevent creepage and/or buckling during focusing and is polymerized against a plastic sheet rather than a second glass plate, since it is easier to detach from the polymerized gel after casting.
3. For a usable focusing pH range of 4–9, use 0.4 mL pH 3–10 ampholytes and 0.1 mL of pH 4–6 ampholytes. Note that the electrode solutions described are for this pH range and are not applicable to all ranges.
4. The polymerization of acrylamide mixtures is inhibited by oxygen and impurities. Shaking up the solution to mix it must be avoided, particularly after degassing. If the dispensing syringe is large enough to hold all of the mixture, the process of filling the syringe is probably sufficient to mix the final additions. If the gel does not set, try degassing

longer or try a different bottle of ammonium persulfate, and if still unsuccessful, double the amount of TEMED included. If this does not work, the reagents are not pure enough. If the gel sets unevenly, the glass plate was not clean.

5. The small sample volume allowed makes precipitated samples desirable. Chloroform/methanol precipitation (4) is a good way of doing this, and has the additional advantage of removing lipid. The description given here is for small volumes, such as might be processed for electrophoretic analysis, but the volumes may be scaled up in proportion 10-fold, using 15 mL Corex tubes. All steps are performed at room temperature. Take 150 μ L of protein solution in a 1.5-mL microfuge tube, add 600 μ L methanol, mix by three or four inversions of the capped tube, and centrifuge for 10 s. Add 150 μ L chloroform, blend briefly on a vortex mixer, and centrifuge for 10 s. Add 450 μ L of distilled water and blend thoroughly on a vortex mixer. Centrifuge for 10 s to separate the two liquid phases that formed on addition of the water. Carefully remove the upper phase and discard. Do not attempt to remove it all, but leave a residual layer 1–3 mm deep, because most of the protein is at the interface. Add 450 μ L methanol, vortex mix briefly, and centrifuge for 4 min. After centrifugation, a white protein pellet should be visible. Aspirate off and discard (as chlorinated solvent) the bulk of the supernatant. Remove remaining supernatant and dry the pellet in a vacuum desiccator or under a stream of dry N_2 . Pellets dissolve readily and without heating in sample buffers used for electrophoresis or IEF and may be stored indefinitely at -20°C .

If the sample contains proteins that are susceptible to oxidation, clear resolution may only be obtained if the samples are made 2M in dithiothreitol (see Figs. 1 and 2). There is a need for this very high concentration to prevent migration of protein(s) and thiol reagent in different directions during focusing from depleting the protein(s) entirely of reducing agent.

6. The best position for sample loading strips has to be determined by trial and error. The samples we have run all contain proteins that precipitate at acidic pH, so it has been preferable to place the applicators in the pH 8 region of the (prefocused) gel.

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Chapter 20

Computer Analysis of 2-D Electrophoresis Gels

Image Analysis, Data Base, and Graphic Aids

H. H-S. Ip

1. Introduction

Although two-dimensional (2-D) gel electrophoresis has been used for the past ten years, a large amount of potentially useful information is still generally unused, since accurate and rapid methods for extracting, managing, and analyzing the vast quantity of data that can be derived from the gels are not widely available. The difficulty is partly caused by the large number of spots as well as artifacts present in a 2-D gel and the non-uniformity in the electrophoresis process that causes distorted spot patterns and makes comparison among gels difficult. With the advent of data base management systems (DBMS) and computer graphics, 2-D gel data can be organized and manipulated with relative ease.

This chapter describes work that has been done toward automating the process of extracting useful quantitative information from 2-D gels and comparing spot data derived from separate gels. The design and implementation of a data base and a data analysis system for 2-D gels are also outlined.

An approach to the problem is to present a computer with an image of a 2-D gel and automate the process of locating and analyzing protein spots seen on the gel image. Several systems have evolved along this line (for examples, *see* refs. 1-4).

The computational problems one faces in this approach are as follows:

1. Elimination of background level and "noise" from the gel image.
2. Delineation of areas in the image corresponding to protein spots on the gel.
3. Measurements of protein spot attribute values, e.g., the position of the spot on the gel, the size of the spot, and so on.
4. Pairing of corresponding spots seen on separate gel images.

Each of the above subtasks involves the processing of a large amount of data. Our work takes advantage of the processing power of a 2-D array computer, CLIP4S (5), developed at University College London. An array computer that processes large quantity of data "in parallel" reduces the time and cost of performing data-intensive tasks compared with a conventional computer, which carries out computations serially.

We have also investigated the applicability of a commercially available data base management system (DBMS) to storing and managing data relevant to 2-D gel analysis. The data base is linked to the program

modules, which provides various utilities for visualizing, manipulating, and analyzing protein spot patterns by means of computer graphics. This includes procedures for reconstructing the protein spot pattern as seen on the original gel on a computer terminal or on paper, procedures for matching spots across gels, and graphics aids for manipulating and editing spot data.

2. System Overview

The computing hardware for abstracting protein spot data from a 2-D gel image comprise the CLIP4S array computer and a LSI11-23TM minicomputer. Apart from controlling the operations and data movements in and out of the array computer, the minicomputer also provides the tools and environment for software developments under the UNIXTM operating system. Mass data storage is provided by magnetic disks.

Briefly a 2-D gel is back-illuminated from a lightbox with a Vidicon television camera mounted directly above it. The television signals of the 2-D gel image are converted into a digitized image of 512 x 512 picture elements (pixels): each pixel can have one of 256 (8 bits) gray values or intensity (0 = black, 255 = white).

The CLIP4S array computer consists of a 512 x 4 strip of processing cells. Each processing cell is a simple computer in its own right, and each cell is connected to its eight nearest neighbors, as shown in Fig. 1, so that information can be passed between them. The array computer operates on successive strips of 512 x 4 pixels in the gel image, starting from the top of the image and working downward. This scanning process is completely automatic. The processing power of the array computer comes from the design; all available 2048 processing cells operate on their assigned pixels simultaneously. For example, 2048 additions can be completed in the

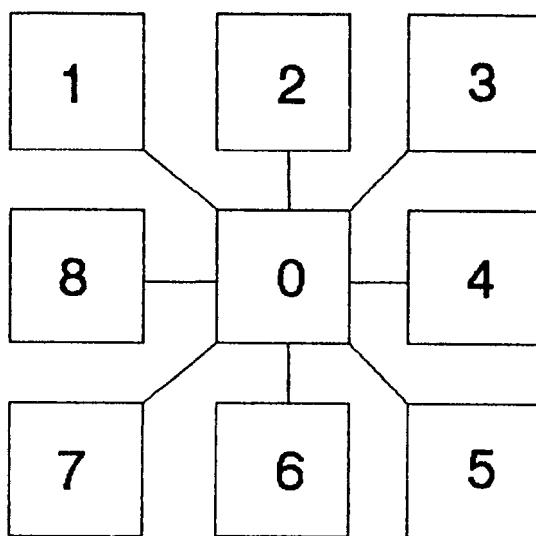


Fig. 1. Interconnections of the CLIP4S processing cells.

time it takes one processing cell to complete one addition.

Protein spot measurements derived from the gel image by the array computer are transferred via magnetic tapes to the DEC2060 mainframe computer at the Imperial Cancer Research Fund. This Mainframe computer holds the 2-D gel data base and the relevant data analysis programs. The user interacts with 2-D gel data using a SIGMEX 6130 graphics workstation and a Summagraphic data tablet. The former is connected to the mainframe computer. The system configuration is summarized in Fig. 2.

3. Processing of 2-D Gel Electrophoresis Images

3.1. "Noise" Reduction

To reduce digitization (electrical noise in the gel image), the camera scans the gel eight times, producing

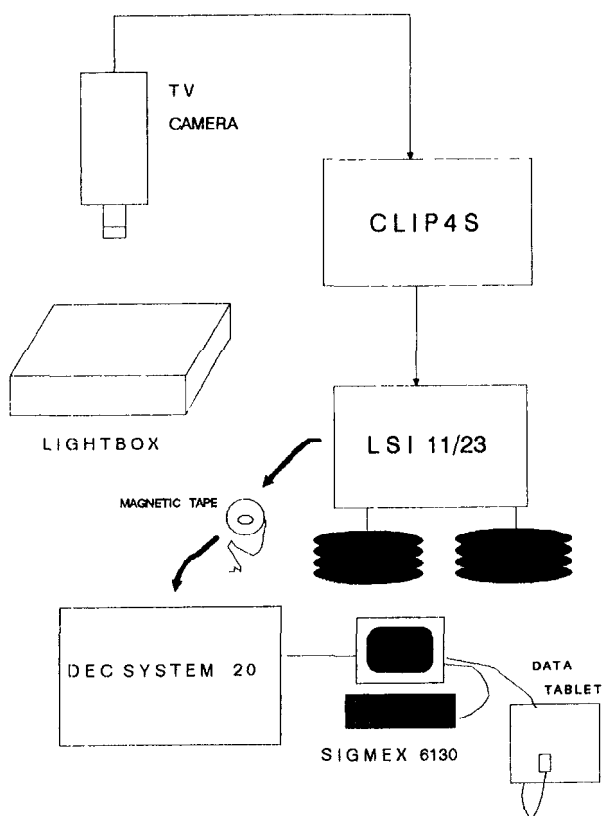


Fig. 2. System configuration and overview.

eight digitized images of the gel. These images are added, and the resultant image is divided by eight to obtain an averaged image. Each pixel gray value in the averaged image is, therefore, an average of the corresponding pixel gray values in the eight input images.

Data in the averaged image can be "smoothed" in a statistical sense by fitting them with a high-ordered polynomial function. Figure 3 shows the result of fitting the image data with a cubic B-spline function. The advantage of using B-spline functions over other data-smoothing functions, such as weighted local average or

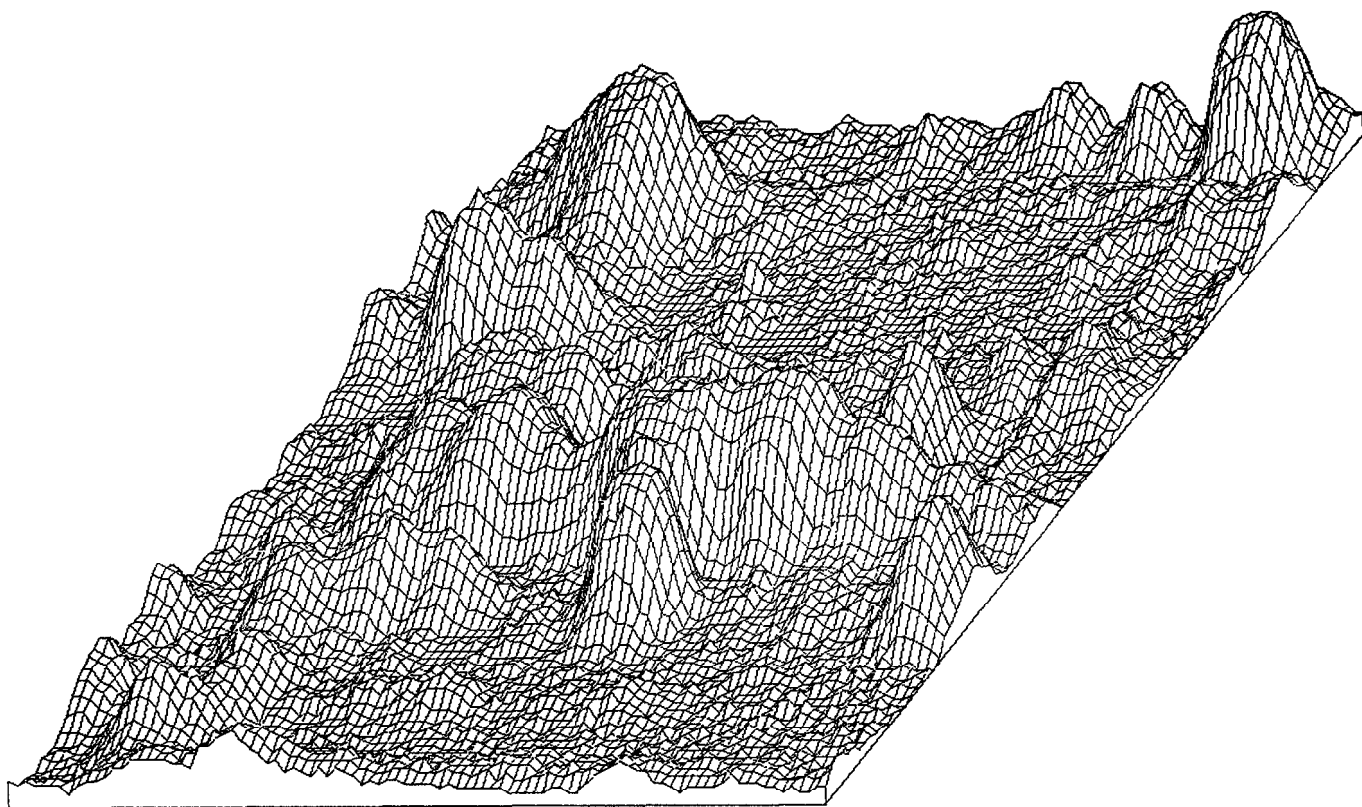


Fig. 3. Noise reduction. (a) 3-D intensity plot of the original gel data.

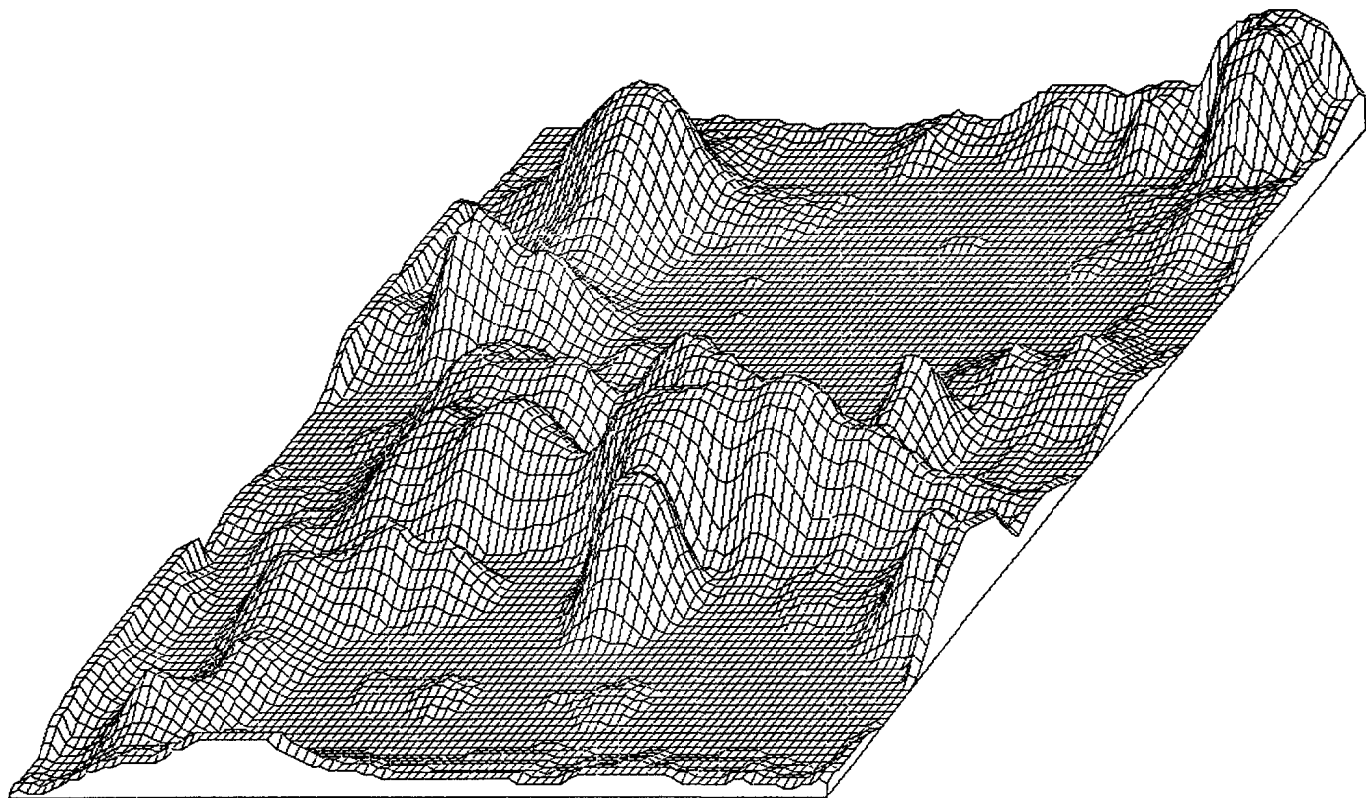


Fig. 3b. Results of fitting a cubic B-spline function to (a).

a median filter, is that it gives better control over the amount of detail we would like to retain in the final image. This is particularly important for 2-D gel images in that small, faint protein spots are not "smoothed out" by the operation that aims to reduce the effects of noise.

3.2. Background Estimation and Subtraction

This process estimates the background levels at different parts of the gel image and subtracts them from the original image. A spatial filtering technique called "opening" (6) is used. Opening consists of two basic morphological operators, namely EXPAND (or dilation) and SHRINK (or erosion). The operation EXPAND/SHRINK replaces each pixel in the gel image by the maximum/minimum gray value of itself and those of its eight nearest neighbors.

Without going into detail of morphological operators, in general, expanding an image n times followed by shrinking the expanded image n times effectively filter out any objects whose areas are less than $(2n + 1)$ by $(2n + 1)$ pixels, i.e., opening acts as a low-pass filter. The size of the objects to be removed from the image is determined by the number n of EXPAND and SHRINK operations that apply to the image. With the objects removed from the image, whatever is left behind in the resultant image is considered to be background level.

Since background levels in the gel vary smoothly from region to region, the present implementation set n large enough to remove any dark objects in the gel image. The value of n ranges from 20 to 40 pixels, depending on the size of the largest dark area (typically corresponding to the area in which a number or touching or overlapping spots can be found) in the image. Since overestimating the value of n is better than underesti-

inating it (the latter may result in the elimination or reduction of genuine spot signals in the image), the value of n is usually set to around 40 pixels for most gel images. Figure 4 illustrates this background subtraction process.

3.3. Spot Delineation

There are two basic components in the spot-delineation process: the detection of the spot and the determination of spot boundaries. The former aims to detect the location of a protein spot in the image with emphasis on resolving overlapping spots; the latter estimates the boundary of the detected spots. Ideally, the boundary should include most of the protein loading of the detected spot and, at the same time, take into account the effects of overlapping spots.

The spot-detection process models the intensity profile of a protein spot on a Gaussian function and locates the "central core" of the spot using a method described by Lemkin and Lipkin (2). Our method, however, differs from that described in ref. 2 in the forms of the Laplacian or second derivative masks that are used to locate the core of protein spots and the way spot boundaries are determined. Instead of determining the spot boundaries by propagating out from the central cores according to a set of heuristic rules that are based on observations on the shape of the Gaussian function and its derivatives, the boundaries of the spots are computed by simply EXPANDING the central cores a number of times. Possible merging of separate central cores during this EXPANDING phase is prevented by the "exoskeletons" which will be described in detail later.

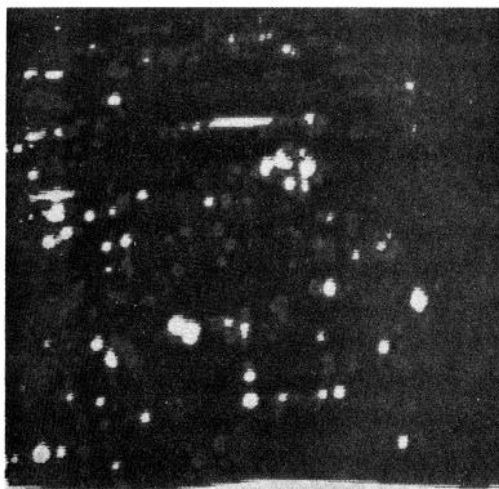


Fig. 4a. Original gel image.

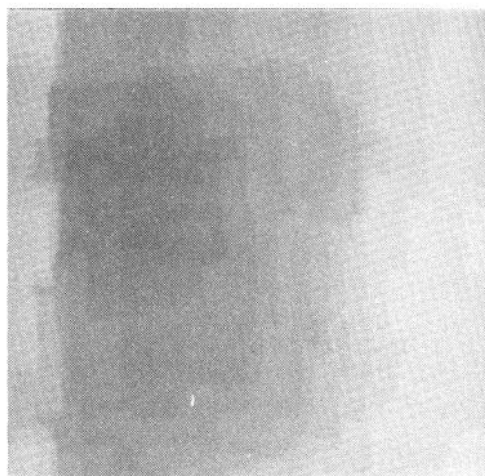


Fig. 4.b. Background.

3.3.1. Central Core Detection

The central core detection algorithm can be summarized as follows. The image gray values are "convolved" using the set of one-dimensional masks:

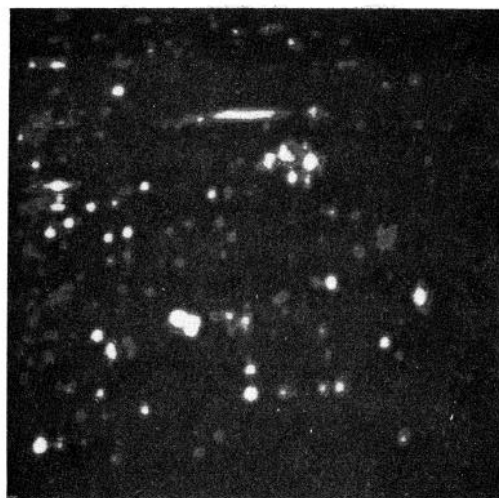


Fig. 4c. Background subtracted gel image.

$$[1\ 0\ 0\ 0 - 2\ 0\ 0\ 0\ 1] \quad [1\ 0\ 0\ 0 - 2\ 0\ 0\ 0\ 1]^T$$

The convolution operation forms a weighted sum, for each pixel in the image, of pixel values with weighting factors and neighborhood defined by the above masks. Points on the image that give rise to negative values for both masks are defined as the central cores of the spots. Isolated points are considered as noise and are removed. Each group of eight-connected points remaining is taken as the central core of a spot. Figure 5 shows the central core image as computed by this process.

The two masks are variations to the second derivative masks used in Lemkin's method. It was observed that there are fluctuations in the intensity values within a protein spot that cause local departure of the protein spot intensity distribution from the Gaussian function that is being modeled. Consequently more than one central core may be generated from a single spot when using the local 3 by 3 difference formula of Lemkin. The

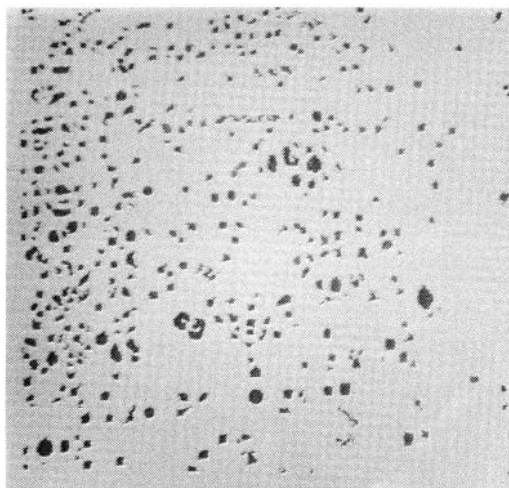


Fig. 5. Central core image of Fig. 4c.

effects of this local fluctuation can be reduced by using the modified second derivative masks that effectively "step over" such local fluctuations.

This fluctuation of intensity values within a spot may be attributed to local variations of polypeptide loading within the spot as a result of the nonuniformity of the electrophoresis process. Other possible contributing factors are the digitization and staining processes.

The size of the modified masks, however, determines the resolution at which overlapping spots can be resolved. The trade off between using different size masks is that, if the mask is relatively too small (e.g., 3 by 3), a single large spot may result in several distinct central cores; if the mask is large (e.g., 15 by 15), overlapping spots may give rise to a single central core. These observations suggest the use of multi-resolution masks for different size spots; the research problem in this case would be to find a suitable control strategy for the application of the different-sized masks and a way to interpret the resulting image.

Central cores arising from artifacts in the gel can be removed by deleting any central cores smaller than a given size, e.g., isolated points, and those with intensities below a given threshold value.

Although streaks in the gel image can be filtered out using directional filter, e.g., EXPAND followed by SHRINK in the direction perpendicular to the direction of the streaks, they are sometimes retained in order to account for the total protein loads of the spots. In the presence of a streak, the above delineation method would produce a series of central cores along the length of the streak.

3.3.2. *Determination of Spot Boundaries*

Central cores generated from the above process typically cover only the central portion of the spots. Some way of propagating the central core outward in order to account for all the protein load attributes to the spot is needed.

This can be done by computing the "exoskeleton" of the central cores in the image. Exoskeleton is a planar graph that marks the midlines of the space between the set of objects (central cores) in an image. Each arc in the exoskeleton forms an equidistance boundary between a pair of nearest objects. The exoskeleton of a set of central cores can be computed by finding the medial axes of the background in the image. The following set of thinning masks (7) are used:

00X	X00	X1X	X1X
011	110	110	011
X1X	X1X	X00	00X
000	1X0	X11	0XX
X1X	110	X1X	011
11X	XX0	000	0X1

where 1 = pixel, which must have value 1 (white)
 0 = pixel, which must have value 0 (black)
 X = don't care

This set of masks or bit patterns is matched successively to all the points in the "inverse" of the central core image. When a match is found, the central pixel is deleted (i.e., switched from a value of 1 to 0). The process is repeated by cycling through these masks until the output image stabilizes. Free ends in the resulting skeletons are "chewed" off using the following thinning masks:

X00	X1X	00X	000
110	010	011	010
X00	000	00X	X1X
000	XX0	00X	000
X10	010	01X	010
X00	000	000	0XX

To determine the spot boundaries, the central cores are EXPANDED K times. After each EXPAND, the expanded central cores image is logically "AND" [i.e., corresponding pixels having values of 1 (white) in both operand images remain white, whereas all other pixels are switched to values of 0 (black) in the resultant image] with the inverse of the exoskeleton image. The latter step has the effect of preventing the expanding spot areas from merging. The value of K is currently set to 5, which is an estimate of the extent that the polypeptide fragment load of a spot may "spread" in the gel. This value also reflects the focusing capability of the run. Figure 6 illustrates the spot boundaries determination process. When the process is completed, two output images for each scanned gel are stored, namely the

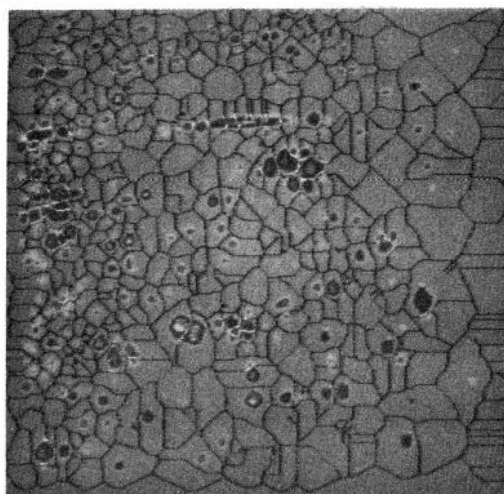


Fig. 6a. Exoskeleton of Fig. 4.

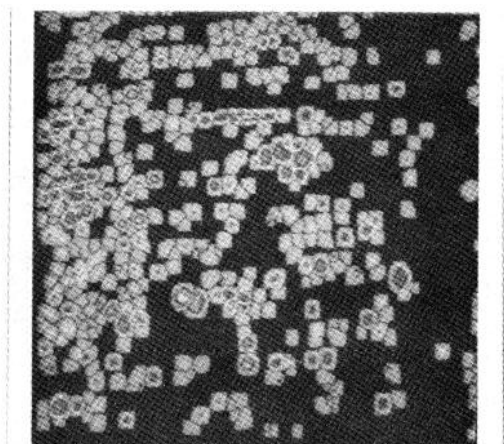


Fig. 6b. Estimated spot boundaries.

background subtracted image and the central core image.

3.4. Measurement of Spot Attribute Values

The set of attribute values to be measured for a protein spot includes:

1. Spot center coordinates with respect to the bottom left corner of the image.
2. Central core area.
3. Integrated density within the spot boundary.
4. Maximum gray value (peak value) of the spot intensity distribution.
5. Minimum gray value of the spot intensity distribution.

Measurements of these attributes are computed automatically from the processed images and grouped into a data record, one for each spot. These data records can be stored inside the data base and form the input to the data-analysis programs in the system.

4. Data Base for 2-D Gels

4.1. Design Issues

For a given experiment, the relevant information in our data base is organized in a hierarchical fashion. At the top of the hierarchy is information on the cell line used in the experiment, followed by information on the different samples derived from that cell line. Each sample data record contains information on the associated gel, such as the amount used in the run and the physical and chemical parameters of the run. Successive levels in the hierarchy contain information on the scanning parameters and the numerous spot data records derived from each scan of the gel. The organization of the data base is shown in Fig. 7.

We have investigated the construction of a data base for 2-D gel data using a commercially available data base management system (DBMS). In this way, we can take advantage of the tools provided by the DBMS, such as facility to construct queries, mechanism for

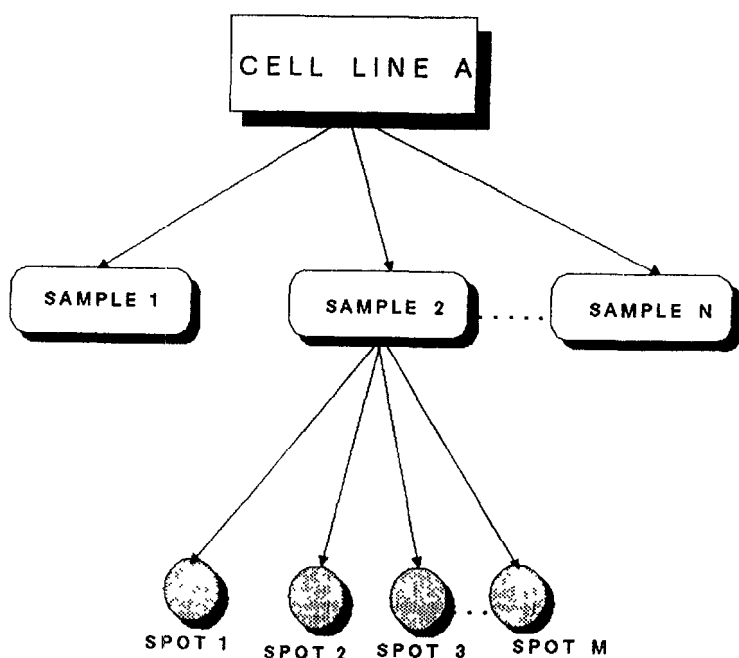


Fig. 7. Organization of a 2-D gel data base.

searching and updating relevant data records, and so on. The initial design was concentrated on the partition of all relevant information into separate entities and the relationships between entities. A specific query is answered by combining information on the relevant entities through relational operators. A clear understanding of the set of most likely queries would be advantageous in the sense that the access paths to the relevant data can be optimized during the early stages of the design.

The system is a 1022 Data Base Management System (8), in which different entities are stored in separate data sets, and information stored in different data sets can be combined using various operators, e.g., MAP, JOIN. The system also provides a simple syntax for

formulating user queries. Complicated queries can be constructed from a set of available commands in conjunction with relational/conditional operators, e.g., the following sequence of queries can be used to select from the data base all cell line data records whose cell line reference begins with "cmt" and whose tumor type is high metastatic, then find all protein spot records that have been found for these cell lines:

```
FIND cell_line_ref BEGIN cmt AND tumor_type  
EQUAL high_metastasis.  
MAP TO spots_data_set VIA cell_line_ref
```

Keyed attributes of entities are indexed for quick access. There are also facilities for manipulating and combining subsets of selected data, called the selection groups. Selection groups are the result of previous queries. Application programs access data stored inside the data base through a set of FORTRAN interface subroutines.

4.2. Implementation Issues

There are a number of limitations in our version of the 1022 DBMS, which makes the implementation of the 2-D gel data base structure entirely within the framework of 1022 impractical. One of them is the fixed length storage structure of 1022, which makes it necessary for the implementor of the data base to predefine the storage space that will be used to store each attribute of the entity. Consequently, more storage space is assigned than is necessary, because each data record takes up the amount of space that will be needed for the worst-case entry, even though most of the assigned space is left empty for a typical record. Another limitation is that

deleted record space cannot be reclaimed until the entire data base is rebuilt (DUMPed). This means for a data base whose entries will be frequently edited (deleted and added), a large amount of storage space will be locked up and frequent "DUMPing" is needed in order to retrieve this space.

In the context of 2-D gel data base data on cell lines, samples are relatively static in the sense that once it is entered, the data record is not likely to be deleted or updated. On the other hand, data on protein spots and across gels matches are highly dynamic. Incorrect matches may be deleted (in the case of no match) or updated (in the case of mismatch); spot data records may also need to be deleted or updated because of possible errors in computer delineation. More importantly, since each gel may produce thousands of spots, using a fixed amount of storage for each protein spot data record is not suitable. This is because slight overestimation of the storage space required for a spot data record will result in significant wastage of storage space when multiplied by thousands of spot records. Under the circumstances, it is more appropriate to use only the minimum amount of storage space required by each spot record. This means using a variable-length record structure. Moreover, the storage overhead of each spot record resulting from entry in the key table for fast data base searches is also significant when multiplied by several thousands.

As a result of the above considerations, it was decided that data on cell lines, samples, and runs will be stored using 1022 data record structures, and data on protein spots and matches will normally be held in flat ASCII files that can be loaded into 1022 data record structures when needed. Application programs are designed to access spot data from these flat files, and con-

version routines are provided to convert spot data from 1022 record format to flat file format and vice versa.

Details of the implementation will not be covered here, except to give a brief description of what information is stored in the data sets. The 1022 2-D gel data base comprises four separate data sets, namely cell lines, samples, scans, and spots. The information stored in each data set is shown in Table 1. Using the 1022 MAP function, attribute values of associated data records stored in separate data sets can be extracted.

The data base on cell lines, samples, and scans is maintained and updated by the biologists as and when data become available. The spot data base contains data records of spots that are needed for data base queries, e.g., associating a subset of protein spots with data on the cell lines or samples from which these spots are derived.

There is no reason, however, to prevent these data from being incorporated into a more flexible DBMS that supports variable size records. Combinations of information stored inside these two types of file can be used in multiple gel analysis. For example, corresponding spots found across a series of gels can be compiled by "chaining" through the corresponding match files and having their respective attribute values collated, e.g., the variation of densities of corresponding spots across a series of exposures of the same autorad gel.

5. Multiple-Gels Analysis

The computation of individual spot data has already been covered above: here we describe the process of matching protein spots across gels.

Table 1
Attributes for the Four Data Sets in the 2-D Gel Data Base

Data set	Attribute name	Example
<i>Cell lines</i>	Cell line reference	CMT167/1
	Hybrid	Yes
	Hybrid partner	CMT170
	Metastasis	High
	Tumor	Yes
	Comments	None
<i>Samples</i>	Cell line reference	CMT167/1
	Sample	20/E
	Sample date	25/6/85
	Run date	10/2/86
	To use (microliter)	19.6
	Optical density	0.12
	Photo	Yes
	Comments	Exposure at 1,2,3 min
<i>Scan</i>	Cell line reference	CMT167/1
	Sample	20/E
	Scan date	5/5/86
	Scan file name	CMT167F21
	Comments	None
<i>Spot</i>	Scan file name	CMT167F21
	Spot id	148
	Centroid X-coordinate	204
	Centroid Y-coordinate	392
	Area	148
	Integrated intensity	784
	Maximum intensity	9
	Minimum intensity	2

5.1. Spot-Pairing Algorithm

The spot-pairing algorithm works on the assumptions that (a) the two sets of spots are roughly aligned and (b) the distortion of the position of each spot can be reduced using a linear transformation applied globally to each spot on one of the matching gels. Assumption (a) can be satisfied by manually aligning the gel with an image of another gel that has already been processed. The camera system of CLIP allows an image from the video camera to be superimposed with another stored image. Alternatively, some form of "flickering" system may be used to roughly align the images. For the purpose of multiple gel analysis, it is better to roughly align the set of gels in an experiment against a standard or reference gel. This gel is usually the one considered by the experimenter as the cleanest gel of all those used in the experiment in terms of spot clarity and occurrence.

During gel matching, the set of spots delineated from the gel images is compared, and the pairing algorithm proceeds as follows:

1. Compute an area threshold above which $k\%$ of the spots in gel 1 lies.
2. Extract the subset of spots from gel 1 whose area lies above the area threshold.
3. Repeat steps (1) and (2) from gel 2.
4. Pair spots in the two subsets using a nearest-neighbor scheme.
5. Compute the linear transformation parameters from the matched pairs.
6. Apply the transformation to all the spots in one of the gels.
7. Pair the rest of the spots using the same nearest-neighbor scheme.

The rationale behind the algorithm is that, given that two gels are derived from samples that are in some way related, e.g., low and high metastasis states of the same cell line, their gene products should be similar in terms of variety and abundance. Consequently, the set of proteins that is synthesized in abundance should be similar for the two samples. This implies that the pattern of the set of dominant spots (large/dark spots) should also be similar for the two gels. Matching these two patterns initially has the following advantages: (a) it provides a means of estimating the transformation parameters reliably because artifacts that can potentially distort the correction transformation are not likely to show up as major spots on the two gels, and (b) the distribution of major spots is normally less clustered than that of all the spots seen in a gel: consequently, pairing based on nearest-neighbor criterion can be carried out more reliably.

The value of k in step (a) above is predefined in the program and is chosen such that a reasonable number of major spots are selected for the transformation parameters estimation. The percentile method is used in order to account for the variation of overall spot size in different gels. It effectively chooses a suitable area threshold for each gel depending on the overall statistics of spot sizes in the gel.

The set of matched major spots plays the role of the "landmark set" in subsequent pairing. This landmark set is usually defined manually in other interactive pairing algorithms (e.g., 3,9). For widely distorted gels, instead of using the relatively computationally cheap nearest-neighbor criterion, the set of major spots can be matched using more expensive and sophisticated matching algorithms (e.g., 10,11). It is expected that, for the above reasons, instead of applying an expensive

matching algorithm to all the spots in the gels, matching the set of major spots only using the expensive algorithm would result in highly accurate matches that can then be used to guide the match for the rest of the spots using a less sophisticated, but computationally cheaper, algorithm.

The estimated transformation is applied to the rest of the spots in one of the gels, and the set of transformed spots is then matched against the other set of untransformed spots. When the distortion in the two gels is small, nearest-neighbor match produces reliable results. Confusion usually occurs in areas in which the local distribution of spots is highly clustered.

For highly distorted gels, it is possible to use the system to partition the spots into separate subsets. Corresponding subsets of spots in the two gels are then matched separately.

The results of the matches are output onto another file that contains pairs of the matched spot indices. These values enable the associated spot data to be retrieved from the spot data files. Mismatches may be corrected by means of various graphics aids, which are described below.

6. Graphics Aids

The top-level menu provides access to the various modules in the 2-D gel analysis system, e.g., data base, data manipulation, hard copy service, and so on. These modules are accessed by moving a cross-wire to the appropriate slot in the displayed menu. Once inside the selected module, different utilities, e.g., gel display, can be invoked similarly via submenus. The set of utilities available for studying spot data is shown in Table 2. During gel display, which reconstructs the spot pattern in a gel, each of the spots detected in the gel is drawn as

Table 2
Menu for Analyzing Spot Data

READ FILE	Read in a set of spots from a file and make it "current"
WRITE FILE	Store current group in a file
LIST	List all sets of spots currently held in the system
GETPAIR	Pair two sets of spots
AND	Logical AND of two sets of spots
NEWNAME	Give a new name to a set of spots
EXOR	Logical exclusive OR two sets of spots
DISPLAY	Graphics display of current group submenu: Display data of a spot Define a calibration set Cut out part of current group
STORE	Store current group in the working memory
GSEARCH	Pick out spots satisfy given constraints from current group
TOP	Return to TOP level, e.g., to exit
PRINT	Print numerical data of current group
SETCURRENT	Define a new current group

an "equivalent circle" (circle having the same area as the original spot) on the terminal screen. The equivalent circles can be color-coded according to the intensity of the associated spots in the gel.

To display numerical data such as attribute values of a spot, the user simply moves a cross-wire using a "mouse" to the spot on the display. This facility can also be used to select and highlight a subset of spots that is of particular interest to the researcher. These user-selected spots can be stored and analyzed separately from the rest of the spots in the same gel.

Graphics facilities for multiple gel analysis include superimposition of a pair of gel displays. If a match file

exists for the pair of gels, vectors linking corresponding spots are also drawn. Numerical data on a pair of matched spots can be displayed similarly by moving a cross-wire to the spots. This provides a simple way of inspecting and comparing quantitative data on any pair of spots. Matches can also be edited (deleted or altered) using the "mouse," in which case the corresponding match vector will disappear when the match is cancelled or point to another spot in the case of a match correction. The updated match information is written out onto another file on exit.

Hard copies of the displays are drawn using a Hewlett-Packard plotter (Fig. 8). An option is available to plot spots on a gel that is common with other gels in one color and the rest of the spots in another color. When this form of display is plotted on a transparency (acetic paper) and is scaled such that the equivalent circles superimposed exactly with the spots on the gel, gel comparison results as suggested by the computer can be confirmed or rejected by the investigator by overlaying the plot on the gels concerned.

Experience has shown that it is not always easy to directly relate a given equivalent circle plotted on a terminal screen to the actual spot on the original gel, mainly because of the lack of intensity information (shadings) in such a display.

The graphics facilities in the system as it stands provide a means by which the investigator can easily index spot data, to combine or subtract subsets of spots from different gels and to edit and study matches in multiple gels analysis. The various graphics display utilities serve to highlight results of the computer analysis that are to be studied in conjunction with the original gels.

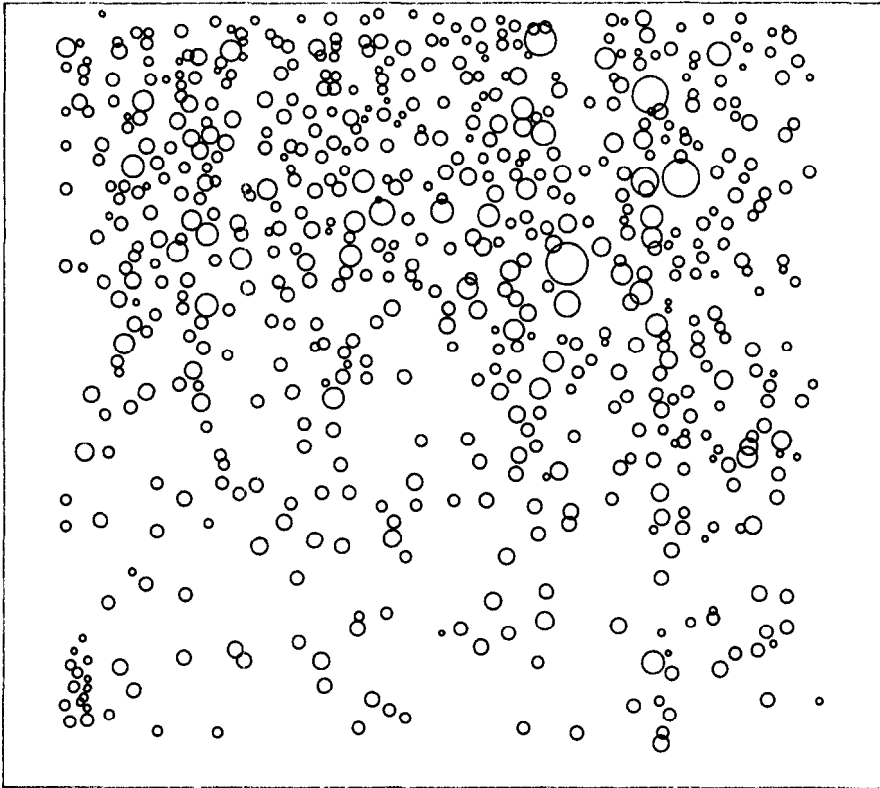


Fig. 8. A computer-reconstructed spot pattern.

7. Applications

The system is currently used in a number of separate feasibility studies. They are (a) a study of differences in the spot patterns associated with gene products produced by cells in high and low metastatic states, (b) a study of protein rates of synthesis, and (c) a study of relationships between radioactivities and spot intensities in autoradiographs. The system is also being assessed for the automatic detection of qualitative and quantitative differences in the analysis of a cDNA recombinant library by differential screening (12).

8. Discussion

The success of computerized 2-D gel analysis depends on close collaboration between researchers working on computer development and those in the laboratory. The complexity of the programs (algorithms) for extracting spot data and consistent matchings across multiple gels depends ultimately on the reproducibility of the 2-D gel process. It has been observed that highly reproducible gels in terms of spot density and pattern can be achieved if the run conditions are carefully monitored and controlled. Gels in the past have been run mostly for qualitative assessments by human eyes. It is important, in this case, for the person who runs the gels to realize that the gels will be inspected by a computer, which is much less capable of discriminating artifacts and ambiguities in the gel image, yet more sensitive to subtle quantitative variations. Consequently, the quality of the data depends very much on the quality of the gels and the way the electrophoresis process is monitored and controlled.

In quantitative gel analysis, as observed by other workers in the field, to account for systematic variations in and across the gels, some form of data normalization is carried out on spot data prior to data base searches or statistical analysis. The choice of data-normalization method varies depending on the kind of analysis being performed. More importantly, different results may be obtained for different types of data normalization and the subset of spot data used in the normalization process. Different forms of data normalization (prior to data base searches) have been discussed by Lemkin and Lipkin (13). In this system, the matched spots data can be output in such a way that they can be readily analyzed using standard statistical packages, such as

GLIMTM and MINITABTM, to take advantage of the vast number of statistical tools available in these packages.

The system is organized so that new program modules (e.g., an implementation of an improved spot-pairing algorithm) can be added to the system at a latter date and with few side effects on other unrelated program modules.

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Chapter 21

Two-Dimensional (Crossed) Immunoelectrophoresis

John M. Walker

1. Introduction

Two-dimensional (2-D) immunoelectrophoresis, also known as crossed immunoelectrophoresis, is a particularly useful technique for the quantitation of mixtures of proteins and the analysis of the composition of protein mixtures. The method consists of two sequential electrophoretic steps:

1. The first dimension, during which the protein mixture to be analyzed is separated by electrophoresis in an agarose gel.
2. The second dimension, during which the separated proteins are electrophoresed at right angles into a freshly applied layer of agarose containing a predetermined amount of antibody.

During the second electrophoretic step, the proteins migrate into the gel until they are completely precipitated by the antiserum, each protein forming a separate precipitate peak. This technique exploits the fact that at pH 8.4 gamma globulins migrate toward the cathode. Antibody and antigen therefore migrate toward each other and form precipitation peaks of antigen-antibody complexes. The area under any peak is directly proportional to the concentration of that protein in the sample and inversely proportional to the concentration of specific antibody in the gel.

This technique was originally introduced as a method for quantitatively measuring changes in serum protein patterns in human disease and has proved particularly successful in this respect (1-6). As many as 40 different serum proteins may be quantitated in a single experiment. As a demonstration of this technique, therefore, this chapter will describe the 2-D immunoelectrophoresis of a serum sample. This method can be applied, however, to a range of problems in the field of molecular biology. It is important to remember that the choice of protein(s) to be quantitated is controlled by the antiserum used in the second dimension. For example, isoenzymes will normally separate during electrophoresis in the first dimension. Using the appropriate antiserum (which will recognize all isozymes) in the second dimension will result in overlapping peaks corresponding to each isozyme, the area under each peak being proportional to the concentration of each isozyme. Similarly, the relative proportions of different proteins (e.g., lens crystallin proteins) at different stages of tissue development can be quantitated with this technique. Other examples of where this technique has been used include the measurement of protein complex formation (7), the differentiation of morphologically similar spe-

cies (8), the determination of the antigenic composition of cell-wall extracts (9), the measurement of protein expression in different tissue types (10), the detection of variant forms of proteins (11), the measurement of enzyme-inhibitor complexes (12), screening for the presence of, and quantitation of, specific proteins in different species (13), the binding of radiolabeled ligands to carrier proteins (14), the study of the structure and morphogenesis of viruses (15), and the quantitation of protein synthesis during larval development (16).

2. Materials

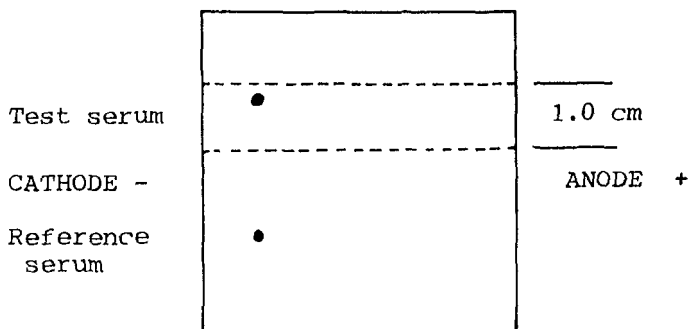
1. Barbitone buffer (0.06M, pH 8.4) is prepared from sodium barbitone and barbitone. Dilute this buffer 1:1 with distilled water for use in the electrophoresis tanks.
2. Agarose, 2% w/v (aqueous).
3. Protein stain: 0.1% Coomassie Brilliant Blue in 50% methanol/10% acetic acid. (Dissolve the stain in the methanol component first, and then add the appropriate volumes of acetic acid and water.)
Destain: 10% methanol/7% acetic acid.
4. Glass plates, 5 x 5 cm, 1.0–2.0 mm thick.
5. Bovine serum.
6. Reference serum: bovine serum containing 1 drop of concentrated bromophenol blue solution.
7. Anti-bovine serum.
8. Flat-bed electrophoresis apparatus with cooling plate.

3. Method

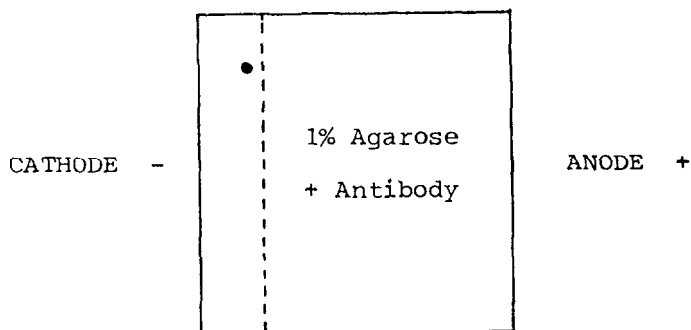
1. Dissolve the 2% agarose by heating to 100°C, then place the bottle in a 52°C water bath. Stand a 5-mL

pipet in this solution and allow time for the agarose to cool to 52°C. At the same time the pipet will be warming.

2. Place a test tube in the 52°C water bath and add 2.8 mL of 0.06M barbitone buffer to the test tube.
3. Leave for 3 min to warm, then add 2.8 mL of agarose solution. This transfer should be made as quickly as possible to avoid the agarose setting in the pipet. Briefly Whirlymix the contents and return to the water bath.
4. Thoroughly clean a 5 x 5 cm glass plate with methylated spirit. When dry, put the glass plate on a leveling plate or level surface.
5. Remove the test tube from the water bath and pour the contents of the tube onto the glass plate. Keep the neck of the tube close to the plate and pour slowly. Surface tension will keep the liquid on the surface of the plate. Alternatively, tape can be used to form an edging to the plate.
6. Allow the gel to set for 5 min. While the gel is setting, pour 0.03M barbitone buffer into each reservoir of the electrophoresis tank and completely wet the wicks. The wicks are prepared from six layers of Whatman No. 1 filter paper. They should be cut to exactly the width of the gel to be run, and should be well wetted with barbitone buffer.
7. Make two holes in the gel 0.8 cm in from one edge of the plate and approximately 1.5 cm in from the side of the plate. This is most easily done by placing the gel plate over a predrawn (dark ink) template (such as the one shown in Fig. 1), when well positions can easily be seen through the gel. The wells can be made using a Pasteur pipet or a piece of metal tubing attached to a weak vacuum source (e.g., a water pump).

First dimension electrophoresis

After electrophoresis the gel is cut where indicated by the dotted line.

Second dimension electrophoresis

Strip cut from first dimension

Fig. 1 Diagram showing sample loading and gel-slicing positions.

8. Place 0.5 μL of serum (bovine) in the test well and 0.5 μL of reference serum in the other well. "Top up" the wells with electrode buffer ($\sim 1 \mu\text{L}$).
9. Place the gel plate on the cooling plate of an electrophoresis tank, and place the electrode wicks over

the edges of the gel. Take care not to overlap the sample wells, and ensure that these wells are nearest the cathode. Hold the wicks in place with glass blocks (*see* Note 4 in section 4).

10. Immediately pass a current of 20 mA (~ 250 V) until the blue marker dye (which is bound to the albumin) in the reference sample reaches or just enters the anode wick. This will take approximately 40 min. (This is about 10 V/cm across the plate.)
11. Toward the end of this run, repeat steps 2 and 3, using a new test tube, but this time use only 2.2 mL each of buffer and agarose. (You will be covering a smaller surface of glass plate with this second dimension solution.)
12. When the plate has run, return the plate to the leveling table and cut the gel as shown in Fig. 1. Transfer the gel slice (containing the test sample, *not* the reference sample!) to a second clean glass plate (*see* Note 5 in section 4).
13. To the tube in the waterbath, now add 150 μ L of rabbit anti-bovine serum antibody and briefly Whirlmix to ensure even dispersion of antibody. Return briefly to the water bath to allow any bubbles to settle out.
14. Pour the agarose and antibody mixture onto the second plate adjacent to the strip of gel and allow the gel to set for 5 min.
15. Place the plate in the electrophoresis apparatus as before and run at 3.0 mA/plate overnight (1 V/cm), making sure that the sample strip is at the cathode end (i.e., protein separated in the first dimension now move toward the anode).
16. At the end of the overnight run, precipitation peaks will be seen in the gel. These are not always easy to

visualize and are best observed using oblique illumination of the gel over a dark background (e.g., using a dark background illuminator).

17. A more clear result can be obtained by staining these precipitation peaks with a protein stain. First the antiserum in the gel, which would otherwise stain strongly for protein, must be removed. One way of doing this is to wash the gel with numerous changes of saline (0.14M) over a period of 2–3 d. The washed gel may then be stained directly or dried and then stained, to give a permanent record. To dry the washed gel, place it on a clean sheet of glass or gel bond, wrap a piece of wet filter paper around the gel, and place it in a stream of warm air for about 1 h. Wet the paper again with distilled water and remove it to reveal the dried gel fixed to the glass plate. Any drops of water on the gel can be finally removed in a stream of warm air.
18. A more convenient and quicker method of preparation is to blot the freshly run gel dry. With the gel on a sheet of clean glass or gel bond, place eight sheets of filter paper (Whatman No. 1) on top of the gel and apply a heavy weight (lead brick or beaker of water) for 1 h. After this time, carefully remove the now wet filter papers to reveal a flattened and nearly dry gel. The last sheet of filter paper may be stuck to the gel and should be thoroughly wetted with distilled water before removing. Complete the drying process by heating the gel in a stream of hot air for about 1 min. When completely dry, the gel will appear as a glassy film.
19. Whichever method of gel preparation is used, the gel can now be stained by placing in stain (with shaking) for 10 min, followed by washing in de-stain. Blue peaks will be seen on a clear back-

ground. Additional peaks that were not apparent when viewed as immunoprecipitates should now be evident.

4. Notes

1. When possible, the gel plates used should be as thin as possible to maximize the effect of the cooling plate. This is particularly important in the first dimension run when considerable heat is evolved.
2. The amount of antiserum to be used in the gel depends of course on the antibody titer and the amount of protein run in the first dimension, and is best determined by trial and error. If peaks go off the top of the gel, then either the antiserum concentration must be increased in the second dimension or the amount of protein run in the first dimension reduced. The reverse applies if the peaks produced are too small for accurate measurement. The amount of antiserum quoted in the section 3 (150 μL) is the volume of commercially available anti-bovine serum used in our laboratory when producing 2-D immunoelectrophoresis gels from bovine serum. This is an ideal test system for someone setting up the technique for the first time.
3. The working temperature for the agarose gel (52°C) is just above the setting temperature of agarose, and great care must be taken to ensure that agarose solutions are not allowed to drop below this temperature prior to pouring the gel. This temperature is chosen because we wish to minimize the change of denaturing the antiserum, which would be considerably less stable at higher temperatures, but of course the agarose must stay in solution. In any case, once the antiserum has been added to the

agarose, the gel should be poured immediately to prevent the antibody from being exposed unnecessarily to high temperatures.

4. Because of the thickness of the electrophoresis wicks used, they are quite heavy when wet and have a tendency to "slip off" the gel during the course of the run. This can be prevented by placing thick glass blocks on top of the wicks, where they join the gel, or a heavy glass sheet across both wicks. This, by also covering the gel, has the added advantage of reducing evaporation from the gel.
5. When cutting the gel strip from the first dimension, make sure that sufficient gel remains adjacent to the sample track to allow for the placing on a wick on this gel in the second dimension, i.e., cut a gel slice with the sample to one side of this slice (*see* Fig. 1). It is important that the wicks do not cover the sample when running the second dimension.
6. A uniform field strength over the entire gel is critical in rocket immuno-electrophoresis. For this reason the wick should be exactly the width of the gel.
7. When setting up the electrophoresis wicks, it is important that they be kept well away from the electrodes. Since the buffer used is of low ionic strength, if electrode products are allowed to diffuse into the gel, they can ruin the gel run. Most commercially available apparatus is therefore designed such that the platinum electrodes are masked from the wicks and also separated by a baffle. To minimize these effects, the use of large tank buffer volumes (800 mL per reservoir) is also encouraged.
8. The time quoted for electrophoresis is suitable for most samples. For proteins with low electrophoretic mobility, however, the times given for the elec-

trophoretic runs may be too short. In the protocol described, the bromophenol blue spot that is visualized in the first dimension is not "free" bromophenol blue, but the dye bound to albumin. Since albumin is one of the fastest-running proteins in the first dimension (only a few minor prealbumin proteins run faster), when the blue dye reaches the anode wick we know that the majority of the serum proteins are now spread along the full length of the gel. When studying other protein mixtures, however, the appropriate time for the first dimension run should be determined in initial trial experiments. Free bromophenol blue can be used as a marker dye to indicate the status of an electrophoresis run at any given time. Similarly, in the second dimension, the time of electrophoresis given may not be long enough to allow for complete development of precipitation peaks for proteins with low electrophoretic mobility. Do note that it is not possible to "overrun" the second dimension electrophoresis. Once the precipitation peaks are formed, they are quite stable to further electrophoresis.

9. If no results are obtained, check that the proteins being analyzed run to the anode under the conditions used. This assumption is implicit in the method described here.
10. Some workers include 3% w/v polyethylene glycol 6000 in the gel, which is said to enhance precipitation of antigen-antibody complexes and thus increase sensitivity.
11. For extreme sensitivity, a silver stain method for detecting immunoprecipitates in agarose gels has been described (17).
12. An excellent review, with appropriate photographs, of problems that can arise with immuno-

electrophoretic techniques is given in Axelsen (1983) (see Further Reading).

Further Reading

Axelsen, N.H., Kroll, J., and Week, B., eds. (1973) *A Manual of Quantitative Immunelectrophoresis—Methods and Applications* Universitetsforlaget, Oslo.

Axelsen, N.H., ed. (1983) *Handbook of Immunoprecipitation-in-Gel Techniques* Blackwell Scientific, Oxford, London.

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Chapter 22

Peptide Synthesis

Brian Austen

1. Introduction

Peptides are required for many different aspects of biotechnology. Since DNA sequences are generally functional only in the protein for which they code, the explosion in sequences that are known has led to an increase in the requirement for synthetic peptides. Quite often, the peptide will be haptenized to a protein carrier and then used to raise antibodies that will then recognize the parent protein. In other instances, synthetic peptides with biological activity will be required in high purity for testing in functional assays. The synthesis of peptide analogs that differ from the naturally occurring sequence in key residues are also often in demand. The methods described here are suitable for making peptides up to about 20 residues in length and do not require sophisticated equipment or highly toxic chemicals such as hydrogen fluoride.

A rapid method of synthesis of peptides on a solid-phase support will be described. The essential feature of

this method is the use of a polymer that functions to protect the C-terminal carboxyl and at the same time renders the growing peptide completely insoluble. The polymer is permeable to the various reagents and solvents used so that excess reagents may be removed by filtration and washing. The method was first developed by Merrifield (1). A number of different variations of methods of solid phase synthesis are in use in many laboratories throughout the world. These methods have a common strategy in that the peptide is assembled from the C-terminal end, and the C-terminal amino acid with its α -amino group protected is first attached through an ester linkage to the solid-phase polymer. The protecting group is then chemically cleaved off, and the next amino acid, again with its α -amino group protected, is coupled via peptide linkage to the α -amino group of the insolubilized C-terminal residue. This cycle of deprotection and coupling an additional amino acid residue is repeated until the peptide is fully assembled on solid phase. Since the side chains of many amino acids would react with the reagents used in making the peptide bonds, they are first protected with chemical groups that are stable to the conditions used to deprotect the α -amino group, but may be removed at the termination of the synthesis.

In many laboratories, a strategy initially developed by Meienhofer and coworkers (2), and also by Sheppard and colleagues at the MRC laboratories in Cambridge (3), is widely used. In this, temporary protection of the α -amino group is achieved by the base-labile 9-fluorenylmethoxycarbonyl (Fmoc) group, first described by Carpino and Han (4). Side chain protecting groups are chosen so that they may be removed by mildly acidic conditions at the end of the synthesis. The growing peptide is attached to the solid-phase support by a variety of

different types of linkers, one of which is cleaved under the same acid conditions as side chain protectants to yield C-terminal acids, and another which is cleaved with ammonia to yield a C-terminal amide or hydrazine to yield C-terminal hydrazides.

The Cambridge group has developed the use of functionalized polyamide gel as a solid-phase support. This support has the advantage that it is fairly hydrophilic and swells dramatically in dimethylformamide (DMF), a solvent favored for peptide synthesis. Kent (5) has shown, however, that polystyrene is equally good as a solid-phase support providing the correct choice of solvent (DMF) is made. Polyamide has the disadvantage that it is more difficult to handle, sticks to glass, tends to be difficult to wash, and has a tendency to become granular as the synthesis proceeds. Moreover, only polystyrene is suitable for the T-bag method (see next chapter). In consequence we use polystyrene with an acid-sensitive linking group such as that present in the *p*-alkoxybenzyl alcohol resin of Wang (6), in which the peptide-resin bond is labilized by the electron-donating ability of the para-ether oxygen. Peptide amides or hydrazides may be synthesized on 4-(hydroxymethyl)-benzoylmethyl resin in which the peptide-to-resin bond is labilized to nucleophiles by the *para*-ester group. Norleucine, included by Sheppard as an internal linker to aid analysis of the degree of substitution, may be dispensed with since it is easy to weigh small samples of polystyrene prior to amino acid analysis.

2. Materials

2.1. Chemicals

1. 4-Alkoxybenzyl-resin. (4-(Hydroxymethyl) phenoxymethyl-copolystyrene-1% divinylbenzene)

(200–400 mesh) resin is suitable for preparation of peptide acids (0.7 mEq/g reactive groups). This resin may be obtained from Nova Biochem, Peninsula Labs, Bachem, or Pierce.

This resin may also be more cheaply prepared from Merrifield resin (chloromethylated copoly-styrene-1% divinylbenzene, about 0.9 mEq/g, 200–400 mesh), also obtained from these suppliers (6). Merrifield resin (5.1 g; 4.6 mmol) in 25 mL of dimethylacetamide is allowed to react with 4-hydroxybenzyl alcohol (0.74 g; 5.9 mmol) and sodium methoxide (0.32 g; 6.1 mmol) at 80°C for 24 h. The resin is collected and washed with dimethylformamide, dioxane, dichloromethane, and methanol.

2. 4-(Hydroxymethyl)-benzoyloxymethyl-copoly (styrene-1% divinyl benzene) resin is suitable for preparation of peptide amides or hydrazides (0.7 mmol/g). It may be obtained from Nova Biochem.
3. Fmoc amino acids may be obtained from Cambridge Research Biochemicals (CRB), Bachem, or Peninsula. It is possible to purchase preactivated derivatives (pentafluorophenyl esters) that are now available from CRB. Reaction with these esters are, however, reportedly less rapid than using carbodiimide, but are necessary for coupling Fmoc-Gln, Fmoc-Asn, and Fmoc-Lys (Boc). As more chemical companies produce Fmoc amino acid derivatives, it is hoped that prices will fall. Alternatively, they may be readily synthesized from free or side chain-protected amino acids by the method of Paquet (7), using 9-fluorenylmethyl succinimidyl carbonate and an organic extraction at low pH to isolate the product. The purity of the products may be established by reverse-phase HPLC on a C8 column (e.g., Brownlee RP-300) in 0.1% TFA with

an acetonitrile gradient; they are eluted at quite high concentrations (about 40%) of acetonitrile. The following side chain-protected amino acids are recommended:

Fmoc-Arg(Mtr) (mesitylene trisulfonyl)

Fmoc-Asp(OtBu) (*O*-*t*-butyl ester)

Fmoc-Glu(OtBu) (*O*-*t*-butyl ester)

Fmoc-Cys(Trt) (Trityl). This protecting group comes off with trifluoroacetic acid to give the free sulfhydryl, which may then be used to couple directly to protein carriers via the bifunctional linker *m*-maleimidobenzylsuccinimide (8).

Fmoc-Cys(Acm) (Acetamidomethyl). This group is stable to the trifluoroacetic acid used to cleave the peptide from the resin and is useful if extensive purification of the peptide is required. After purification the Acm group may be removed by treatment with mercuric acetate (see below).

Fmoc-Lys(Boc)-Pfp ester. The pentafluorophenyl ester of ϵ -*N*-*t*-butoxycarbonyl-Lys). Obtained from CRB.

Fmoc-Asn or Gln-Pfp ester. Obtained from CRB.

Fmoc-Ser(tBu) (*O*-*t*-butyl ether)

Fmoc-Thr(tBu) (*O*-*t*-butyl ether)

Fmoc-Tyr(tBu) (*O*-*t*-butyl ether)

Fmoc-His(Bum)[*N*(π)-*t*-butoxymethyl] (Nova Biochem)

All other Fmoc amino acids are available from CRB or Bachem. In our experience both companies produce high-purity products. These derivatives are very slightly cheaper from Peninsula, but our experience with this supplier is very limited.

4. α -*N*-Boc-amino acids. These are added as the *N*-terminal amino acid only. They are cheaper than

- Fmoc-amino acids, and may be obtained from Fluka, Bachem, or Sigma.
5. Dimethylformamide (DMF). This solvent must be free of dimethylamine, an impurity that is often present. The quality of DMF may be assessed by a colorimetric reaction with dinitrofluorobenzene (FDNB). Mix equal volumes of DMF with a fresh FDNB solution (1 mg/mL) in 95% ethanol, let stand for 30 min, and read the absorbance at 381 nm. Blank FDNB (0.5 mg/mL) usually reads 0.2. Good quality DMF should have an OD of no more than 0.15 higher than the blank. Reagent grade DMF may be purified by distillation from ninhydrin under vacuum from a water pump, using a nitrogen bleed, and stored over molecular sieve 4A (BDH Chemicals). Alternatively, purchase the Puriss grade from Fluka, add about 1 in of molecular sieve 4A to the bottom of a 21 bottle, flush with nitrogen, and store sealed in the cold room for at least 4 d before use.
 6. Dichloromethane (DCM). Obtained from Fisons. Reagent grade may be used without purification.
 7. Piperidine. Obtained from Fluka. Distill this reagent from potassium hydroxide in a good fume-cupboard.
 8. Trifluoroacetic acid (TFA). This should be free of aldehydes; distillation should be avoided because of the danger of forming anhydrides. An acceptable grade can be purchased from Kali-Chemie, West Germany, or Peboc Ltd., Llangefni, Anglesey.
 9. Diisopropylcarbodiimide (Fluka).
 10. 1-Hydroxybenzotriazole (BDH).
 11. Dimethylaminopyridine (BDH).
 12. *N*-Methyl-morpholine (Fluka). Distill from ninhydrin.
 13. Methanol (UV-grade) (Fisons).

14. Siliconizer (BDH).
15. Pyridine (BDH). Distill from ninhydrin in a fume cupboard.
16. Benzoyl chloride (Fluka).
17. Diisopropylethylamine (Fluka). Distill from ninhydrin.
18. *t*-Amyl alcohol (BDH).
19. *N*-Acetylimidazole (BDH).

2.2. Apparatus

1. Synthesis glass vessel fitted with No. 3 glass sintered frit and No. 14/20 Quickfit ground-glass joint (see Fig. 1). The dimensions suitable for 2 g of resin would be about 40 mm across and 75 mm in length.
2. Motorized shaker capable of rotating the shaker through about a 105° arc. This can be made by any competent workshop using a 24 rpm motor attached by a crank to a disc that is itself attached to a laboratory clamp. Commercial shakers are apparently available from Sequemat, 109 School St., Watertown, Massachusetts, 02172.
3. General apparatus, i.e., oil pump, water pump, rotary evaporator, fraction collector, centrifuge, spectrophotometer.
4. Access to an amino acid analyzer and HPLC.

2.3. Chromatographic Materials

1. Ion-exchange cellulose (CM52 or DE52) from Whatman.
2. Sephadex G-15 and SP-Sephadex (C-25) from Pharmacia.
3. Reverse-phase column (4.6 × 250 mm). Brownlee RP-300 (Anachem).
4. Amberlite MB-3 mixed ion-exchange resin from BDH.

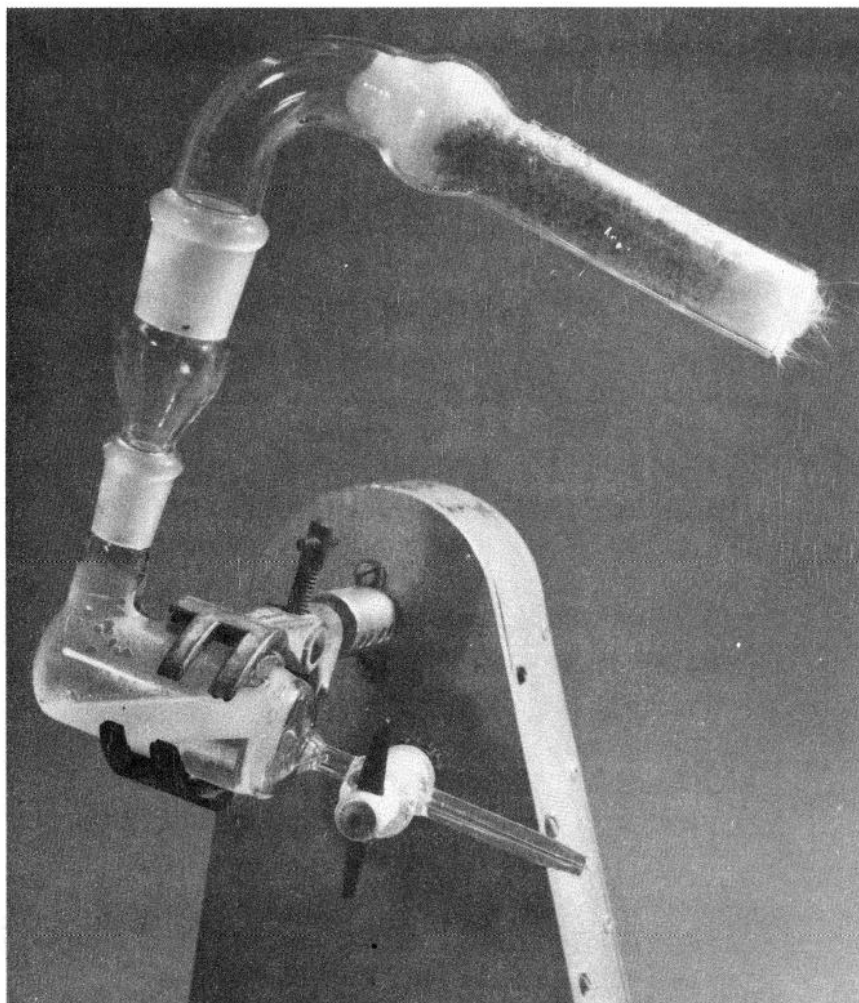


Fig. 1. Glass synthesis vessel attached to a motorized shaker. The vessel contains polystyrene resin and is fitted with a silica gel drying tube.

3. Ninhydrin Monitoring

The extent of coupling may be monitored by the color developed by the resin with a ninhydrin reagent, which is indicative of the amount of free amino group present. The conditions are described by Sarin et al. (9).

Ninhydrin reagents—stock solutions are prepared as follows:

Solution A. Mix 40 g of reagent grade phenol with 10 ml of absolute ethanol. Warm until dissolved. Stir with 4 g of Amberlite MB-3 resin for 45 min. Filter. Add this to a second filtered solution made by adding 2 mL of 100 mL of water containing 65 mg of KCN to 100 mL of pyridine, and also stirring with 4 g of Amberlite MB-3 resin.

Solution B. Dissolve 2.5 g of ninhydrin in 50 mL of absolute ethanol. Stopper and store in the dark under nitrogen.

Take enough suspended resin to half fill the stem of a small Pasteur pipet. Transfer to a plastic tube and wash three times with methanol. Dry the resin in a vacuum desiccator for 15 min, then weigh small samples of about 2 mg into plastic tubes, add 125 μ L of solution A ninhydrin reagent and 32 μ L of solution B. Heat in a boiling water bath for 5 min. Cool. The blue color present at this stage is generally sufficient for a visual qualitative assessment of coupling, but for quantitation proceed as follows. Add 1 mL of 60% ethanol and elute from a Pasteur pipet plugged with a small amount of glass wool. Rinse with 0.2 mL of 0.5M tetraethylamine hydrochloride in DCM, and then rinse with 1 mL of 60% ethanol. Dilute with 10 mL of 60% ethanol. Calculate the amount of unreacted group using the extinction coefficient of a free amino group of 1.5×10^4 /M/cm as a percentage of the amount of free amino group measured on piperidine-treated resin after coupling the C-terminal residue as described below.

4. Methods

4.1. Attachment of the First Residue

This method is appropriate for both acid- and base-labile resins. Calculate the amount of resin required for the amount of peptide you hope to get. About 0.6 mmol of peptide per g of resin is the maximum to expect from a synthesis after cleavage from the resin, but this may be reduced to 0.1 mmol after extensive chromatographic purification of a difficult sequence. A convenient amount of resin to work with is 2 g.

1. First clean the synthesis vessel with chromic acid, wash extensively with water, dry, and coat with a siliconizing agent. Dry again in an oven. Place the vessel on the shaker and add 2 g of resin.
2. Fully suspend the resin with DCM (about 30 mL for 2 g), shake for 2 min, and with a drying tube containing silica gel on the quickfit socket (Fig. 1) and the vessel in the vertical position, draw off the solvent by connecting the tubing from a water pump vacuum to the outlet tube via a trap consisting of a 21 Buchner flask with a bung and outlet. This is repeated for all further washes.
3. Wash twice more with DCM and once with DCM/DMF (1/1 mixture) and draw off the excess solvent.
4. Molar amounts of protected amino acids, coupling agents, and catalyst for addition of the first residue added are calculated from the number of moles of the alkoxy-substituent on the resin. Dissolve a sixfold molar excess of the Fmoc derivative of the C-terminal residue in a volume of DMF sufficient to swell the resin, i.e., for 2 g of the Bachem resin with 0.7 mmol/g substitution, 8.4 mmol of the Fmoc amino acid in 6 mL DMF, and add to the resin.

5. Add a sixfold molar excess of diisopropylcarbodiimide in 5 mL of DCM, shake for 2 min, and add a sixfold molar excess of *N*-methylmorpholine and 0.6-fold molar amount of dimethylaminopyridine in 1 mL of DMF, which act as catalysts for formation of the C-terminal ester with the resin, and shake for 1 h.
6. Wash once with DCM:DMF (1:1), (12 mL) and four times with DCM (16 mL each wash). If Asn, Gln, or Lys(Boc) is the C-terminal residue, they should be incorporated as the Fmoc pentafluorophenyl ester, using sixfold molar excess without the carbodiimide, and adding first a sixfold molar excess of *N*-hydroxybenzotriazole in 10 mL of DMF to the resin and shaking for 2 min, then adding a sixfold molar excess of the pentafluorophenyl ester in another 2 mL of DMF to the resin and shaking for another 2 min, and finally adding *N*-methylmorpholine (sixfold excess) and dimethylaminopyridine (0.6-fold amount) as catalysts and reacting for 3 h.

4.2. Measuring the Substitution of the C-Terminal Residue

1. To measure the extent of incorporation of the C-terminal amino acid, take a small amount of the resin, enough to half fill the stem of a short Pasteur pipet, transfer to a tube, remove excess DCM, and add about 0.5 mL of 20% piperidine in DMF.
2. Leave for 5 min, then remove excess reagent, and wash several times with DCM, then methanol.
3. Perform the monitoring ninhydrin reaction as described above.
4. To check by an independent method, hydrolyze a weighed sample of the deprotected resin with a 1:1 mixture of concentrated HCl and propionic acid in an evacuated tube at 130°C for 3 h, dry in vacuo, and

subject the residue to amino acid analysis. If the extent of substitution is less than 0.3 mmol/g, then repeat the coupling reaction with the first residue.

5. Finally, block remaining reactive groups having washed the resin three times with DCM by a 1-h incubation with 10-fold molar excesses of benzoyl chloride and pyridine in DCM. Wash out three times with DCM. The molar excesses of reagents described in the coupling methods below are relative to the amount of C-terminal residue incorporated.

4.3. Coupling Successive Residues

1. Deprotection and coupling cycles are initiated for addition of further residues, building toward the N-terminal end of the peptide. A general procedure suitable for most types of amino acid residues except Gln, Asn, and Lys(tBoc) will be described. It would be preferable to transfer the apparatus to a fume cupboard for handling DCM and piperidine, which are toxic.
2. The volume of each wash used below is enough to fully suspend the resin.
 - Step 1. Wash with DMF, 3 × 2 min.
 - Step 2. Wash with DCM, 3 × 2 min.
 - Step 3. Wash with DMF, 3 × 2 min.
 - Step 4. Deprotect with 20% piperidine, 2 × 5 min.
 - Step 5. Wash with DMF, 3 × 2 min.
 - Step 6. Wash with DCM, 3 × 2 min.
 - Step 7. Wash with DCM:DMF (1:1 by vol), 2 × 2 min.
 - Step 8. Add fivefold molar excess of Fmoc-amino acid over the amount of C-terminal residue on the resin in 6 mL of DMF, shake for 2

shake for 2 min, then add fivefold excess of diisopropylcarbodiimide in 6 mL of DCM. Shake for 1 h. Test for coupling with ninhydrin reagent. The reaction may be performed qualitatively, judging any blue color formed by eye.

Step 9. If the test gives no color, wash with DCM/DMF (1/1 by vol), 2 × 2 min, and proceed with the next cycle. If the test gives a blue color, then refer to Note 1 in section 5.1.

4.4. Coupling *Fmoc-Asn*, *Fmoc-Gln*, or *Fmoc-Lys(Boc)*

1. Side reactions occur if these residues are coupled by carbodiimide, so couple them instead as pentafluorophenyl esters (fivefold excess) (obtained from CRB).
2. At step 8 on the coupling cycle, first shake the resin for 2 min with fivefold excess of 1-hydroxybenzotriazole in about 10 mL of DMF, then add the pentafluorophenyl ester in 1–2 mL of DMF, and shake for 3 h.
3. Check for degree of coupling with the ninhydrin reagent, and repeat with a second batch of ester if the resin produces blue color.

4.5. Coupling the *N*-Terminal Amino Acid Residue

1. Since Boc-amino acids are cheaper than Fmoc-amino acids, savings can be made by adding the *N*-terminal amino acid residue at the final coupling as the α -*N*-Boc-amino acid with acid labile (e.g., Boc, *O*-*t*-butyl) side chain protecting groups. If Arg is the *N*-terminal residue, use Boc-Arg unprotected on the side chain.

2. Couple these residues using diisopropylcarbodiimide as described for the Fmoc-amino acid residues, and if Boc-Gln or Boc-Asn are at the N-terminus, equilibrate the resin first with fivefold excess *N*-hydroxybenzotriazole, for several minutes before adding Boc-Gln or Boc-Asn (fivefold excess) followed by diisopropylcarbodiimide. There are no problems with coupling Boc-Lys(Boc).

4.6. Cleavage and Deprotection of the Peptide

4.6.1. Acid-Labile Linker (Alkoxybenzyl-resin)

1. When the last residue has been coupled, and the resin washed, if a Fmoc-amino acid residue was added at the N-terminus, treat with 20% piperidine to step 4 to remove the Fmoc group on the N-terminal residue as described above, then commence washing as follows (*see* section 4.2.2). If a Boc-amino acid residue was added, wash the resin after the final coupling as follows.
2. Wash three times with DMF, three times with DCM, three times with *i*-propanol, and three times with ether.
3. Leave the vessel with a drying tube in place on the water-pump vacuum for 10 min, then evacuate in a desiccator for 30 min on an oil pump.
4. Replace on the shaker and shake for 2 h with 50% (v/v) TFA, 5% (v/v) anisole in DCM (total 20 mL), and if the peptide contains Met, 0.5 mL of ethyl methyl sulfide. If the peptide contains Trp, include 0.9% ethanedithiol along with anisole.
5. If Arg(Mtr) is present, then the cleavage/deprotection conditions should be more drastic. Deprotect with 50% TFA in DCM for 6 h, or if the peptide has

Arg(Mtr), but not Met or Trp, deprotect with 50% (v/v) TFA plus 5% (v/v) thioanisole in DCM (total vol 20 mL) for 90 min. Collect the filtrate, add another 20 mL of 50% TFA (v/v) in DCM, shake for another 2 min, and combine the filtrates. TFA causes nasty burns, so wear gloves and goggles.

6. Rotary evaporate at 30°C and add ether to the remaining oil to give the peptide as a white solid. Wash the solid several times with ether, recovering the solid in a spark-free centrifuge. Vacuum desiccate over NaOH pellets and weigh the recovered peptide.

4.6.2. *Base-Labile Linker 4-(Hydroxymethyl)-benzoyloxymethyl Resin*

1. The peptide linked to the resin by the base-labile group must be treated with ammonia after TFA to yield a C-terminal amide. Treat the peptide-resin with 50% TFA as described in section 4.6.1. to remove protecting groups. Discard the filtrate and wash again with 50% TFA. Discard filtrate.
2. Wash the resin with DCM (three times), *t*-amyl alcohol (three times), 10% diisopropylethylamine in DCM (three times), DMF (three times), and ether (three times).
3. Evacuate first on a vacuum pump with the drying tube in place to let the ether evaporate, and then vacuum desiccate for 90 min over P₂O₅ on an oil pump.
4. In a fume cupboard, collect ammonia gas from a cylinder in a three-necked, round-bottomed flask containing 30 mL of HPLC-grade methanol cooled in ice, passing the gas first over KOH pellets in a gas bottle. The methanol should be protected from the

atmosphere by a KOH drying tube. Pass the gas until it no longer dissolves in the methanol, i. e., bubbles are seen.

5. Transfer the resin to a pressure bottle equipped with a magnet, pour on the ammonia-saturated methanol, stir for 24 h at room temperature. Cool again in ice before opening the bottle in a fume cupboard.
6. Filter off the resin, wash with further 10 mL of methanol, and rotary evaporate the combined filtrates. In case the peptide has precipitated, wash the resin further with 5% acetic acid, and rotary evaporate.

The use of the base-labile linker makes it possible to cleave the peptide with protecting groups still attached. This peptide may be then coupled to a free amino group in a second peptide, protein, or other substance, and the product finally deprotected by treatment in 50% TFA. This strategy, known as "fragment condensation," is often used to prepare peptides too long to put together by repetitive attachment of individual residues. The C-terminal fragment peptide with free amino group must be suitably protected on other reactive groups. This second peptide may be synthesized by the general procedure described in this chapter on the base-labile linker, using an Fmoc-amino acid for the N-terminal residue instead of a Boc residue. The N-terminal Fmoc group is removed by treatment with 20% piperidine and washed as described in section 4.6.1. The peptide is then cleaved as a C-terminal amide by treatment with ammonia as described in the preceding paragraph, but without prior treatment with TFA.

7. To obtain from the base-labile resin a protected peptide fragment activated at its C-terminus for

coupling to an amino group in the C-terminal fragment, couple the α -N-Boc-protected amino acid derivative as the protected N-terminal residue in place of an Fmoc-derivative as described above. Then instead of treating with piperidine or TFA, wash the resin several times with DCM, then DMF.

8. Suspend the resin in DMF (5 mL/g), and add anhydrous hydrazine (Pierce) (wear gloves) (30 mol/per mol of peptide). Stir in a closed container for 2 d.
9. Filter and wash with DMF.
10. Rotary evaporate the combined filtrate and washings under vacuum from a good oil pump.
11. The protected hydrazide formed may be converted to an active azide before coupling to the amino group of another peptide by the following procedure. To a solution of the hydrazide (100 mg in about 4 mL DMF) cooled to -10°C , in an ice-salt mixture, a solution of 1M HCl in DMF (2 mL) is added, followed by isoamyl nitrite (0.26 mL). After 5 min at -10°C , the mixture is cooled to -30°C , in a methanol-dry ice mixture, neutralized with triethylamine (0.26 mL), and added to the amino component in minimal volume of DMF. Leave for 2 d at 4°C . The product may be isolated by addition of water. Protecting groups are then removed by incubation in 95% TFA and 5% water for 2 h. Rotary evaporate at 30°C and precipitate the product with ether.

4.7. Characterization and Purification

1. Some criteria of purity of the product must be obtained. Amino acid analysis after acid (6M HCl) or enzymic hydrolysis (aminopeptidase M) will check the composition, and high-voltage electro-

phoresis of the product on Whatman 3MM paper will ascertain the heterogeneity in charge (*see* vol. 1 of this series). If HPLC is available, analysis by reverse-phase separation on a C8 300A silica column, eluting in 0.1% TFA with a gradient of acetonitrile, will indicate the degree of heterogeneity. Readers should refer to chapters (in vol. 1 of this series) on protein sequencing for detailed methods of peptide analysis and characterization.

2. If required, the peptide may be purified by ion-exchange chromatography, using either Whatman CM52 if the peptide is basic, eluting with ammonium acetate (pH 5.0), or if the peptide is acidic, chromatograph on Whatman DE52 resin, with ammonium bicarbonate (pH 8.0). Use the following procedure:
3. For 100 mg of crude peptide, place 40 g of CM52 on a sinter and slurry with 0.6M ammonium acetate (strong buffer). Adjust to pH 5.0 with acetic acid. Draw off and wash with 0.01M ammonium acetate (pH 5.0) (2.0 L) (dilute buffer). Slurry with this dilute buffer and pack into a 1.5 cm diameter column. Run the eluate through a UV monitor at 280 nm if the peptide contains Tyr or Trp, or 240 nm if not.
4. Pump dilute buffer at 30 mL/h until the conductivity of the eluate is close to that of the dilute buffer, then dissolve the peptide in water (5 mL), adjust pH to 5 with dilute acetic acid or ammonium hydroxide, and load onto the column. Collect fractions.
5. Pump dilute buffer for 1 h, then start a linear gradient with 200 mL of dilute buffer to 200 mL of strong buffer. Peptides are recovered by lyophilization and again checked for purity.
6. Acidic peptides are purified on DE52 in the same

way, using 0.01M ammonium bicarbonate (pH 8.0) as the dilute buffer and 0.6M ammonium bicarbonate as the strong buffer at pH 8.0.

7. If the peptide is insoluble at pH 5 or 8, it may be dissolve in 30% acetic acid and subjected to ion-exchange chromatography on SP-Sephadex (C25) suspended in 30% acetic acid, eluting with a gradient of NaCl up to 0.6M in 30% acetic acid using the same amounts and volumes as described for CM52. The resin is allowed to swell in excess 30% acetic acid (three times the expected suspended volume) overnight before use, and fines are removed by swirling up, allowing to settle for 20 min, and removing the excess supernatant. The resin is packed into a column and washed with 3 volumes of 30% acetic acid before loading the peptide. The peptides are desalted by gel filtration on Sephadex G-15 (4 x 60 cm) in 30% acetic acid, and the acetic acid removed by rotary evaporation.

4.8. Deprotection of Cysteine

If the acetamidomethyl group was chosen for protection of cysteine, it should be removed after purification. Dissolve the peptide in a minimum amount of water and adjust the pH to 4.0 with acetic acid or ammonia. Add a 10-fold molar excess over the acetamidomethyl content of mercuric acetate (BDH) and incubate for 3 h at room temperature. Then pass hydrogen sulfide from a gas bottle (obtainable from BDH) through the solution in a fume cupboard for 1 h. Centrifuge the suspension and isolate the product from the supernatant by chromatography on Sephadex G-15 in 2M acetic acid.

5. Notes

5.1. Incomplete Coupling

1. If the ninhydrin test gives unacceptable blue color (>2% of total amino group prior to coupling), try adding fivefold molar excess of *N*-hydroxybenzotriazole in 1 mL of DMF, which will form an active ester *in situ*; incubate for a further hour and test again. If the resin still produces color, try adding 1 mL of 0.5M diisopropylethylamine in DCM to neutralize any protonated and therefore unreactive amino group. Shake for a further 15 min, then wash twice with DCM:DMF (1:1 by vol) (2×2 min). The peptidyl resin should not be left any longer with diisopropylethylamine for fear of removing the Fmoc group. If color with ninhydrin is still produced, then repeat steps 5–9 (section 4.3). If total coupling is still not achieved, then block by *N*-acetylation any remaining amino groups by 15-min incubation with 10-fold excess of *N*-acetylimidazole in DCM and proceed with the next cycle. This blocking reaction will terminate any unreacted peptide, which will thus not participate in further coupling steps. The resulting terminated peptide will be easier to separate out at the end of the synthesis than a form differing from the required peptide by the deletion of only one residue.

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Chapter 23

Synthesis of a Series of Analogous Peptides Using T-Bags

Brian Austen

1. Introduction

To obtain a number of analogous peptides differing only in side chains in one or two positions, it may be convenient to perform the synthesis on resin confined to a number of porous bags (which are described as T-bags), so that when residues that are to be changed are coupled, a bag containing the peptidyl resin can be isolated from other bags and coupled to a different protected amino acid. At the completion of coupling, the bag may be returned to a bottle containing the other bags, and the synthesis continued. The principal and usefulness of this method was originally described by Houghton (1), who synthesized several hundred peptides at once in one step in order to explore the contribution of individual side chains in the specificity of antibody binding sites. The use of Fmoc chemistry and acid-labile resin means that peptides may be cleaved from the resin

samples by individual reactions with TFA. If a method using tBoc chemistry is used as described by Houghton (1), then special apparatus to handle many individual reactions with HF to cleave the peptides from the resin at the end of synthesis is required. The main characteristics of solid-phase peptide synthesis are as described in the previous chapter, which should be referred to for description of methods and source of many of the materials, and solvents, and so on. Since it is not possible to monitor the completeness of coupling at each step in T-bags, the peptides obtained are unlikely to be very pure, but should be suitable for screening of the epitopes of antibodies.

2. Materials

1. *p*-Alkoxybenzyl resin (100–200 mesh) is obtained from Bachem or prepared from Merrifield resin (chloromethylated copolystyrene–1% divinylbenzene) (100–200 mesh) obtained from Pierce, as described in the previous chapter.
2. Fmoc-amino acids (CRB) (*see* previous chapter).
3. Polypropylene mesh (PP 74) for preparing T-bags can be obtained in meter lengths from Henry Simon Ltd., PO Box 3, Stockport, Cheshire, UK.
4. Heat-sealer may be obtained from Jencons Ltd.
5. Glass bottle with polypropylene cap; 100–200 mL (e.g., Sorval Corex centrifuge bottle).
6. For nature and source of solvents and reagents for solid phase synthesis, *see* previous chapter.

3. Methods

Before carrying out this procedure, please refer to previous chapter for a description of the main characteristics of solid phase peptide synthesis.

3.1. Preparing the Bags

1. Prepare the number of bags that you require, one for each peptide that you wish to make. For 20 bags, cut a strip of polypropylene mesh 30 x 4 cm. Draw with a pencil a line down the middle in the long dimension and at 1.5-cm intervals across the strip. Experiment with your heat sealer to find the correct time setting to completely seal two pieces of polypropylene together without destroying it. About 5 s heating followed by 10 s cooling is suitable with the Jencons heat-sealer.
2. Place identifying numbers using a graphite pencil in the middle of each 1.5 cm section, but close to the long edge so that when sealed down the long edge, the number will be sealed too. Fold the polypropylene down the center line, and heat seal the long edges together about a third of a centimeter from the edge. It helps to place clips along the fold to hold it in place while sealing. Take care to allow to cool before removing the mesh from the heat-sealer.
3. Heat-seal several more times along the length of the mesh toward the outside edge so that the sealed area extends over the full third of a centimeter and the number is completely sealed in. Heat-seal across the short dimensions of the folded strip, starting along close to one edge, then just below each pencil line. Cut along each pencil line across the short dimension. This should give 20 bags, each open along one edge.

3.2. Filling the Bags

Weigh out 30 mg of *p*-alkoxybenzyl-polystyrene into each bag and heat-seal along the remaining side

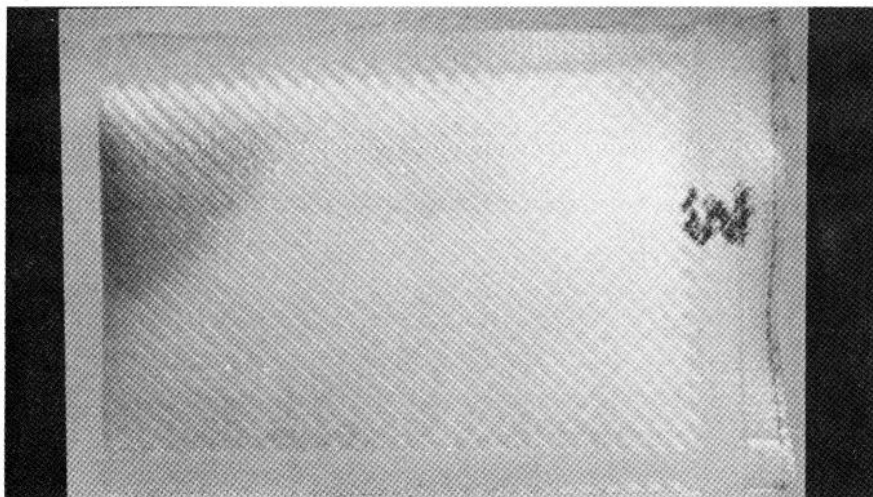


Fig. 1. A sealed polypropylene T-bag containing resin.

(Fig. 1). The yield from each bag should be about 10 μ mol of peptide per bag.

3.3. Coupling the C-Terminal Residue

1. Place the bags in a beaker. Couple the C-terminal amino acid in at least 10 mL of reagents, which will take six bags. Scale up the volumes if more bags are used so that all the bags are covered with reagent.
2. The reagents consist of 0.25M Fmoc-amino acid in DMF and 0.25M diisopropylcarbodiimide in DCM added in equal amounts. If Fmoc-Asn, Fmoc-Gln, or Fmoc-Lys (Boc) are to be coupled (*see* Chapter 22) then first add 0.25M *N*-hydroxybenzotriazole in DMF, then an equal volume of an 0.25M solution of the Fmoc-amino acid pentafluorophenyl ester in DMF and omit the carbodiimide.
3. Cap the beaker with foil, shake on a vigorous rotary shaker for 2 min, then add *N*-methylmorpholine, so

that its concentration in the total volume is 0.125M, and 4-dimethylaminopyridine dissolved in a small volume of DMF so that its concentration in the reagents surrounding the T-bags is 0.0125M. Cover again and shake at room temperature for 2 h. If different residues are to be coupled at the C-terminus, then first sort bags according to their coded number, into different beakers, and couple with different protected amino acids.

3.4. Coupling Successive Residues

1. Transfer the bags together (up to 200 bags) to the centrifuge bottle and perform the following steps, shaking the bottle after placing the plastic seal and cap in place with each solvent, and decanting through a stainless steel sieve placed over the top to keep the bags from falling out. Use enough solvent to cover all the bags, i.e., between 30 and 100 mL per wash, and try not to let the bags dry out or there may be loss of resin from the bags. Perform the operations in a fume cupboard.
2. Carry out the following steps:
 - Step 1. DMF, 3 × 2 min wash.
 - Step 2. DCM, 3 × 2 min wash.
 - Step 3. DMF, 3 × 2 min wash.
 - Step 4. 20% Piperidine in DMF, 2 × 7 min wash.
 - Step 5. DMF, 3 × 2 min wash.
 - Step 6. DCM, 3 × 2 min wash.
 - Step 7. DMF, 4 × 2 min wash.
3. After the washing in step 7, for each step in the synthesis, either the same amino acid residue will be coupled onto the resin in one or several of the bags

according to the sequences required. Sort the bags out according to number, then place in beakers appropriately marked for the amino acid residue to be added. Couple with Fmoc-amino acids using volumes of reagents that cover the bags. As before, the reagents consist of 0.25M Fmoc-amino acid in DMF and 0.25M diisopropylcarbodiimide in DCM added in equal volumes, but no tertiary amine catalyst is required. Add the Fmoc-amino acids first to the bags in a beaker followed by the diisopropylcarbodiimide, cover with aluminum foil, and leave stirring with a magnet for 2 h. For Fmoc-Asn, Fmoc-Gln, or Fmoc-Lys (Boc), add 0.25M 1-hydroxybenzotriazole in DMF first, then an equal volume of 0.25M solution of the pentafluorophenyl ester in DMF and omit the carbodiimide.

4. Wash with 100 mL of DCM:DMF (1:1 by vol) for 3 x 2 min. Repeat from step 1.

3.5. Deprotection and Cleavage from the Resin

1. As explained in the last chapter, Boc-amino acids can be coupled as the N-terminal residue using the same conditions as described above for Fmoc-amino acids. After the last residue has been added, wash three times each with DCM:DMF (1:1 by vol), DMF, DCM, and *i*-propanol.
2. Cut open the bags and transfer the resin in each to individual marked glass vials.
3. Wearing gloves and goggles and using a fume-cupboard, add 1 mL of 50% TFA in DCM with the appropriate added chemicals (to protect amino acids from side reactions as described in the last chapter) to each vial and leave for 2 h at room temperature.

4. Filter off each peptide from the resin through small sintered glass filters with side arms attached to a marked quickfit tube with slight vacuum from a water pump. Wash each resin with a further 2 x 1 mL of 50% TFA and also filter.
5. Evaporate the filtrates with streams of nitrogen in a fume cupboard and precipitate the peptides by addition of 3 mL of ether to each tube.
6. Vortex each tube, centrifuge in a spark-free centrifuge for 2 min, and dispose of the supernatant.
7. Wash another three times with ether and leave the tubes open on the bench for the remaining ether to evaporate. Then evacuate the tubes in a desiccator over KOH pellets.

4. Notes

1. Peptides may be characterized by HPLC or amino acid analysis as described in the last chapter, and tested for binding to antibodies by ELISA assay, measuring the competition exerted at various dilutions of peptide of binding of antibodies to immobilized antigens on the wells of PVC plates.

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Chapter 24

Production of Antisera to Synthetic Peptides

W. J. Gullick

1. Introduction

The objective of immunization is to produce polyclonal or monoclonal antibodies that react with a chosen molecule. Immunization may be *in vivo* by injecting antigen into animals to produce antisera or to provide B-lymphocytes for immortalization as hybridomas, or alternatively, *in vitro* using splenocytes from untreated animals (1) for subsequent fusion with myeloma cells to produce monoclonal antibodies (MAbs).

This chapter will be confined to the production of antibodies to proteins using synthetic peptides as immunogens. Conventional immunization may involve the injection of impure, semipure, or pure proteins to produce polyspecific or monospecific antisera. Since monospecific antisera are generally more useful, having a pure antigen is desirable. The problem of impure anti-

gen can, however, be overcome by producing MAbs. This method relies on having an unambiguous screen in order to select the appropriate hybridomas. Frequently, this means a rather complicated analysis such as immunoprecipitation rather than by ELISA, which, when using impure immunogen, again requires purified protein for screening. Additional disadvantages are potential immunodominance by other antigens in the mixture over the desired immune response, and, importantly, there may be an immunodominant epitope on the antigen itself. If MAbs are required that recognize other regions of the molecule, this may involve time-consuming elimination of undesired MAbs of the same specificity produced against the dominant epitope. An example of this problem is work involving the acetylcholine receptor in which one region of the native molecule elicited many different MAbs and was subsequently called the main immunogenic region (2). In addition, mapping the binding sites of such MAbs on the surface or sequence of a protein is time-consuming and complicated (3).

Within the last few years, a different approach to generating protein-reactive antibodies has been developed that has several advantages over conventional immunization. This involves synthesizing short peptide sequences, coupling them to immunogenic carrier molecules, and immunizing animals with the conjugates. An immune response is usually generated both against the carrier and to the peptide acting as a hapten. The antipeptide antibodies produced, frequently, but by no means always, react with the protein from which the peptide sequence was selected. This technique is of course dependent upon having primary sequence information, either derived directly by sequencing a purified protein or quite commonly by using an amino acid

sequence predicted from an open reading frame of cloned genomic or cDNA molecules. The latter situation does not require that any protein be purified, and in fact, in most cases the protein may never have been identified or characterized at all.

Immunization with synthetic peptides has the advantages that pure immunogens are used and an unlimited quantity of pure stable antigen is available for screening, often by ELISA, which is particularly useful in the production of MAbs when large numbers of hybridoma supernatants need to be screened quickly (*see* vol. 1 of this series). Perhaps more importantly, however, the region within the primary structure of the protein of interest to which the antibody binds is clearly defined with high resolution. The antibodies are, in fact, "site-directed" and can be generated against regions of proteins that are not normally immunogenic when the whole protein is employed as an antigen. The main problem of synthetic peptide immunogens is that although one almost always can make anti-peptide antibodies, perhaps on average only one in three or four of these sequences generates antibodies that react with this structure in the context of the whole molecule. It is therefore very important to choose a peptide immunogen carefully so as to improve the success rate, thereby minimizing the cost and the number of animals employed, and several techniques have now been developed for this end.

2. Choosing the Right Peptide

If, for instance, one only has the sequence of the 10 amino-terminal amino acids of a protein, or if two protein sequences are highly related except for a short sequence and one wishes to make antibodies that dis-

criminate between them, the choice is already made. If, however, a longer sequence is available and you have no predisposition to select a particular region, there are several predictive methods that are considered helpful but not infallible.

2.1. *Hydropathy Plots*

Hydropathy is a measure of the polarity of a molecule. Each amino acid is assigned a hydropathy value, defined by its propensity to dissolve in the organic or aqueous phase of a two-phase solvent system. A plot of the local hydropathy of the sequence is generated by selecting groups of usually six or seven contiguous amino acids (the "window") and averaging their hydropathy value. By starting at the N-terminus of the protein and iteratively moving the window by one residue each time and performing the same calculation, one obtains a plot of the varying hydropathy along the sequence. The immunogenic epitopes of some proteins have been defined experimentally and are quite often composed of charged or hydrophilic residues that appear as peaks of local hydrophilicity on the plot (4,5). For a detailed discussion of this method, *see* ref. 6.

2.2. *Protrusion Index*

Water-soluble proteins are folded such that amino acids with hydrophilic side chains are generally disposed on the surface of the protein and are thus sterically available to interact with antibody combining sites. A different method of calculation called the "protrusion index" has been developed (7) that attempts to predict the relative protrusion of a sequence from the protein's globular surface. This method predicts that

the more superficial the region, the more likely it is to be a good candidate as a synthetic peptide.

2.3. Secondary Structural Predictions

There is some evidence that turns in proteins are immunogenic, and the position of these within a sequence can be predicted mathematically (8). The great advantages of the above methods are that they only require primary sequence information and that the calculations involved are fairly simple, particularly that for hydropathy plots. Biochemical computer software is now commercially available that can perform one or more of these analyses.

2.4. Sequence Mobility

Recently two papers have suggested that the more flexible a stretch of amino acids (the atomic or segmental mobility), the more likely it is to be immunogenic (9,10). This analysis suffers from the fact that a quite detailed X-ray crystallographic structure of the protein is required, and this consequently limits its general usefulness.

All these methods are tested for their predictive ability by comparison with empirically determined antigenic sites of whole proteins. It should be emphasized that this is not the same as immunizing with short haptenic peptides that are attached to a carrier protein, usually at many different sites, and whose conformation will be influenced by their local environment on the surface of the carrier. By using these methods, however, one can hope to improve the chance of producing anti-protein reactive antibodies, and since all essentially lead to the choice of hydrophilic sequences, this produces the more practical advantage that the peptides are likely to

be water-soluble and thus easier to use in aqueous coupling reactions (see below). One good bet if in doubt is to try the N- or C-termini of the protein, which are frequently, but not always, successful, possibly because they may more closely resemble the peptide immunogen than an internal, more conformationally constrained, sequence.

The choice of immunizing peptide may be influenced by other considerations. Functional significance of a particular region of a protein can sometimes be predicted from primary sequence motifs. An example of this is the sequence contributing to the nucleotide binding site of various protein kinases that can be rather well defined based upon conserved residues with characteristic spacing (11). Antibodies to such regions have been produced that are specific inhibitors of enzyme activity (12). Sometimes antibodies have been successfully raised to proteins homologous to that under study. Choosing the cognate region for synthesis may favor success, since, assuming the two proteins are folded in the same way, it is likely that the chosen region is superficial and available for antibody binding.

The length of peptides that have been synthesized has varied considerably, although 12–15 residues is most common. Longer sequences are more costly and do not seem to be more effective at generating anti-protein reactive antibodies, perhaps because they possess tertiary structure of their own that may not resemble that of the same sequence in the whole protein. Shorter peptides sometimes fail to stimulate a sufficiently high affinity and specific immune response for the native protein. One final point is that it is not wise to select a sequence that may be posttranslationally modified in the mature protein, such as a site of N-linked glycosylation or addition of lipid. Phosphoryla-

tion of serine, threonine, or tyrosine residues, however, does not seem to significantly influence the binding of polyclonal antibodies (13), although it might radically affect the binding of some MAbs.

3. Peptide Synthesis

There are two ways to obtain synthetic peptides; by synthesizing them oneself or by buying them. Several companies now perform custom synthesis, and some will also couple peptides to carrier molecules. The main disadvantage of this is that it is expensive. At the time of writing, a rough estimate of the cost of purchasing peptides is about \$150 per residue for a partially purified product. The more peptides that are used, the more likely one is to obtain anti-protein reactive sera, but the cost increases. The average delivery time is 4–6 wk, and this may represent an undesirable delay. The great advantage of purchase is that no skill or effort is required from the experimenter, and the product is reliable.

Peptide synthesis, however, is not too daunting for the non-organic chemist. There are two basic chemistries in use. The more common is that developed by Merrifield (for which he won the Nobel prize), which employs t-BOC amino acid derivatives, and a newer approach, developed by Sheppard, using f-MOC amino acid derivatives, which is presently gaining in popularity. For references to the detailed procedure *see* chapters 22 and 23 and refs. 14 and 15. One very considerable advantage of the new f-MOC chemistry is the avoidance of liquid hydrogen fluoride, which is used in t-BOC syntheses to cleave off protecting groups and to remove the peptide from the solid support after synthesis is complete. This compound is potentially very hazard-

ous, and although contained reaction vessels are available, great care must still be taken when using it.

Synthesis can be performed in instruments varying in cost from about \$5–10 for a simple glass tube consisting of a scinter and two taps, to an essentially fully automated machine at about \$100,000. There are now, however, more semiautomated machines appearing on the market in the \$15,000–75,000 range. The advantages of the more automated machines are that, obviously, less time is required from the operator and, perhaps, a greater degree of reproducibility and purity of product. Clearly, however, the purchase of the more automated systems can only be justified over purchase of the peptides if many sequences are required. Peptide synthesis is described in detail in Chapters 22 and 23.

4. Materials

4.1. *Glutaraldehyde Coupling of the Peptide to the Carrier Protein*

1. Keyhole limpet hemocyanin. Use as a slurry in 65% ammonium sulfate from Calbiochem (catalog no. 374811). It usually comes at about 30–40 mg/mL protein concentration.
2. Phosphate buffered saline. (PBS): 0.14M NaCl, 20 mM sodium phosphate buffer, pH 7.2.
3. 50 mM Sodium borate/HCl buffer, pH 9.0.
4. 100 mM Sodium phosphate buffer, pH 8.0.
5. 25% Solution of glutaraldehyde. It is not critical to use high-quality material such as EM grade, but this is quite cheap anyway. I have found Sigma (catalog no. G-6257) satisfactory, but almost any variety would do.
6. 1 M Glycine/HCl buffer, pH 6.0.